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作　　者：萨吉德

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二维材料的光学、电学、光电与热电特性

Optical, Electrical, Photoelectric and Thermal Electric Properties of 2D Materials

研究生：萨吉德

指导教师：孙萌涛

北京科技大学数理学院

北京 100083，中国

Doctor Degree Candidate： SAJID ALI

Supervisor： Mengtao Sun

School of Mathematics and Physics

University of Science and Technology Beijing

30 Xueyuan Road， Haidian District

Beijing 100083， P.R. CHINA

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研究生： 萨吉德

指 导 教 师： 孙萌涛 **单位：** 北京科技大学 **职称：** 教授

指导小组成员： 程宇清 **单位：** 北京科技大学 **职称：** 讲师

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摘 要

本论文采用第一性原理计算方法，对代表性二维（2D）材料的基本物理性质进行了系统研究。基于密度泛函理论（DFT），并结合电子输运、声子以及光学响应计算，旨在揭示材料微观结构与宏观性能之间的内在联系，评估二维材料在光电子器件及能源相关应用中的潜在价值。

1. 研究了生长于 Cu（111）衬底上的双层 β_{12} 类相硼烯的电子、光学及热电性质。基于 DFT 的计算结果表明，该体系表现出金属性导电特征、强可见光吸收能力以及异常大的塞贝克系数，显示出优异的热电性能。研究发现，衬底诱导应变可显著增强载流子的各向异性输运特性，同时保持近红外波段的光学透明性。本研究突出了衬底耦合在调控硼烯光学与热电性能中的重要作用。

2. 系统研究了单层 γ-graphyne 的电子结构、热电性能及电输运特性。第一性原理计算揭示了其具有类狄拉克锥的能带色散关系以及高度各向异性的载流子输运行为，从而产生超高的本征载流子迁移率。同时，γ-graphyne 表现出优异的热电性能，其特征包括较高的塞贝克系数和较低的晶格热导率，从而获得较高的热电优值。研究结果表明，γ-graphyne 是一种极具潜力的高性能热电与电子器件材料。

3. 基于第一性原理光学计算研究了 γ-graphyne 的层数依赖光学性质。结果表明，其带隙随层数变化显著，可实现从可见光到近紫外波段的可调光学跃迁。多层结构中表现出明显的各向异性光学吸收特征，反映出强烈的光-物质相互作用。这种层数可调特性使 γ-graphyne 在光电子与光子器件领域具有重要应用前景。

4. 研究了 MoS_2-SiS_2 和 WS_2-SiS_2 范德华异质结构的结构、电子及光电性质，计算方法基于 DFT 并结合 HSE06 杂化泛函修正。研究发现，该类异质结构呈现 II 型能带对齐特征，有利于光生电子与空穴的有效分离。通过应变调控，可实现直接带隙及光学吸收性质在较宽光谱范围内的可调。上述结果表明，TMD-SiS_2 异质结构在先进光电子与能量转换器件中具有重要应用潜力。

综上所述，本论文构建了一个统一的理论框架，用于理解二维材料及其异质结构的光学、电学、光电和热电性质。研究结果为通过层数调控、应变工程和界面耦合实现材料性能优化提供了清晰的理论指导。为新一代二维材料在光电子、热电及能量采集技术中的理性设计与应用奠定了理论基础。

关键词： 二维材料； 石墨炔（Graphyne)； 范德华异质结构（vdW 异质结构）；光学性质；密度泛函理论（DFT）

Optical, Electrical, Photoelectric and Thermal Electric Properties of 2D Materials

ABSTRACT

A systematic first-principles investigation of the fundamental physical properties of representative two-dimensional (2D) materials is presented. Using density functional theory (DFT) combined with electronic transport, phonon, and optical response calculations, the work aims to establish structure–property relationships and evaluate the potential of 2D materials for optoelectronic and energy-related applications. The main work is as follows.

1. Bilayer β_{12}-like phase borophene on a Cu (111) substrate using DFT-based electronic, optical, and thermoelectric modeling is investigated. The results demonstrate metallic conductivity, strong visible-range optical absorption, and an unusually large Seebeck coefficient, indicating superior thermoelectric performance. Substrate-induced strain is shown to significantly enhance anisotropic charge transport while preserving optical transparency in the near-infrared region. This work highlights the role of substrate coupling in tuning the optical and thermoelectric properties of borophene.

2. The electronic, thermoelectric, and electrical transport properties of single-layer γ-graphyne are studied. First-principles calculations reveal Dirac-cone-like band dispersion and highly anisotropic carrier transport, leading to ultrahigh intrinsic carrier mobility. γ-Graphyne exhibits excellent thermoelectric performance, characterized by high Seebeck coefficients and low lattice thermal conductivity, resulting in a high thermoelectric figure of merit. These results identify γ-graphyne as a promising candidate for high-performance thermoelectric and electronic devices.

3. The layer-dependent optical properties of γ-graphyne are investigated using first-principles optical calculations. The bandgap is found to be strongly dependent on the number of layers, allowing tunable optical transitions across the visible and near-ultraviolet regions.

Pronounced anisotropic optical absorption is observed in multilayer structures, indicating strong light–matter interaction. This layer-dependent tunability makes γ-graphyne attractive for optoelectronic and photonic applications.

4. The structural, electronic, and optoelectronic properties of MoS_2–SiS_2 and WS_2–SiS_2 van der Waals heterostructures are investigated using DFT combined with hybrid HSE06 corrections. The heterostructures exhibit type-II band alignment, enabling efficient separation of photo-generated charge carriers. Strain engineering allows modulation of direct bandgaps and optical absorption across a wide spectral range. These findings demonstrate the potential of TMD–SiS_2 heterostructures for advanced optoelectronic and energy-conversion applications.

In summary, this thesis provides a unified theoretical framework for understanding the optical, electrical, photoelectric, and thermoelectric properties of 2D materials and their heterostructures. The results offer clear design principles for tuning material performance through layer control, strain engineering, and interfacial coupling. This work contributes to the rational design of next-generation 2D materials for optoelectronic, thermoelectric, and energy-harvesting technologies.

Key Words: 2D Materials; Graphyne; vdW Heterostructures, Optical property; DFT

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1 Introduction

1.1 Background

The research community in technologically advanced societies is actively exploring efficient materials and investigating ways to manipulate them at the atomic scale for diverse technological applications[1, 2]. As a result, the demand for novel, cost-effective, highly efficient, and durable materials have surged in the modern era of nanotechnology[3, 4]. Among these, two-dimensional (2D) materials have garnered significant attention due to their exceptional properties, including high carrier mobility, ultralight weight, superior conductivity, remarkable mechanical strength, and extended spin diffusion length. Consequently, extensive research has been conducted on both pristine and modified forms of 2D materials[5]. Techniques such as melding, stacking, and structural engineering have led to the development of new material configurations, such as nanotubes, nanoribbons, and nanowires, thereby expanding their potential applications. The following sections provide an in-depth discussion of the properties and applications of various 2D materials. The exploration of these materials continues to open new frontiers in electronics, energy storage, and biomedical applications[6, 7]. Their versatility makes them promising candidates for next-generation technologies, including flexible electronics and quantum computing. Ongoing research aims to further enhance their performance and discover innovative ways to integrate them into commercial and industrial applications. The accumulation of scientific knowledge and technical breakthroughs appears to have followed an increasingly exponential and self-reinforcing trajectory throughout human history. The pace of technological revolutions has accelerated in recent decades and is expected to accelerate considerably more in the twenty-first century and beyond, providing universal conditions remain stable. We are currently on the verge of another spectacular scientific and technical revolution known as the nanotechnology era. This new technique holds the possibility of allowing us to influence matter at the nanoscale scale, leading to the development of inconceivable tools and

technologies, such as nanoparticle-based cancer medicines to high performance composites[8, 9], nanotechnology opens up new avenues for innovation in health, electronics, materials, and so on[10, 11]. Along with the emergence of developing technologies, a new class of low-dimensional systems is playing an important role in the advancement of nanoscale sciences. This comprises the zero-dimensional (0D) class, which includes quantum dots, the one-dimensional (1D) class, which includes nanoribbons, nanotubes, and nanowires, the two-dimensional (2D) class, which includes single-atom thick materials, and the three-dimensional (3D) class, which includes nanoballs and nano cones[12]. Dimensionality is one of the main features that describe different classes of nanostructures. It is a very important material property in that it determines the atomic structure of a material, as well as, has a significant impact on the constituent properties of the material. A compound or a monatomic species can be said to show divergent behavior when constrained or extended to different dimensional regimes. Carbon nanostructures are crucial components in field of nanotechnology.

1.2 Two Dimensional Materials

Modern science and technology have significantly improved with the discovery and exploration of the carbon-based materials. After the discovery of fullerene and single-walled carbon nanotubes (CNTs) as zero and one-dimensional carbon nanomaterials, respectively, scientists started to work on the isolation of two-dimensional graphitic materials or the creation of one-dimensional nanoscale ribbons with two-dimensional crystals. These endeavors bore fruit in 2004 when Geim and Novoselov reported the initial isolation and electrical characterization of a graphene transistor. The majority of these nanostructures consist predominantly of sp^2 hybridized carbon atoms, giving rise to a diverse array of structures. These range from the spherical C_{60} fullerene, resembling a soccer ball, to the ultra-thin, single-atom-layer carbon sheet known as graphene, and further extend to the cylindrical, rolled-up configurations seen in carbon nanotubes, as depicted in Figure 1-1.

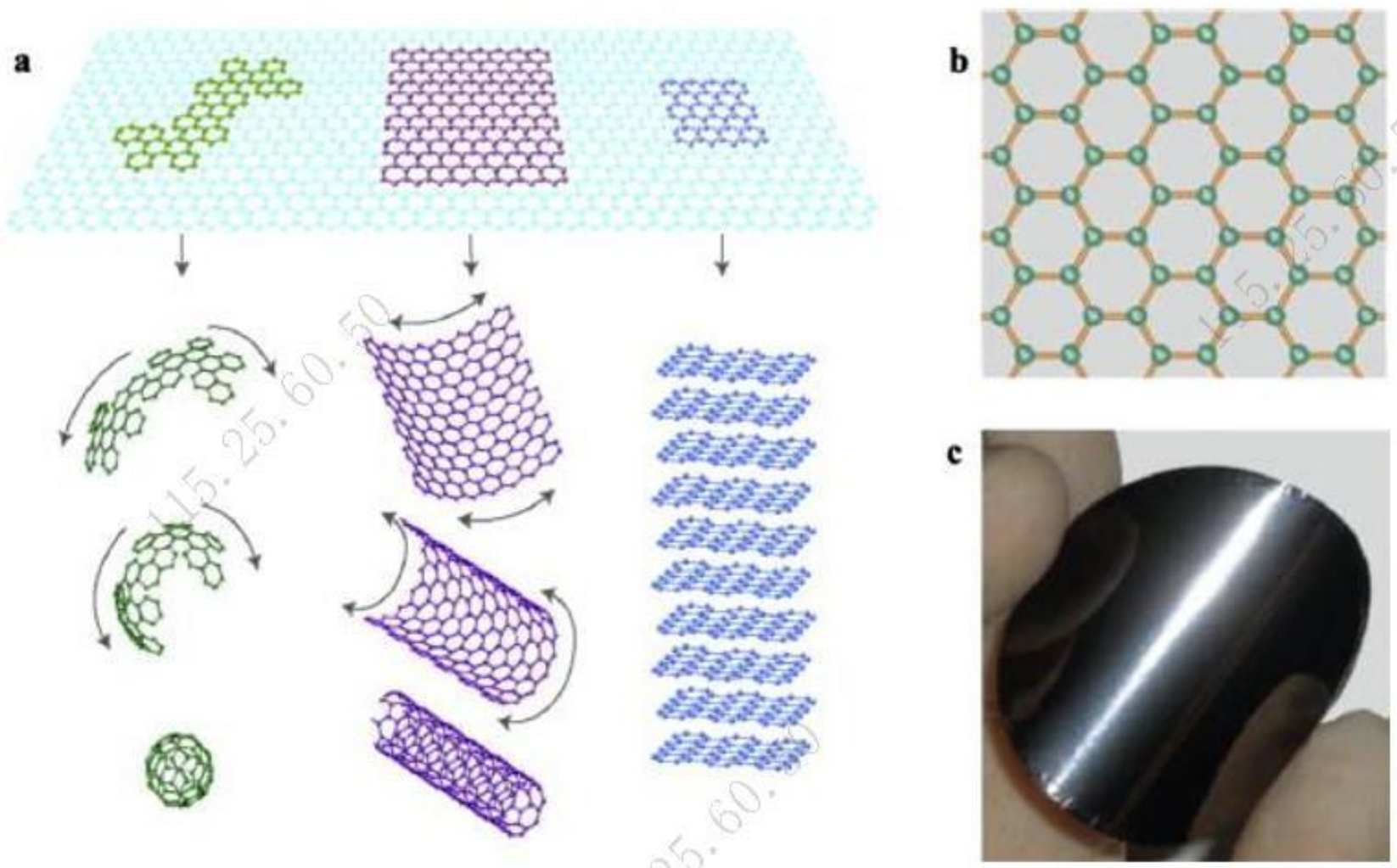


Figure 1- 1 (a) A C60 fullerene molecule, a carbon nanotube and graphite are in the order from left to right. Graphene, a single sheet of graphite, can be considered as building block of all these carbon structure (image taken from Ref. [9]). (b) Honey-comb lattice

The discovery of graphene, with its remarkable properties such as optical, electrical, photoelectric and thermal electric has paved the way for a new category of materials known as "2D materials." Inspired by the success of graphene, researchers have turned their attention to both layered and non-layered 2D materials, driven by their distinctive physical and chemical attributes. These unique properties are primarily attributed to the dimensional effect and modifications in their band structure. This review underscores the recent advancements in cutting-edge research concerning the synthesis, characterization, and isolation of single and few-layer nanosheets, along with their assembly. Additionally, it delves into the electronic, magnetic, optical, and mechanical properties of 2D materials, showcasing their potential applications in catalysis, electronics, optoelectronics, spintronic devices, sensors, high-performance electrodes, and nanocomposites[13].

Table 1-1 provides a comprehensive list of all existing members within the 2D layered materials family[14].

Table 1- 1 2D materials family (table produced from Ref. [13])

Graphene family	Graphene	*hBN* ("white graphene")	BCN	Fluorographene	Graphene oxide
2D chalcogenides	MoS_2, WS_2, $MoSe_2$, WSe_2	**Semiconducting dichalcogenides:** $MoTe_2$, WTe_2, ZrS_2, $ZrSe_2$, and so on	**Metallic dichalcogenides:** $NbSe_2$, NbS_2, TaS_2, TiS_2, $NiSe_2$, and so on	**Layered Semiconductors:** GaSe, GaTe, InSe, Bi_2Se_3, and so on	—
2D oxides	Micas, BSCCOLayered Cu oxides	MoO_3, WO_3TiO_2, MnO_2, V_2O_5, Ta_2O_5, RuO_2, and so on	**Perovskite-type:** $LaNb_2O_7$, $(Ca,Sr)_2Nb_4O_{10}$, $Bi_4Ti_3O_{12}$, $Ca_2Ta_2TiO_{10}$, and so on	**Hydroxides:** $Ni(OH)_2$, $Eu(OH)_2$, and so on	others

The properties of 2D materials can be dynamically adjusted through various in situ techniques[15] to enhance the capabilities of optoelectronic devices. Among these, the tunability of their optical properties has garnered significant attention in recent decades. Researchers have devoted their efforts to uncover a multitude of methods for achieving desired outcomes[16]. Table 1-2 provides a concise analysis of the advantages and drawbacks of various optical tuning methods for both 2D and non-2D materials. There exists immense potential for transforming optoelectronic devices through the integration of 2D materials with adjustable optical properties[17,18]. As the demand for enhanced device functionality continues to grow, the exploration of optoelectronic characteristics in 2D materials and devices has expanded beyond the conventional examination of optical intensity, photo-response, and frequency[19-21]. It now encompasses the harnessing and manipulation of polarized light and the spin states of particles, including phenomena like valley-polarization photoluminescence[22, 23], anisotropic photo responses, and spin-valley tronic devices[24, 25]. Investigating the polarized-light-dependent optical properties of 2D materials proves invaluable in unraveling the inherent physics governing strong interactions between light and matter. Furthermore, it holds the potential to play a pivotal role in the development of sophisticated optoelectronic devices within the realm of nanotechnology[26-28].

Table 1-2

Table 1- 2 Comparison of 2D materials and non-2D materials' pros and cons in terms of different tuning methods.

Tuning Method	Advantages	Weaknesses
Electrical	Strong quantum confinement in the low-dimensional layers. Low density of state.	Non-2D materials Energy loss Weak refractive index change. Unreliability of long-term use Small Pockels coefficients.
Strain	Ultrahigh Young's modulus in a single atomic thickness. High crystallinity. Strong intrinsic piezoelectric effect in the monolayer. Synergetic combination of piezophototronic effect and other properties in 2D materials. Easy integration with traditional electronic technologies and modern semiconductor processes.	—
Magnetic field	Giant Faraday rotation (e.g., monolayer graphene). Low intrinsic losses. Possess interesting magneto-optical properties such as valley Zeeman splitting in monolayer TMDCs. Broadband tunability of magneto-optical response in microwave, terahertz, and visible ranges (e.g., monolayer phosphorene).	Large footprint that limits realization of large Faraday rotation. Inflexible tunability of magneto-optic effect.
Acoustic wave	Ultrahigh Young's modulus in a single atomic thickness. Large surface area. High frequency sensitivity to absorbed molecules. Excel at both elastic-optic coefficient and refractive index. Flexible operability.	Frequency bandwidth limited to hundreds of MHz due to large structural sizes of bulk materials. Single crystals such as GaAs and $LiNbO_3$ with anisotropy are difficult to tune simultaneously in acoustic-optical effects.
Thermal heating	High thermal conductivity. Controllable and broadband optical absorption. Direct contact with photonic platforms without lattice mismatch.	Low thermo-optic effect. Low thermal conductivity. High power consumption due to large heating volume and low thermo-optic coefficient.

In recent times, the rapid expansion of the global economy has brought about substantial implications to address the ever-increasing demands of industrial production. The current daily consumption patterns on a global scale warrant special attention, as they cater to a burgeoning world population, leading to a significant upsurge in the consumption of conventional fossil fuels. This surge has, in turn, given rise to persistent environmental issues, including but not limited to global warming, acid rain, and water pollution[29-32]. The extraordinary structural, mechanical, and physical attributes inherent in 2D materials have sparked significant interest across the domains of fundamental science and practical engineering applications[33]. These materials, due to their remarkable mechanical properties, hold immense potential across a wide spectrum of applications, encompassing ubiquitous electronics, flexible displays, smart health diagnostics, and wearable computing[34-36]. As an illustrative example, take the case of single-layer MoS2, which serves as a semiconductor featuring a substantial band gap[37]. It has garnered acclaim as a highly promising contender for utilization in field-effect transistors, offering an impressive on/off ratio and heightened carrier mobility. Moreover, by enhancing the quality of samples and making astute choices in electrode materials, there exists substantial room for a significant improvement in low-temperature carrier mobility[38]. Furthermore, a profound comprehension of the thermal properties is directly linked to a pivotal application of 2D materials—namely, the conversion of thermal energy into electricity and efficient cooling. Exploring the structural attributes, distinctive features, and unique benefits of 2D carbon materials for targeted hydrogen production applications serves as a roadmap for enhancing 2D materials to achieve superior photoelectric performance and effective catalysts, ultimately contributing to the mitigation of energy challenges.

1.2.1 Novel Two-Dimensional Material Families

Researchers have looked into a number of different van der Waals (vdW) heterostructures because they can change their electrical properties and could be useful in nanoelectronics and photocatalysis. A study that stood out showed quantum coupling effects

between P-ZnO and SiC-ZnO heterostructures. It showed that when an external electric field was applied, the band alignment changed from an indirect type-I to a direct type-II. This ability to change the band gap makes these heterostructures good candidates for use in nano-electronic devices.

Another study looked at the electrical and structural properties of a bismuthine/SiC heterostructure. Researchers showed that they could effectively manipulate the electrical structure by changing the gap between layers and applying biaxial strain. Adding spin–orbit coupling (SOC) made the band gap much smaller, and the decreased effective mass of electrons suggested that they would be able to move more easily, which is an important trait for high-speed nano-electronic parts. More theoretical research looked at the ZnO/GeC heterostructure using first-principles calculations. This system had a type-II band alignment and a band gap of about 2.73 eV. Its conduction and valence band locations, as well as its ability to change the band gap depending on strain, make it good for uses like nano-pressure sensing and hydrogen evolution photocatalysis. We also looked at the photocatalytic activity of a heterostructure made of GaN and BSe. Because of its staggered band alignment and strong built-in electric field at the interface, the system showed that it could stay stable at high temperatures and separate photogenerated charge carriers in space. The fact that the band structure can be changed under strain, that the band edges are in good places for splitting water at pH levels of 0 and 7, and that it absorbs a lot of light all promote its use in solar-powered catalytic applications. BSe monolayers are known for their good physical properties. They have also been combined with ZnO to make a heterostructure with staggered band alignment, which is good for separating charges. The big drop in potential across the interface generated a strong internal electric field. This hetero-bilayer looks like a good candidate for photocatalytic water splitting since its band edges line up well with water redox potentials and its optical absorption properties can be changed by strain. A lot of people are also interested in blue phosphorene mixed with TMDCs. A comprehensive look at how they stacked, how their charges were spread out, how their bands lined up, and how they behaved optically showed that the heterostructures, like BlueP/MoS_2 and BlueP/$MoSe_2$, had type-II band alignment, which helped separate charges efficiently. They were able to convert energy

with efficiencies of 1.16% and 0.98%, respectively. When an outside electric field was applied, the originally indirect band gap changed to a direct one, which improved the optical response in both the UV and visible bands. In the same way, combining MXenes with Blue Phosphorene has led to heterostructures that can perform well in both optical and electrical applications. Because the valence and conduction band boundaries are at different places between layers, these materials can change from type-I to type-II band alignment. These kinds of heterostructures are not only flexible (they can handle strain up to 16%), but they also absorb a lot of UV-visible light, which suggests that they could be useful in flexible energy devices. First-principles methods have also been used to look at the Blue Phosphorene/BSe heterostructure. Researchers looked at several ways to stack the materials and used binding energy and phonon dispersion analysis to make sure they were stable in terms of energy and motion. Their appeal of staggered band alignment and interlayer charge transfer was seen, with BSe layers giving electrons to phosphorene because they have a lower potential. Strain engineering improved the optical and photocatalytic characteristics even more, with good band edges at pH 0 for processes that split water. Researchers have also looked at how graphene and phosphorene interact with one other from the beginning. These studies showed that the formations were stable and that the band structures changed when electric fields were applied at right angles. This opens up the possibility of using them in electronic devices that can do more than one thing. Janus monolayers have opened up new options for heterostructures. Because Janus materials are polar, blue phosphorene and Janus TMDCs make unusual hetero-bilayers with two different stacking modes. External electric fields and mechanical strain changed the alignment of their type-II bands, making it easier to combine devices with multiple functions. Moderate strain levels (up to 6%) could even convert an indirect to a direct band gap.

Finally, density functional theory (DFT) was employed to look at heterostructures made of Black and Blue Phosphorene and ZnO. Black Phosphorene had a direct band gap with type-I alignment, but Blue Phosphorene made an indirect type-II heterostructure. Notably, Black Phosphorene exhibited pronounced piezoelectric anisotropy, making it well-suited for high-performance piezoelectric devices.

1.2.2 Alternative Two-Dimensional Materials ZnO, BSe, and Blue Phosphorus Monolayers

Recent experimental advancements have produced atomically thin zinc oxide (ZnO) monolayers exhibiting thermodynamic stability by pulsed laser deposition techniques[39]. Figure 2[40] shows that the C_{6v} symmetric sp^2-hybridized network has in-plane ZnO bond lengths of 1.93 Å and buckling parameters of less than 0.08 Å. This means that it can stay stable in the environment for more than six months without surface passivation. 2D-ZnO is different from other transparent conductive oxides because it has a direct quasi-particle gap of 3.34 eV (GW-corrected) and anisotropic exciton binding energies of 120–180 m_eV. This makes it perfect for deep-UV electroluminescence applications, where it can reach external quantum efficiencies of 9.7% at 275 nm[40]. ZnO monolayers are important because they have two photocatalytic functions: (1) valence band maxima at 7.1 eV versus vacuum allow the production of hydroxyl radicals (OH•) for breaking down pollutants, and (2) conduction band offsets (4.2 eV) allow H+ reduction (HER overpotential = 0.33 V). Experimental transient absorption spectroscopy demonstrates ultrafast hot-carrier transfer ($\tau = 150$ fs) from ZnO to co-catalysts such as WS_2, attaining solar-to-hydrogen efficiencies of 8.1% under AM 1.5G[41]. To overcome the inherent constraints of broad gaps, our group's doping investigations demonstrate a 12% reduction in the bandgap (to 2.94 eV) through nitrogen substitution (ZnO0.88N0.12), along with a threefold increase in charge separation durations achieved by mitigating Coulomb scattering[42]. When looking at III-VI binary systems, BX (X=O, S) monolayers that have been mechanically exfoliated have unique trilayer structures (Fig. 2) with ABCA-type stacking and interlayer shear moduli that are higher than 25 GPa[43]. First-principles phonon dispersion shows that BSe is dynamically stable up to 800 K, and that direct bandgaps can be changed from 3.8 eV (BO) to 2.4 eV (BSe) by replacing anions. These materials show valence band edge positions (-6.2 ± 0.3 eV vs vacuum) that are higher than the OER potential (+1.23 VRHE), and the photocurrent densities for BSe/RuO_2 heterojunctions approach 3.7 mA/cm^2 at 1 sun illumination[44]. Our new angle-resolved photoemission spectroscopy (ARPES) data shows that GaS monolayers have very high hole mobility anisotropy ($\mu = 4.1$). This is because the effective mass is reduced in an anisotropic

way ($m^*h = 0.38m_0$ along Γ-M)[44]. Group III-VI binary monolayers include BO, BS, BSe, GaS, and AlTe. These materials are particularly crucial for developing nanoelectronics and optoelectronics that work well. These monolayers are stable throughout time and have a hexagonal honeycomb shape. They belong to the D3h space group. Figure 2 shows that they normally grow in bilayer shapes and that their unit cell is made up of four atoms stacked in a Y-X-X-Y arrangement. These materials are semiconductors that can absorb UV (ultraviolet) radiation. They have the proper band edge positions for both the oxidation and reduction potentials that are essential for photocatalytic water splitting.

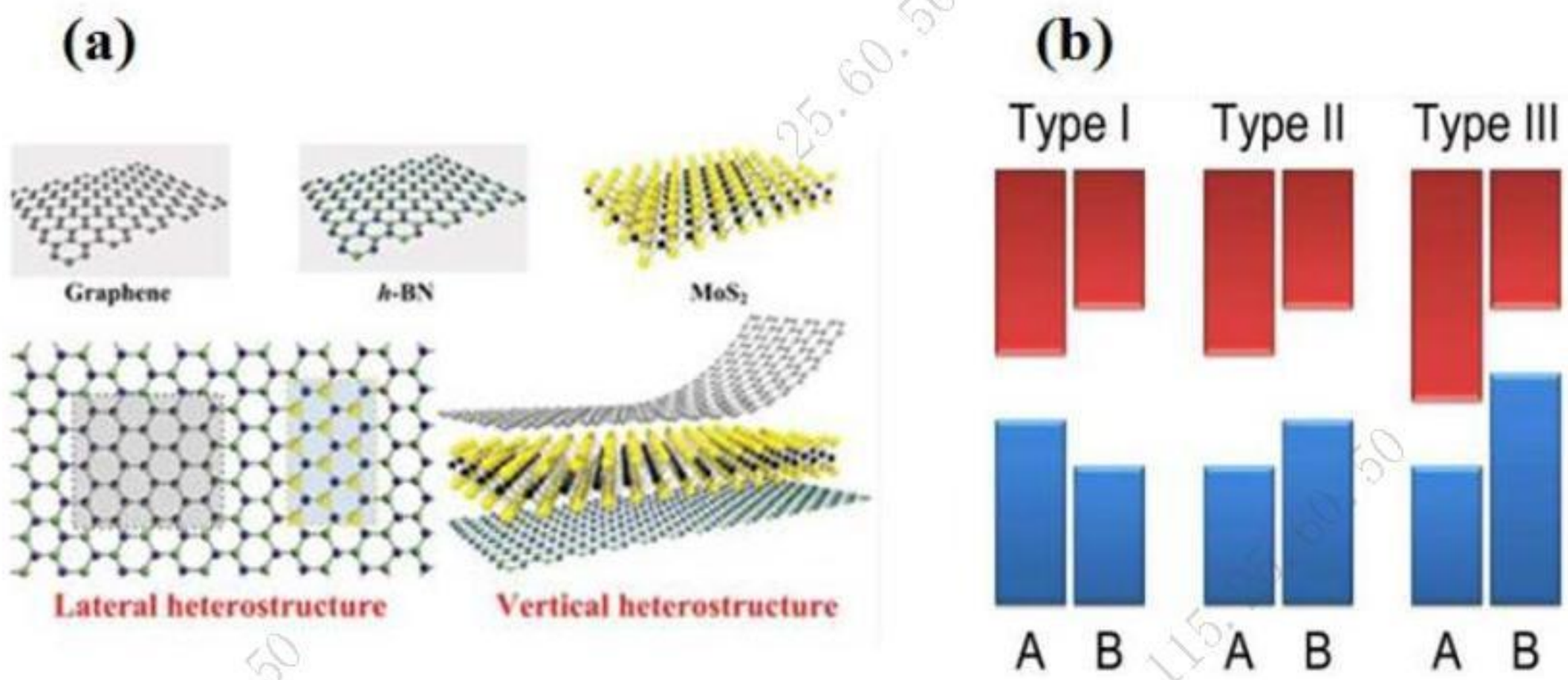


Figure 1- 2 Side and top view of group III-VI monochalcogenides monolayers.

Other two-dimensional materials, like phosphorus, have fantastic structural properties that make them operate better. For example, they are better at conducting electricity, absorbing light, and being hydrophilic[45]. There are different types of elemental phosphorus, such as red, black, blue, white, and violet. The colour of each allotrope reveals how wide its band gap is. Black phosphorus has been shown through experiments to be the most stable allotrope. There are four atoms in each unit cell of black phosphorus. Figure 1-2(a)[45, 46] shows that it contains armchair ridges and a well-defined anisotropic zigzag puckering structure. But it has a narrow fundamental band gap (1.10 eV), which makes it less effective for optoelectronic uses. On the other side, blue phosphorus (blue-P) is another allotrope with a stable shape that is practically the same as black phosphorus. Figure 1-2(b)[46] shows that Blue-P has a hexagonal shape that looks like graphene. This is because the P atoms are reversed from

down to up, which retains the same local bond angles and demonstrates isotropic buckling symmetry. There are several good things about the single-layer blue-P. For example, it has a wider fundamental band gap (2 eV), it absorbs a lot of light in both the visible and ultraviolet (UV) ranges, it has an ideal band edge position for redox reactions in water splitting, and it has weak van der Waals (vdW) interactions that make it easy to exfoliate. These traits make blue-P a great material for building quasi-2D hybrid structures for novel optoelectronic devices.[47]

1.3 Tailoring of Physical Behavior of Two-Dimensional Materials

The remarkable physical characteristics of the aforementioned 2D materials, coupled with specific challenges, create a compelling motivation to concentrate on altering their physical behavior through diverse techniques such as doping or introducing defects[48], alloying, strain engineering[49], stacking into van der Waals (vdW) heterostructures[50], and the formation of superlattices[51]. The following is a short summary of these tailoring methods, which are meant to help create new gadgets with certain features.

1.3.1 Van der Waals Heterostructures

Recently, stacking 2D materials into van der Waals (vdW) heterostructures (see Figure 1.4) has shown promise as a method that leads to fascinating new phenomena due to the quantum interaction between the stacked layers. The amazing properties of vdW heterostructures, such as their wide surface area, extremely thin layers, lack of lattice matching limitations, and capacity to create interlayer excitons, emphasise their great potential for the improvement of novel and commercial devices[52]. VdW heterostructures are often divided into three groups: type-I, type-II, and type-III, as shown in Figure 1.4. This is because of how the conduction and valence bands are set up in relation to each other. There is a gap in the first two categories that is not endless. Type-III, on the other hand, has no gap and is often called the "broken gap" structure[53]. Type-I vdW heterostructures have a "straddling gap" shape because the same layer makes up both the conduction band minimum (CBM) and the valence band maximum (VBM). This indicates that they function well with

lasers[54]. Type-II vdW heterostructures, on the other hand, are made up of different layers that make up the VBM and CBM. People often call these structures "staggered gap," and they are especially useful for modifying how holes and electrons come together. This means they can be used in nanoscale and photovoltaic devices[55, 56]. Type-II van der Waals heterostructures can change the properties of each 2D monolayer and affect how electrons and holes come together. This is why they are so crucial for the future generation of electronic and optoelectronic devices[57-59]. Another type is planar heterostructures. They put 2D layers on top of each other or next to each other, with no seams between them[52]. Nonetheless, some challenges, such as achieving atomically sharp and coherent junctions, robust atomic bonding at interfaces, edge contamination, and the absence of interlayer excitons, hinder the application of lateral heterostructures in advanced nanoelectronics devices. The main goal of this study is to create new van der Waals heterostructures by combining two-dimensional materials, such as Janus TMDC, ZnO, BSe, and blue-phosphorus monolayers. This is because vdW heterostructures have certain very special physical properties. These mostly semiconducting monolayers will help us figure out how to make new devices work in the fields of nanoelectronics, optoelectronics, and renewable energy[60].

1.3.2 Strain Engineering

Strain engineering refers to the modification of a material's physical behavior by altering its mechanical or structural properties through stretching or compression[61]. However, certain limitations, such as lattice defects and the low strain failure threshold (around 0.1%), restrict the ability to apply external strain to bulk materials for property tuning[60]. In contrast, external strain has a significant impact on the bond length, bond angle, bond strength, and the interaction of electronic orbitals in 2D materials. This ability to control the physical behavior of 2D materials opens up possibilities for their use in a wide range of optoelectronic and renewable energy applications. Strain is generally classified into two types: compressive and tensile strain, each of which influences the physical properties differently[62]. Tensile strain typically enhances carrier mobility, reduces the band gap, weakens bond strength, and causes a red shift in lattice dynamics. Conversely, compressive strain increases bond strength,

widens the band gap, decreases carrier mobility, and induces a blue shift in lattice vibrations. Additionally, uniaxial or biaxial strain predominantly affects the structural and electronic properties of materials, enabling the development of flexible devices for logic and electronic circuits[63]. Strain engineering also significantly alters electrical, optical, spin splitting, and chemical properties. The large surface area and high stretchability of 2D materials make them ideal candidates for applying strain, often with only minor modifications. Furthermore, strain engineering can be used to tune the properties of vdW-stacked 2D materials, facilitating the fabrication of advanced devices with enhanced functionalities[63].

1.4 Advancements in Two-Dimensional Materials and Their Van Der Waals Heterostructures

Atomically thin, isolated layers of layered materials are reassembled through weak van der Waals (vdW) interactions to form vdW heterostructures (vdW-HS), which exhibit remarkable properties and new functionalities for applications in electronics, optoelectronics, and energy harvesting[64].

1.4.1 Monolayers of Transition Metal Dichalcogenides

Graphene is a two-dimensional semimetal characterized by the absence of a band gap and a significant ambipolar electric field effect. Using scanning tunnelling microscopy, Boubekeur et al. identified silicene[65]. It has a honeycomb shape like graphene and is particularly valuable in nanotechnology. Mak et al. discovered that a single layer of MoS2 exhibits direct band gap characteristics and a luminescence efficiency of 10^4, much surpassing that of its bulk counterpart[66]. Wu et al. conducted experiments that confirmed MoS2 thin films have direct band gap properties, signifying an n-type conduction mechanism[67]. First-principles calculations have been employed to analyze single, bi-, and tri-layered MoS2 and WS2 synthesized by various chemical methods. Ramakrishna et al. confirmed the stability of MoS2 monolayers using first-principles calculations, revealing a direct band gap of 1.7 eV at the K-point of the Brillouin zone[68]. The vapor phase transport method was used for the first time to manufacture MX2 monolayers[69, 70]. The arrangement

of the metal atoms in these monolayers caused them to have diverse structural phases. The 1T and 2H phases were the most common. A study found that the photoluminescence intensity was quite high and that the size of the transition and chalcogen atoms had an effect on the band gap[71] [84]. Cheng et al. utilized ab initio calculations to demonstrate that all Janus monolayers (MoSSe, WSSe, WSTe, and WSeTe) are stable, with the exception of MoSTe and MoSeTe[72]. Lu et al. employed chemical vapor deposition to replace the Sulphur atom with selenium in the MoS2 monolayer, whereas Zhang et al. substituted selenium with Sulphur, thereby confirming the stability of the Janus MoSSe monolayer[73, 74]. Zhang's group also used density functional theory (DFT) simulations to link the electrical structures and Raman vibration modes of the MoSe monolayer to what they witnessed in the lab. Yao et al. demonstrated that the significant spin-orbit coupling (SOC)-induced Rashba spin splitting in MXY monolayers (where M = Mo, W and X ≠ Y = S, Se, Te) renders them suitable candidates for prospective spintronic applications[75]. Xia et al. investigated the structural, optoelectronic, and photocatalytic characteristics of MXY monolayers utilizing DFT, demonstrating that the band transition (from direct to indirect) and the induced dipole moments are associated with the atomic radius and electronegativity disparities of the X and Y elements[76]. Chaurasiya et al.'s simulations of phonon band dispersion showed that Janus WSSe and WSe_2 monolayers are stable. They also noted that biaxial compressive strain shortens bond length and makes these monolayers unstable. They also found that biaxial strain changes the way band gap transitions happen, from indirect to direct[77]. Additionally, in the WSSe monolayer, due to the broken symmetry of the Janus MXY structure, Rashba spin splitting was observed at the Γ-point, which is enhanced by strain and external electric fields[78]. Blue phosphorene (BP) is a type of phosphorene that has a hexagonal, buckled structure. It has a large band gap of about 2 eV[79]. Both experimental and DFT studies have shown that BP could be useful in optoelectronic and nanoelectronics applications. Sahoo et al. utilized DFT simulations to validate the stability of single-layer planar ZnO monolayers[80, 81]. In the same way, a single layer of SiC monolayer was made using sonication and X-ray diffraction (XRD). Photoluminescence (PL) showed that it had hexagonal symmetry and great light-emitting properties[82].

1.4.2 Van der Waals Heterostructures of Transition Metal Dichalcogenides

The quantum interaction among stacked layers in van der Waals (vdW) heterostructures gives rise to novel phenomena and facilitates the modulation of the properties of individual monolayers, thereby categorizing them as new materials. First-principles calculations, preserving unstrained lattice constants, have been employed to investigate the structural and optical properties of van der Waals heterostructures, including MoS_2, various transition metal dichalcogenides (TMDCs), graphene, and boron nitride (BN)[47]. Because the interactions at the Γ-point are so strong, these vdW heterostructures have an indirect band gap. In van der Waals heterostructures made of MoS2, BN, and graphene, the layers don't interact much, and the band structure works as if the layers are independent. MoS_2 and WS_2 heterojunctions, on the other hand, have optically active band gaps that are defined by the location of the valence band maximum (VBM) and conduction band minimum (CBM) on different monolayers[83]. Scientists have also used first-principles calculations to learn how to experimentally find heterostructure configurations by looking at how stacking and thickness change the phonon spectra of MoS_2 and WS_2 monolayers[84]. The 2D elastic moduli of bilayer MoS_2-WS_2 were lower than those of their respective monolayers but comparable to those of homo-bilayers, indicating robust interlayer contacts within the vdW heterostructures[85]. Researchers employed chemical vapor deposition (CVD) to fabricate MoS_2-hBN vdW-HS characterized by minimal lattice misalignments, pristine interfaces, and excellent interlayer contacts[85]. First-principles calculations have also examined the stacking sequence, band gaps, and optical spectra of various TMDC vdW heterostructures. It has been demonstrated that MoS2-WS2 and $MoSe_2$-WSe_2 van der Waals heterostructures are effective for photocatalysis[86].

1.4.3 The Optical and Electrical Characteristics of vdW Heterostructures

Optical absorption spectroscopy revealed that the $MoSe_2$/WSe_2 vdW-HS exhibited a significant spectrum broadening of 20 to 35 m_eV, with charge separation occurring at the band boundaries due to A and B excitons lasting from 20 to 35 fs[87]. Amin et al. employed first-principles calculations to elucidate the direct (indirect) band gap characteristics of MoS_2-

WSe_2 (WS_2-$MoSe_2$) vdW-HS[88], demonstrating that the attenuation of optical spectra relative to the individual monolayers was a consequence of interlayer charge transfer. Wang et al. investigated Li-intercalation in single and hetero-bilayers of MX_2 (M = Mo, W; X = S, Se) and demonstrated that the MoS_2/WS_2 van der Waals heterostructure exhibited superior electrical conductivity[89]. They showed that lithium atoms move easily through MoS_2/WS_2 and MoS_2/$MoSe_2$ van der Waals heterostructures.

1.4.4 Photocatalytic Applications of vdW Heterostructures

Lan et al. recorded a significant UV photo-response obtained by superimposing WS_2 on ZnO film, thereby improving the transport of photogenerated carriers, with ZnO acting as a light-harvesting element in the ZnO/WS_2 vdW-HS[90]. People say that SnSe/MoS_2 vdW-HS can move electricity and light very well. They have a type-II band alignment, a high on-off current ratio (1×10^5 A), a strong photoresponsivity, and a quick response time, which makes them good for self-powered applications[91]. Strain engineering and external electric fields significantly influenced the optoelectronic properties of MoS_2/SiC vdW-HS, rendering them advantageous for nanoscale applications[92]. First-principles simulations on the energy and dynamic stability of SiC-MX_2 vdW heterostructures investigated the transition from indirect to direct type-II band alignment under mild tensile strain[93]. These heterobilayers showed better optical responses and photocatalytic effectiveness, which shows that they could be useful in optoelectronic and water-splitting applications.

1.4.5 Photocatalysis utilizing ZnO-based van der Waals heterostructures

ZnO has been getting a lot of attention in new optoelectronic and renewable energy technologies because it has great physical properties. Wang et al. examined the optoelectronic characteristics of MoS_2/ZnO van der Waals heterostructures and discovered that the inclusion of a type-II bands gap and an electric field enhanced charge separation. ZnO/WS_2 vdW-HS was acknowledged as a remarkable photocatalyst for water splitting due to its effective charge separation and optimal band edge positioning. Additionally, the band gaps of ZnO/WX_2 (X = S, Se) were shown to stay the same regardless of the angle of rotation[94].

1.4.6 How to Make and Use vdW Heterostructures

Li et al. utilized DFT to examine the transformation of a p-type Schottky contact in SiC/graphene to an n-type Schottky contact under the influence of an external electric field[94]. The direct type-II SiC/MoS_2 vdW-HS had very good optical absorption in the visible and far-infrared ranges. The tri-layer graphene/SiC/MoS_2 vdW-HS with different stacking arrangements had Schottky barriers that could be changed. Zhang et al. investigated WS2/BSe vdW-HS, characterized by an indirect type-II band alignment, which prolonged the lifespan of photoinduced charge carriers by facilitating charge transfer from BSe to WS_2 monolayers[95]. The band edges met the usual redox potentials, which meant that the hydrogen evolution reaction (HER) could happen at a wide range of pH values[96]. Luo et al. demonstrated that the $MoTe_2$/$MoSe_2$ type-II heterojunction is an excellent candidate for future nanoscale optoelectronic and photovoltaic devices due to its effective charge separation and the ability to modulate its photoresponsivity by varying the gate voltage[97].

1.5 Strain Engineering in Two-Dimensional Materials and Their Van Der Waals Heterostructures

Strain engineering is a useful method for altering the characteristics of 2D materials, such as their band structure, band gap, and carrier mobility. We can use epitaxy on small amounts of material to test out strain and make devices work better. For example, a compressive strain of 0.5% changed the indirect band gap of MoS_2 into a direct band gap[98]. WSe_2 monolayers can be changed by a compressive strain of 1%, and WTe_2 can be changed by a compressive strain of 2%[49]. Calculations based on first principles have shown that graphene monolayers are stable under strains of up to 14% and tri-layer graphene is stable under strains of 12.5%[99]. In van der Waals heterostructures, engineering the band gap is crucial for achieving effective carrier separation in optoelectronic, photocatalytic, and photovoltaic devices. Pak et al. used DFT simulations to look into the excitonic behavior, band gap, and photoluminescence (PL) intensity modulation of MoS_2/WS_2 vdW-HS when it was under uniaxial tensile and compressive strains[100]. Additionally, it was reported that

strain could alter the band gap in InSe/MoS_2 vdW-HS. When the strain was compressive, the band gap switched from indirect to direct behavior[101]. Farkous et al. theoretically demonstrated an indirect type-II band alignment in MoS_2/WS_2 van der Waals heterostructures. A substantial biaxial strain of 14% prompted a semiconductor-metal transition, markedly affecting the optical properties[102]. Zhang et al. investigated the ZnO/BP van der Waals heterostructure, which exhibited charge separation and enhanced optical absorption across the ultraviolet to infrared spectrum under applied strain[103]. Zhen et al. say that the MoSSe-ZnO vdW heterostructure worked as a semiconductor with indirect type-II band alignment. The band gap changed from semiconductor to metallic when tensile and compressive loads were applied[104].

1.6 Making Two-Dimensional Materials to Work

Two-dimensional (2D) materials are becoming important in many disciplines, such as electronics, energy storage, and photonics, since they exhibit certain qualities[105, 106]. These materials usually work better than bigger ones because they are smaller, have a greater surface-to-volume ratio, and quantum confinement effects. But sometimes you need to tweak the natural features of 2D materials to make them work better for a certain job. Functionalizing 2D materials is one of the finest ways to make them work better. In certain cases, it means modifying the chemicals or the surface so that they perform better[107]. There are two basic ways to functionalize 2D materials: by changing the chemical structure and the edges and surface. Both are incredibly crucial for how 2D materials look, move, react with other things, and carry electricity.

1.6.1 Chemical functionalization to make things work better

Adding new atoms, molecules, or groups to the surface or structure of a 2D substance is called chemical functionalization[108]. This approach can improve the material in a number of ways, such as by making it more stable, better at conducting electricity, and better at functioning as a catalyst. You can chemically functionalize a substance by adding something to it, making covalent and non-covalent bonds, or attaching it to a surface[109]. These improvements make it work better in a number of ways, like storing energy and in

optoelectronics.

(i) Various forms of chemical functionalization

Adding foreign atoms (dopants) to the lattice of a 2D material is a typical approach to modify its electrical characteristics[110]. Doping can vary the number of carriers, which changes how well the material conducts electricity. For example, adding boron or nitrogen atoms to graphene can modify its band structure and the quantity of charge carriers by a lot. Adding other elements to transition metal dichalcogenides (TMDs) like MoS_2 or modifying their bandgap can also make them better for some optoelectronic uses by making them more catalytic. Chemical vapour deposition (CVD), atomic layer deposition (ALD), and wet chemical methods all let you add impurities to items before or after they are made[111]. When chemical bonds form between the atoms of a 2D material and the functional groups or molecules that are added to it, this is called covalent functionalization. This technique makes a change that lasts longer than alterations that don't include covalent bonds. Adding hydroxyl or carboxyl groups with oxygen to graphene or TMDs via oxygen plasma or chemical oxidation makes them better at functioning with electrolytes. That's why they're perfect for batteries and supercapacitors, which store energy. Covalent functionalization of TMDs can also modify how the electrons in them are organized. This is why they are good catalysts and sensors[112].

(ii) Non-Covalent Functionalization

Non-covalent functionalization[113] uses weaker forces, like hydrogen bonding, van der Waals forces, or π-π stacking interactions. This method enables you change how 2D materials seem on the outside without changing their natural qualities. Non-covalent functionalization is an excellent approach to modify materials without changing their electrical characteristics too much[114]. For example, you can combine graphene oxide with organic compounds or polymers to make it helpful for producing sensors, delivering medicine, and cleaning water. Adding tiny molecules or polymers to TMDs can also make them more stable or help them operate better in devices that alter energy. Chemical functionalization can make the electrochemical properties of 2D materials much better. This is why they are good for things that store and change energy, including batteries,

supercapacitors, and fuel cells[115]. For example, nitrogen makes graphene operate better in supercapacitors. On the other hand, adding covalent functional groups to MoS_2 makes it work better in lithium-ion batteries. Adding functional groups to TMDs like $MoSe_2$ and WS_2 can also make them better at catalyzing hydrogen evolution processes (HER), which break down water. This is a way to make renewable energy perform better[116]. Functionalized 2D materials are also commonly used in sensors and photodetectors. You can make graphene and TMDs more sensitive to light, biomolecules, or gases by adding certain compounds or nanoparticles to them. For example, sensors consisting of functionalized graphene can discover gases like NO_2 and NH_3 even when they are very low in concentration. Functionalization can also vary the bandgap of TMDs, which makes them better for use in photodetectors that work at different wavelengths[117], such those in the infrared and visible ranges. Chemical functionalization can make 2D materials work better as catalysts, especially for hydrogenation, oxygen reduction, and carbon dioxide reduction. Adding catalytic groups to the surface of 2D materials or linking them together with covalent bonds can modify their chemistry. This can help their active sites function better for some reactions[118]. TMDs, in particular, have shown a lot of promise as electrocatalysts for fuel cells and other energy-related uses.

(iii) Changes to the edges and surfaces of two-dimensional materials

Chemical functionalization is not the only way to affect the characteristics of 2D materials; modifications to the edges and surface are just as essential[119]. Most of the time, these adjustments are made to the surface or edges of the material to make it operate better in particular conditions. 2D materials have a lot of surface area, therefore changing the edges and surface can vary how reactive, stable, and well they perform[108].

(iv) Etching and Patterning

Etching is one of the most common ways to modify a surface. It means slowly and carefully taking items off the surface[120]. Common ways to modify the shape of 2D materials on the nanoscale are plasma etching, laser etching, and chemical etching. This method lets you construct surfaces or edges with accurate shapes, including sharp points or facets that point in a certain direction[121]. For instance, etching can be used to add edges to TMDs or

graphene that are full of defects. This can help them change energy into various forms. Using plasma treatment is another good approach to modify the surface of 2D materials. You can utilize plasma to add groups to the surface that do things, such amine, carbonyl, and hydroxyl[122]. This can modify the material's chemical reactions, how much water it can store, and how much charge is on its surface. People often utilize plasma treatments to help polymers or solvents mix better with two-dimensional materials. This makes them better for use as electrodes in energy storage devices or in composite materials. People routinely utilize tiny chemicals and polymers to improve the way surfaces work[123]. This is especially true for 2D materials that are employed in electronics and optoelectronics. Putting conductive polymers on top of graphene or TMDs, for instance, can make them better at moving charge. This makes them good for sensors and transistors. Adding organic molecules to the surface of TMDs can also help them mix with solvents, which can make them better for giving drugs or taking pictures[124].

(v) Catalysis and Reactivity

To make 2D materials better at catalyzing processes, it's highly vital to modify their edges and surfaces. Changing the surface or adding reactive edge sites can help the material react better with reactants, which can make it a better catalyst[125]. The edges of MoS_2 and WS_2 that are full of defects are great at hydrogen evolution reactions (HER) and other electrocatalytic activities.

(vi) Electronics and Optoelectronics

The edges and surfaces of 2D materials are also very significant for electronics and optoelectronics. Changing the surface of a material can make it better at conducting electricity, let charge carriers move around more readily, and make it shine more brightly. For instance, adding certain molecules to the edges of graphene or TMDs can make field effect transistors (FETs) or photodetectors work better[126-128].

2 Calculation Principles and Methods

2.1 Many Body Problem

One of the hardest parts of computational materials research is figuring out how the electrons in a system's atoms or molecules are arranged. This is true for both bulk materials and structures with few dimensions. Wave functions explain how electrons act when they are near an atom's nucleus. The Schrödinger wave equation[129] can be used to look at the electrical structure of solids or quantum systems that stay the same across time as shown equation 2.1. The Schrödinger equation is the main way to find out where electrons are in a material and how they interact with each other and with things outside of the material. But this equation is hard to solve for systems with a lot of particles since electron interactions are so complicated. Density Functional Theory (DFT) is one way to solve this problem and make math easier. Researchers can look at things like the optical response, band structure, and transport mechanisms in great detail thanks to these technologies.

$$H\psi = E\psi \tag{2-1}$$

$$H = -\frac{\hbar^2}{2m_e}\sum_i \nabla_i^2 - \sum_{i,j}\frac{Z_l e^2}{r_j - R_l} - \sum_{i,l} + 1/2\sum_{i\neq j}\frac{e^2}{r_i - r_j} - \sum_l \frac{\hbar^2}{2M_l}\nabla_l^2 + 1/2\sum_{l\neq J}\frac{Z_l Z_J e^2}{R_l - R_J} \tag{2-2}$$

In this equation $\hbar, (m_e)\ (M_l), r_i\ (R_l)\ and\ Z_l\ (e)$ represent the plank constant, mss, position of electron (nuclei) and charge of electron (nuclei), respectively Kinetic energy of electron (nuclei) is mentioned in second term, while the third and fourth terms in equation 2.2 is for electron-electron and nuclei-nuclei repulsion respectively. In quantum system electrons and nuclei are present in the order of 10^{23}, which shows that this problem is unsolvable. For solving these quantum systems (many body problem) approximations are required.

2.2 Born-Oppenheimer Approximation

In 1927 M. Born and J. R. Oppenheimer proposed that in a molecular system the wave function can be consider separately for nuclei and electron such as[130].

$$\psi(x)=\psi_{electrons}\,\psi_{nuclei} \tag{2-3}$$

As the mass of nuclei is heavier than that of electrons, hence shows negligible respond to coulomb interactions and have much larger inertia. This realized that, these nuclei are assumed as stationary. Hence neglecting the kinetic energy term of nuclei from Eq. 3.2 (Hamiltonian equation). M. Born and J. R. Oppenheimer gives the new Hamiltonian equation;

$$H = -\frac{\hbar^2}{2m_e}\sum_i i + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_i - r_j} + V_R\alpha\,\mathrm{ext}\,r_i \tag{2-4}$$

$V_R\alpha\,\mathrm{ext}\,r_i$ is for the external potential[130]. As in a solid material (compounds) there are a large number of electrons and also nuclei, showing the fact that by neglecting the kinetic energy term of nuclei is still impossible to solve the many bodies problem. Therefore, we need alternative better method for solving the many bodies problem by choosing suitable variables as input.

2.3 Hartree-Fock Approach

Hartree[131] replaced the wave function of a N numbers of electrons system by orthogonal spin orbital φi(x); such that:

$$\Psi(x) = H(x_1, x_2, x_3, \dots, x_N) = \phi_1(x_1)\phi_2(x_2)\phi_3(x_3)\dots\phi_N(x_N) \tag{2-5}$$

In this Eq. φi(x) represent the single electron orbital with variable x which gives information about spatial and spin coordinates. Drawbacks of Hartree approach is that it fails to deals with fermions. Here the unsymmetrized wave function which contains N number of electrons was proposed, which can be further written in the form of Slater determinant[132];

$$\Psi(x) = H(x_1, x_2, x_3, \dots, x_N) = \begin{pmatrix} \phi_1(x_1)\phi_1(x_2)\phi_1(x_3)\dots\phi_1(x_N) \\ \phi_2(x_1)\phi_2(x_2)\phi_2(x_3)\dots\phi_2(x_N) \\ \phi_3(x_1)\phi_3(x_2)\phi_3(x_3)\dots\phi_3(x_N) \\ \vdots \\ \phi_N(x_1)\phi_N(x_2)\phi_N(x_3)\dots\phi_N(x_N) \end{pmatrix} \tag{2-6}$$

For this equation the effective potential will be:

$$V_{\mathrm{HF,eff}}(r_i) = F(r_i) - T(r_i) = -\sum_\alpha \frac{\alpha Z_\alpha}{r_{i\alpha}} + \left(2J_j(r_i) - K_j(r_i)\right) \tag{2-7}$$

Here $F(r_i)$ and $J_i(r_i)$ ($K_i(r_i)$) represent Fock and coulomb (exchange) operators respectively. HF method neglect the correlated motion of electrons which help us by finding the Hamiltonian directly by using first principle instead of parameterization[133]. But it also fails to predict the properties of materials.

2.4 Density Functional Theory

Generally, different schemes adopted to solve Schrodinger equation for many body particles may be based on wave function. Instead, the electron density was supposed to replace the wave function for interacting and non-interacting under external potential and reduced 3N spatial variables to only 3-variables. Thomas and Fermi (TF) presented a model in 1927 and suggested that uniform electron density may be used for describing kinetic energy of a homogeneous system of electrons. However, TF-model has a major drawback to incorporate the energy for a real inhomogeneous or interacting system[134, 135]. In 1998, Walter Kohn received the Nobel prize for extraordinary work in chemistry[136]. Hohenberg and Kohn proposed many body system comprising electrons with non-degenerate ground states and all information are contained in ground state electron density like many body wave functions. To keep this in view, two theorems have been formerly presented for DFT.

2.5 Hohenberg-Kohn Theorems

2.5.1 First Theorem

HK 1st theorem described total energy functional and states that a system moving in an external potential (V_{ext}) and contained N number of interacting particles (electrons and holes), then V_{ext} can be found by ground state electron density. For that the energy functional $E_{HK}[\rho]$ will be:

$$E_{HK}[\rho] = F[\rho] + \int V_{\text{ext}}(r)\,\rho(r)\,dr \qquad (2\text{-}8)$$

Here $F(\rho)$ is for kinetic energy $T[\rho]$ and also for e--e- interaction energy.

2.5.2 Second Theorem

The HK 2nd Theorem described $E_{HK}[\rho]$ and defined in terms of exact ground state density, which represent any V_{ext}. $E[\rho]$ will be minimum energy only if the density is the real ground energy density, and given as:

$$E_0 \leq E(\rho) \tag{2-9}$$

Therefore, by using this equation one can find the accurate value of ground state density $\rho(r)$ for new Hamiltonian. Hence, here the main problem is for finding the suitable approximation to calculate energy function.

2.6 Kohn-Sham Approximation

KS introduced a new approach in 1965, where he replaced the interacting many electrons problem by a fictitious non-interacting Kohn-Sham system[137]. on the basis of KS approximation, the ground state density of the system will be equal to ground state density of KS system, which is given as

$$E_v^0[\rho] = T_S[\rho] + V_0[\rho] + E_H[\rho] + E_{XC}[\rho] \tag{2-10}$$

Here $T_S[\rho]$, $E_H[\rho]$ and $E_{XC}[\rho]$ represents kinetic energy, classical coulomb interaction and exchange-correlation. In this equation $E_{XC}[\rho]$ term is still unknown, which contained electron correlation term. Exchange-correlation ($E_{XC}[\rho]$) term will be written as;

$$E_v^0[\rho] = T_S[\rho] + V_0[\rho] + E_H[\rho] + E_{XC}[\rho] \tag{2-11}$$

For exact form of $E_{XC}[\rho]$ which is unknown, would be approximated. At $\rho=\rho_0$ the ground state energy for real system and for KS system will be same, and would be written as;

$$V_{KS}[\rho] = V_0[\rho] + \frac{\delta E_H[\rho]}{\delta\rho(r)}|_{\rho=\rho_0} + \frac{\delta E_{XC}[\rho]}{\delta\rho(r)}|_{\rho=\rho_0} \tag{2-12}$$

Or

$$V_{KS}[\rho] = V_0(r) + V_H([\rho_0]; r) + V_{XC}([\rho_0]; r) \tag{2-13}$$

in this equation V_{XC} represent exchange correlation potential (non-local complex function), gives a new wave function when apply to the old wave function. Due to the lack information of explicit function for exchange correlation potential demand its approximated form.

2.7 Jacobs Ladder

For solving V_{XC} different approximations is used to find this unknown term. These methods are named accordingly their level of accuracy, which is known as Jacob's ladder[134]. In the below section we discuss these methods in ascending order of Jacobs ladder.

2.7.1 Local Density Approximation

Kohn and Sham (1965) introduced a well-known approximation to solve the unknown term, called local density approximation (LDA)[137]. In LDA the density of any system is localized around a spatial point (r) for interacting homogenous electron gas system, the exact value for $[\rho]$ can be nearly approximated as;

$$E_{\mathrm{LDA}}^{XC} = \int dr\, \rho(r)\, V_0(r)\, E_{\mathrm{LDA}}^{XC}[\rho] \tag{2-14}$$

And also

$$E_{XC} = \varepsilon_X[\rho] + \varepsilon_C[\rho] \tag{2-15}$$

In eq. 3.14, $E_{XC\,LDA}[\rho]$ is function of ρ(r) at a point r. $\varepsilon_X[\rho]$ was calculated by P. Dirac in 1930 for HEG systems and $\varepsilon_C[\rho]$ is available from accurate quantum 25 Monte-Carlo simulations[138, 139]. LDA successfully explain the ground state properties for materials. But the drawbacks of LDA is that, its overestimate weak vdW forces, binding energies (minimum energy) delocalized the higher state, cohesive energy and also underestimates the band gap.

2.7.2 Generalized Gradient Approximation

To overcome the limitation of LDA, a new approximation known as Generalized gradient approximation (GGA) is introduced. In GGA gradient of density in $E_xc[\rho]$ where added. The new equation for $E_xc[\rho]$ will be;

$$E_{\mathrm{XC}}^{\mathrm{GGA}} = \int \rho(r)V_0(r) + E_{\mathrm{XC}}^{\mathrm{GGA}}[\rho, \nabla\rho]\, dr \tag{2-15}$$

For the GGA approach other functions are used, most famous of them are PW91[140], PBE[141], PRBE[142] and PBEsol[143] functionals. The limitation of LDA to overestimate the binding energy is significantly corrected by using PBE functional. Although for metallic or covalently bond systems the GGA functional gives correct value, but fails to treat vdW forces and also highly localized orbitals (d- and f-orbitals). In meta-GGA functional (M-GGA, 3rd rung)[144] which includes the second order density gradient in $E_xc[\rho]$ gives accurate results as compared to GGA and LDA approximations. The difference between GGA and M-GGA is that GGA is semi-local while M-GGA is fully non-local function of the density.

2.7.3 Hybrid Functional

Hybrid functions gives correct band gap of semiconductors and insulators[145]. In the hybrid function, the HF part is added from LDA (GGA). HSE is the most popular hybrid function using screening exchange potential[146,147]. The screening function uses remote HF precise exchange to describe the Coulomb potential. Therefore, for the calculation of large-scale systems, it is very necessary to use HF accurate exchange. SR exchange is obtained from HF precision exchange, while long-range exchange is obtained from PBE function. HSE hybrid function is written as;

$$E_{\mathrm{XC}}^{\mathrm{HSE}}[\rho] = \alpha E_{\mathrm{X}}^{\mathrm{HF,SR}}(\mu) + (1-\alpha)E_{\mathrm{X}}^{\mathrm{PBE,SR}}(\mu) + E_{\mathrm{X}}^{\mathrm{PBE,LR}} + E_{\mathrm{C}}^{\mathrm{PBE}} \tag{2-17}$$

Over PBE[148], in small to intermediate range, Heyd-Scuseria-Ernzerhof (HSE) hybrid functional shows good results. By and large, HSE outcomes are autonomous of choice of splitting parameter exclusion of bandgap alone and there is no systematized

method to compare bandgap assessed by HSE with experimental one[147, 148]. The 5th bar of Jacob's ladder considers functionals which uses all unoccupied and occupied orbital and utilized in random phase approximation (RPA)[149-151].

2.8 Dispersion Correction

Although the standard XC functions of DFT (including LDA), the different flavors of GGA and HSE describe the ground state properties of solids and molecules, where the accuracy of each approximation is different. However, these functions cannot correctly account for the long-distance correlation, which will lead to vdW interaction and is caused by the dynamic correlation between fluctuating charges. Grimme et al.[152, 153] added the long-range forces to DFT for calculating the dispersion term (E_{disp}) in DFT calculations. Which as;

$$E_{\mathrm{DFT-D}} = E_{\mathrm{KS-DFT}} + E_{\mathrm{disp}} \tag{2-18}$$

While (E_{disp}) is given by;

$$E_{\mathrm{disp}} = -s_6 \sum_{i=1}^{N_{\mathrm{atoms}}-1} \sum_{j=i+1}^{N_{\mathrm{atoms}}} \frac{C_6^{ij}}{R_{ij}^6} f_{\mathrm{dmp}}(R_{ij}) \tag{2-19}$$

In this equation s_6 represent the scaling factor. f_{dmp} is also added to neglect the singularities at short range of distances.

2.9 GW Approximation

It is well known that the DFT calculated the exit value of ground state but failed to describe the excited state properties of any materials[154]. For instance, it produces erroneous or underrates the bandgap in a few cases of semiconductors and insulators and substantially diverges in optical spectra as contrasted with experimental outcomes[155, 156]. Many body perturbation techniques can be employed to surmount the shortcomings of DFT and hence can improve the determined bandgaps[157]. This scheme mostly relies on the notion of green function method and

quasi-particle (QP)[158]. The QP retains the opportunity to ineffectually interact with one another through a screened potential instead of firmly interacting potential i.e., coulombic potential. The behavior QP is depicted by QP equation;

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V_H(\mathbf{r}) + V_{ext}(\mathbf{r})\right]\psi_i(\mathbf{r}) + \int d\mathbf{r}'\,\Sigma(\mathbf{r},\mathbf{r}';E_i)\,\psi_i(\mathbf{r}') = E_i\psi_i(\mathbf{r}) \quad (2\text{-}20)$$

The Self-energy term is calculated by GW approximation, which is presented by Hedin[159], and given as;

$$\Sigma^{GW}(\mathbf{r},\mathbf{r}';t) = i\,G(\mathbf{r},\mathbf{r}';t)\,W(\mathbf{r},\mathbf{r}';t) \quad (2\text{-}21)$$

and for dielectric function;

$$W(\mathbf{r},\mathbf{r}';E) = v(\mathbf{r},\mathbf{r}') + \iint W(\mathbf{r},\mathbf{r}_1;E)\,P(\mathbf{r}_1,\mathbf{r}_2;E)\,v(\mathbf{r}_2,\mathbf{r}')\,d\mathbf{r}_1\,d\mathbf{r}_2 \quad (2\text{-}22)$$

And

$$W(\mathbf{r},\mathbf{r}';E) = \int d\mathbf{r}_1\ \varepsilon^{1}(\mathbf{r},\mathbf{r}_1;E)\ v(\mathbf{r}_1,\mathbf{r}') \quad (2\text{-}23)$$

where P is given as;

$$W(r,r';E) = \int \varepsilon^{-1}(r,r_1;E)\,v(r_1,r')\,dr_1 \quad (2\text{-}24)$$

2.10 Bethe-Saltpeter Equation

DFT is not capable of handling the addition or removal of electrons (quasiparticles), and cannot estimate the basic band gaps of semiconductors and insulators. Instead, the GW approximation method is widely accepted for many bodies electron-electron interactions, and can accurately describe the light emission spectrum and the backlight emission spectrum. However, this method cannot describe optical excitation[160, 161]. In order to solve this problem, two particles Green functions must be used by solving the Bethe-Salpeter equation (BSE)[162, 163]. In wide band gap semiconductors, due to the weak shielding effect, strong electron-hole interaction is expected. BSE accurately describes this electron-hole interaction, which takes into

account the calculation of the light absorption spectrum[163]. The GW approximation must be used to obtain W (X; *!* = 0 and individual particle spectra. The BSE Hamiltonian is given by;

$$H_{vk,v'k'} = (E_{ck} - E_{vk})\delta_{vv'}\delta_{cc'}\delta_{kk'} + \int d^3x\, d^3x'\, c_k^*(\mathbf{x})v(\mathbf{x}-\mathbf{x}')c_{k'}(\mathbf{x}') \quad (2\text{-}25)$$

$$\psi_{ck}(\mathbf{X},t)\psi_{vk}^*(\mathbf{X},t) - \int d^3x\, d^3x'\, \psi_{ck}^*(\mathbf{x},t)\,\psi_{vk}(\mathbf{x}',t) \quad (2\text{-}26)$$

$$W(\mathbf{X},\mathbf{X}';\omega=0)\,\psi_{ck}(\mathbf{X})\,\psi_{vk}^*(\mathbf{X}) \quad (2\text{-}27)$$

where v and ($E_{ck} - E_{vk}$) denotes the bare coulomb kernel[164] and the single particle QP energy, respectively[163].

2.11 Full Potential Linearized Augmented Plane Wave + Local Orbital Method

Augmented plane wave method (APW) provides us the deep understanding of this FLAPW method[165]. In APW method we have two regions of the concern, which are Muffin Tin region (MT) & another one is interstitial region. Only difference between these regions is the treatment of the electrons in these two regions. The electrons which are very far from the nucleus are acting like free electrons, & we associate with them plane waves. These electrons are treated as they are in the interstitial region, however for another region we use the properties of spherical harmonic, but one must keep in mind that this is non overlapping region. This whole region has potential associated with it. Notice that this potential is constant for interstitial region, & for muffin Tin region we take the potential as V(r) as shown below in the equation;

$$V(\mathbf{r}) = \begin{cases} V_{\mathrm{MT}}(\mathbf{r}) & \text{if } \mathbf{r} \in \mathrm{MT} \\ \mathrm{Constant} & \text{if } \mathbf{r} \in 1 \end{cases} \quad (2\text{-}28)$$

However, there is still problem of Eigen state by using APW method, which is also the deficiency of the APW method. This also limits the solution of the density functional

equations. This limitation can be overcome by using linearized augmented plane wave (LAPW) method. This method uses such basis sets which are more flexible and complete. Eigen state problem can be easily solved with this method with high accuracy. As stated earlier that we have two regions i.e., muffin Tin region and interstitial region. This further can be as that the electrons which are in the muffin tin region are core electrons and they don't participate in chemical bonding. Muffin tin region have all these electrons which are not participating in chemical bonding. However, some electrons are leaked out of this MT region so they can participate in chemical bonding these are called valence electrons. There exist other electrons which are not valence as well as core electrons, and may referred as semi core states. Mostly semi core states are in rare-earth and D-block elements (transition metal atoms). As always with the theoretical approach that there are always limitations, so as with LAPW, that its way of method of treatment for semi core states. So, for solving this problem of semi core states the concept of local orbitals was introduced by Singh[166] was introduced. By the introduction of local orbitals this method can be usually written as LAPW+lo. For the further improvement in the accuracy of this LAPW+lo, full potential LAPW method is used, which take out the advantage of spherical harmonics in MT region and expand the potential and charge density by using spherical harmonics, however for interstitial region the concept of plane wave is used.

2.12 Projector Augmented Wave Method

This method uses the concept of local orbitals basis set as well as the concept of plane waves[167]. For the interstitial region plane wave is used but near the atom localized basis is used.so all electron wave function is being accessed, hence more advantageous & transferable. Due to the participation of all electrons in this method results of this approximation is very reliable than frozen core approximation. This method is searching for the smooth wave function which can be related with the all-electron wave function by the linear transformation;

$$|\psi\rangle = T|\psi_n\rangle \tag{2-29}$$

The wave function at left hand side has fast oscillation for all the electrons so it is hard to represent because of fast oscillation of the electrons, on right hand side of the above equation smooth pseudo-wave function which can be produced by plane waves. The transformation factor T in above equation can be written as;

$$T = 1 + \sum_R R\, SR \tag{2-30}$$

S_R in for the core region, having a partial wave basis which can be constructed by solving Schrodinger wave equation.

2.13 META Generalized Gradient Approximation

Similar to GGA functional, Meta GGA is also functional resulting from the gradient expansion. But here first we consider the expression for the total ground state energy. By using the viral theorem one can express the magnitude of total energy in term of kinetic energy. For the first time Thomas and Fermi[168] proposed a model of for the expression of kinetic energy of interacting system dependent on the density. For non-interacting system the Kohn sham formalism is a best way for the expression of the kinetic energy. To estimate the difference between the true kinetic energy and KS kinetic energy is the only requirement of the KS formalism. The latter difference between is assumed to be smaller than true kinetic energy and hence less dependent approximation are required. The functional Meta GGA attempts to enhance the description of the kinetic energy in term of density;

$$T = 2\sum_i |\, v_i\rangle\langle\theta_i| \tag{2-31}$$

Then the expression for the meta GGA will be;

$$E_{\text{meta}} - E_{\text{GGA}}[n] = \int d^3r\, n(\mathbf{r})\, E_{\text{xc}}(n(\mathbf{r}), n(\mathbf{r}), \nabla n(\mathbf{r})) \tag{2-31}$$

In this scheme the self-correlation is free, this leads to the ignore the second correlation effect I,e for one electron density the correlation energy should be vanishes.

One of the effective ways to indicate one electron energy rather than Fermis one, Van Weizacker introduced the kinetic energy term TW given by;

$$TW = \frac{|2n|^2}{8n} \tag{2-32}$$

$$V_{\mathrm{LDA+U}}(\mathbf{r}) = \sum_i v_i(\mathbf{r}) n_i(\mathbf{r}) \tag{2-33}$$

For one electron region the ratio of TTW will be one means that those region in solids for which the electron density is very low will corresponds to only one single state. By using this operator one can developed a functional which make the correlation energy of in one electron region in space an ineffective. Somehow the self-interaction problem slightly reduced.

2.14 Local Density Approximation + U

To make correct treatment of localized states a different approach has been proposed, dependent on non-ab initio-based Hamiltonian. First, we consider Hamil- 31tonian based on Hubbard model[137]

$$H = -t \sum_{\langle i,j\rangle,\sigma} \left(c_{i\sigma}^{\dagger} c_{j\sigma} + c_{j\sigma}^{\dagger} c_{i\sigma}\right) + U \sum_i n_{i\uparrow}\, n_{i\downarrow} \tag{2-34}$$

Where $c_i^{\dagger}$ and $c_j^{\dagger}$ stand for creation and annihilation operators. In equation the first term stand for the kinetic energy in model of tight binding in such a way that electrons hopping from i lattice site to another lattice site j. the restriction is imposed over sum of i , j only for nearest neighbors. The kinetic energy term shows the tendency towards delocalized states and dispersion of bands at origin, while the second term represents the interaction on site. U Stand for the energy cost of two electrons which placed on the same site. To study localized states such as d or f electron states the Hubbard Hamiltonian is well suited. The self-interaction becomes free in Hubbard Hamiltonian only when the opposite spins lie on the same site. Thus, by combining Hubbard Hamiltonian with LDA is very useful for obtaining the treatment of both localized and delocalized states. Following this motivation, Anisimov et al.[169] recommended an expression for total energy of LDA+U;

$$E_{\mathrm{LDA}+U} = E_{\mathrm{LDA}} + E_U - E_{\mathrm{dc}} = E_{\mathrm{LDA}} + 1/2\, U \sum_{i \neq j} n_i\, n_j - U/2\, N_d(N_d - 1) \quad (2\text{-}35)$$

Here the occupations of d orbitals are represented by n_i and $N_d = \Sigma_{i\epsilon}\, n_i$. In above equation the 2nd term EU presents only the d-d interaction while the last term Edc stand for double counting correction. To do real implementation of LDA+U, one can built an orbital dependent exchange-correlation potential

$$V_{\mathrm{LDA}+U}\ ;\ V_{\mathrm{LDA}+U}(\mathbf{r}) = \sum_i v_i\,(\mathbf{r})\, h_i h_i \quad (2\text{-}36)$$

Where I run over all possible principal n and azimuthal l quantum numbers and atomic sites. In the presence of the projected operator that act differently on the delocalized (s and p) and on the localized (d and f) orbitals which make the potential orbital dependent, hence non local $v_i(r)$ is computed through the derivative of the total energy with respect to partial electronic density $\rho(r)$. For s and p electrons I,e i=s and p the potential $v_i(r)$;

$$v(\mathbf{r}) = \frac{dE_{\mathrm{LDA+U}}}{d\rho} = \frac{dE_{\mathrm{LDA}}}{d\rho} = v(\mathbf{r}) \quad (2\text{-}37)$$

Resulting $E_{i(r)} = E^{LDA}$. However, in the case of d electron the potential will be;

$$v_i(\mathbf{r}) = v_{\mathrm{LDA}}(\mathbf{r}) + U\left(\frac{1}{2} - n_i\right) \quad (2\text{-}38)$$

Which leads to the eigen value;

$$i(\mathbf{r}) = i v_{\mathrm{LDA}}(\mathbf{r}) + U\left(\frac{1}{2} - n_i\right) \quad (2\text{-}39)$$

When the d orbital is occupied (n_i = 1), the LDA+U energy is shifted by U/2 with respect to the LDA eigen values. However, when orbitals are unoccu- pied, their level is upshifted by +U/2. LDA+U tends to localize electrons on d orbitals where the occupation is larger than 1/2. LDA+U partially solve the self-interaction encountered in LDA. The above discussion treats only one band Hubbard model. Also, it can be extended to more general cases, in which the exchange interaction between d orbital and polarization of d orbitals can be taken into account. To move d electrons from one atom to another atom, the U parameter taken as the energy cost with renormalization of s and p

orbitals by screening[84]. In order to reduce the energy, cost the s and p orbital moves in opposite direction. Several ways are exists ranging from fully ab-initio to empirical approaches to determine the physical meaning of U, by using density functional theory the U parameter can be implemented, through experimental spectra U can be evaluated,

2.15 Practical Implementation

2.15.1 Plane Wave Basis Set

A specific basis set for expanding KS-wave function can be chosen for practical implementations in many computing codes. In quantum chemistry, localized basis such as Gaussian basis set is implemented in Gaussian code. In contrast, plane- wave basis set is more advantageous to deal with systems containing vacancies or dense periodic systems. It is due to the fact that the plane-waves are delocalized, allowing homogenous precision around the whole area in the simulation box.

For a system with periodicity, the KS wave function is product of the Bravais lattice periodicity $f_i(r)$ and plane wave $e^{i[k.r]}$;

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} f_k(\mathbf{r}) \tag{2-40}$$

The function $f_i(r)$ of the cell is expanded for discrete PW basis sets in order to make identical the wave vector (k) to reciprocal lattice vectors (G).

$$f_i(\mathbf{r}) = \sum_{\mathbf{G}} C_{\mathbf{G},i}\, e^{i\mathbf{G}\cdot\mathbf{r}} \tag{2-41}$$

But G. L=2πm, where is an integer. A secular equation may be obtained using PW basis set;

$$\sum_{\mathbf{G}} \left[-\frac{\hbar^2}{2m}(\mathbf{k}+\mathbf{G})^2 \delta_{\mathbf{G}\mathbf{G_0}} + V_{R\sigma}(\mathbf{G}-\mathbf{G_0}) \right] C_{i,\mathbf{k}+\mathbf{G_0}} = \epsilon_i C_{i,\mathbf{k}+\mathbf{G_0}} \tag{2-42}$$

The cut-off energy is represented by term $-\frac{\hbar^2}{2m}(\mathbf{k}+\mathbf{G})^2$ denoting the possible greater value of kinetic energy of PW in the KS-eigen states. For a periodic lattice, there

may be infinite number of Bloch states to treat the electronic problem due to large number of k points in Brillouin zone (BZ). However, the problem can be solved by choosing the adjacent k-vectors regardless of available infinite Bloch states. A high-level computational accuracy of different physical properties is achieved by adopting wave function at reduced number of k point in the BZ[170, 171]. The mathematical expression for unit cell volume Ω of first BZ is written as:

$$\Omega_R = \mathbf{b_1} \cdot (\mathbf{b_2} \times \mathbf{b_3}) = \frac{2\pi^2}{\Omega} \quad (2\text{-}43)$$

A number of accurate methods for sampling the BZ were proposed by Monkhorst-Pack[172], and Chadi and Cohen[173]. These schemes are much better to calculate the electronic potential or total energy of a semiconductor or an insulator even at finite number of k-points in BZ.

2.15.2 Pseudopotential

In solid materials, the electrons localized in regions closer to nuclei are strongly bounded in the core states and loosely bounded in valence states. The core states being chemically inert, remains unaffected by external environment. In contrast, the valence electrons are crucial for interpretation of many physical properties of solids[174]. However, large number of PW are required to accurately describe the valence states thus, making it computationally expensive. In pseudopotential scheme, the effects of electrons in core states are neglected and the basis set size is reduced by preserving the computational accuracy. In the core region, the core electrons and strong coulomb potential must be replaced by an effective weak pseudopotential and the rapidly oscillating valence electron wave- functions are replaced by smoothly varying pseudo-wave functions[175, 176]. Hence, the PW basis set is effectively used. The pseudopotential (PP) effectively treats the valence electrons and combined effect of valence and core electrons and strong nuclear valence electrons interaction. The pseudopotential (PP) can be generally categorized into norm-conserving soft pseudopotential (NCSPP) and Vanderbilt ultrasoft pseudopotential (USPP)[177]. The electron-ion core interaction is incorporated using pseudo wavefunctions

by adopting nonlocal soft norm conserving PP. The NCSPP are hard and requires a high cut-off energy. To overcome this problem, USPP are used, generating softer PP and reduces the PW basis set size.

2.16 Vienna Ab initio Simulation Package

The materials modelling at atomic level can be achieved by performing the first principles study using the Vienna Ab initio Simulation Package (VASP)[178]. The solution to the Schrodinger equation may be obtained by solving Kohn-Sham equation in DFT or by making a solution to Roothan equations in HartreeFock (HF) approximation. VASP includes the hybrid functionals comprising the mixture of Hartree Fock (HF) approach and DFT, Green's functions methods and many body perturbation theory. The PW-basis set is used to express the local potential, the electron charge density and one-electron orbital. The interaction between electron and ion is described by either pseudopotentials (ultrasoft or norm-conserving) and or PAW approach. The geometrical optimization is obtained until the force and energy is converged. After geometrical relaxation, a number of physical properties like electronic, magnetic and optical behavior of materials can be obtained using VASP package

2.16.1 K-Points Optimization

A wave function in the periodic boundary cell can be displayed by using the Bloch theorem[179];

$$\psi_k(\mathbf{r}) = e^{i\epsilon\cdot\mathbf{r}} U_k(\mathbf{r}) \qquad (2\text{-}44)$$

Here wave function is described by k. The number of k points needed for a calculation depends critically on the necessary precision, smearing method and also the nature of materials (metallic, semiconductor or insulator). For our calculations we used different K-Points (K-Points sampling) due to large number of wave vectors, and calculated the minimum values for that. For all the K-Points 12 × 12 × 1 k-meshes gives the

minimum energy, hence we consider 12 × 12 × 1 K-Points for our calculations. For band structure calculation we chose the high symmetry K-Points which is Γ, M, K and Γ in first Brillouin zone (BZ).

2.16.2 Cut-Off Energy

In our INCAR file we fix cut-off energy to 500 eV, which converged energy(forces) to 10^{-4}eV(10^{-4}eV/Å) in the relaxation of the atomic positions. Also, we set 25 $\AA$ vacuum layer along z-axis (between two layers) to interrupt the artifacts of the periodic boundary conditions are used.

2.16.3 Dynamic Stability

For dynamic stability, we used PHONOPY code which is implemented in VASP to confirm the dynamic stabilities of our studied systems[180]. For PHONOPY code which is based on DFT, in which the atomic displacement is treated as a perturbed potential. We have used 4 × 4 × 1 supercell for dynamic matrix, phonon spectrum and for calculating the accurate force constant.

2.16.4 Thermal Stability

For thermal stability we have used ab initio molecular dynamics (MD) simulation[181]. All the volume, lattice constant and temperature are simulated by algorithm of Nose. We chose 4 × 4 × 1 supercell for test sample, and also the total time are set to 5 × 10s while the time step is set to 1×10^{-12} s while the time step is set to 1×10^{-15} s

3 Superior optical and thermoelectric properties of bilayer $\boldsymbol{\beta_{12}}$-like phase borophene synthesized on the Cu (1 1 1) film

3.1 Introduction

The fantastic two-dimensional (2D) material graphene was discovered in 2004[1, 182]. Although graphene's exhibits properties, it cannot meet more diverse needs, thus it is now more crucial than ever to discover comparable 2D materials[1, 182-187]. Boron is the nearest neighbor to carbon, lighter than carbon, and can theoretically be anticipated to form a 2D material named borophene[188]. Borophene is believed to possess greater lightness, rigidity, and flexibility in comparison to graphene. It can be found in various morphologies and phases due to the rich bonding conformation of boron. Borophene, also has excellent electronic and thermal transport properties[189], and can form theoretical contact with semiconductors to efficiently reduce resistance contact and further enhance the performance of related 2D transistors[189, 190]. Borophene is much lighter and more reactive so that it can act as a catalyst to change hydrogen atoms into ions[191]. And it shows outstanding applications in energy storage devices, particularly magnesium, sodium, and lithium-ion batteries. Borophene possesses exceptional properties, such as metallicity, optical transparency, extremely high flexibility, and superconductivity[192, 193].

The lattice structure of borophene depends on the substrate and synthetic conditions, and boron atoms forms a self-doping model with the substrate when it are added to a boron honeycomb lattice (v =1/3)[194]. Borophene has numerous theoretically predicted phases[195-197], and it depends upon the concentration of hexagonal vacancies of borophene within the range 0-1/3, such as triangular borophene, honeycomb boron sheets, α, β_{12}, δ_6, and χ_3[198, 199]. In response to these forecasts, borophene has been synthesized on heterogeneous substrates such as Ni (1 1 1)[199], Ir (1 1 1)[200], Au (1 1 1)[201], and Ag (1 1 1)[202].

The development of borophene on different metal substrates has mostly stayed limited to monolayer (ML)[203]. The ML borophene or boron sheets have been found experimentally on various metal substrates as boron is the lightest metalloid which has ML and BL polymorphs[204-206]. The stability of BL borophene is inferior to ML, while it exhibits excellent thermal stability[207]. Stable BL borophene has intriguing properties like double Dirac cones, anti-ferromagnetism, Dirac fermions, and Dirac nodal line[208].

BL borophene, compared with ML borophene, is more stable due to the interlayer bonding; while BL borophene is difficult to be synthesized because of the passivation from the metal substrate[204]. Once monolayer borophene has formed on the substrate, continuing deposition could result in the formation of 3D clusters rather than flat, multilayered borophene. In addition, the growth conditions need to be carefully controlled to prevent the formation of other boron-based compounds or structures that hinder the formation of a BL borophene[209]. These challenges make the synthesis of BL borophene more complex compared to monolayer borophene. The BL α-phase borophene is successfully synthesized on Ag (1 1 1) film, providing the basis for borophene-based devices. The interlayer covalent bonding creates two Dirac points in BL borophene, suggesting its potential applications in optical sensors and thermoelectric devices[210]. Experimentally, BL β_{12} -like phase borophene is also successfully synthesized on Cu (1 1 1) film[211]. It is found that BL borophene with different phases can be synthesized on different metal substrates.

BL phase borophene films on Cu (1 1 1) show promise for both optical and thermoelectric applications, such as high-efficiency solar cells and waste heat recovery systems. Borophene's high conductivity and transparency also make it a potential material for transparent conducting electrodes in touch screens, OLEDs, and wearable devices.

Researchers have been exploring the unique properties of borophene. The BL of borophene has superior optical and thermoelectric properties, but borophene films are still challenging to synthesize. This work is a major breakthrough for quinary borophene on Cu (1 1 1). These findings contributed to the development of advanced optoelectronics and thermoelectric based on BL borophene synthesized in Cu (1 1 1).

3.2 Method

The Vienna Ab initio simulation package (VASP)[212] was used to study the energy bands and optimization structure of BL borophene. The calculations utilized the Perdew-Burke-Ernzerhof (PBE) functional and Generalized-gradient approximation (GGA)[211], and the DFT-D3 method was employed to enhance atomic geometry[213]. A vacuum layer in the z direction prevented interlayer interaction. The final force convergence was less than 0.05 eV/Å, the energy convergence was within 1×10^{-5} eV, the k-points was 4x2x1, and the cut-off energy was was set to 300 eV.

The calculation of optical properties is based on the frequency-dependent dielectric function,

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \tag{3-1}$$

And the absorption coefficient is,

$$\alpha(\omega) = \frac{\sqrt{2}\omega}{c}\left[\sqrt{{\varepsilon_1}^2 + {\varepsilon_2}^2} - \varepsilon_1\right]^{\frac{1}{2}} \tag{3-2}$$

where ε_1 is the real and ε_2 is imaginary part of dielectric function[214].

BoltzTraP2 package[215] was used to obtain the Thermoelectric properties of BL borophene by solving the Boltzmann transport equation (BTE) with relaxation time approximation. The Seebeck coefficient and thermal conductivity (κ) can be expressed as:

$$S = \frac{1}{qT}\frac{\mathcal{L}^{(1)}}{\mathcal{L}^{0}} \tag{3-3}$$

$$\kappa_e = \frac{1}{q^2T}\left[\frac{\left(\mathcal{L}^{(1)}\right)^2}{\mathcal{L}^{(0)}} - \mathcal{L}^{(2)}\right] \tag{3-4}$$

while the moments of generic transport coefficients can be written as,

$$\mathcal{L}^{(\alpha)}(\mu;T) = q^2\int\sigma\,(\varepsilon,T)(\varepsilon-\mu)^{\alpha}\left(-\frac{\partial f^{(0)}(\varepsilon;\mu,T)}{\partial\varepsilon}\right)d\mathcal{E} \tag{3-5}$$

where μ represents the chemical potential and T represents the Boltzmann temperature, respectively. The transport distribution function in Equation (5) is,

$$\sigma(\varepsilon, T) = \int \sum_b V_{b,k} \otimes V_{b,k} \tau_{b,k} \delta\left(\varepsilon - \varepsilon_{b,k}\right) \frac{dk}{8\pi^3} \quad (3\text{-}6)$$

The average binding energy (E_b) can be expressed as follows:

$$E_b = \frac{E(A_m B_n) - m*E(A) - n*E(B)}{m+n} \quad (3\text{-}7)$$

Our computations revealed that the binding energy was 6.0557128 eV. Here m and n are1, $E\ (A_m B_n)$ is E (Cu+BL Borophene), E(A) is E(Cu), E(B) is E(BL borophene) and E is the average binding energy. We believe that the high binding energy demonstrates the strong cohesive forces within the BL structure, and indicate a favorable level of stability. The analysis of binding energy unveils crucial insights into the superior optical and thermoelectric properties exhibited by the BL β_{12}-like phase borophene synthesized on the Cu (1 1 1) film[216, 217].

3.3 Discussion

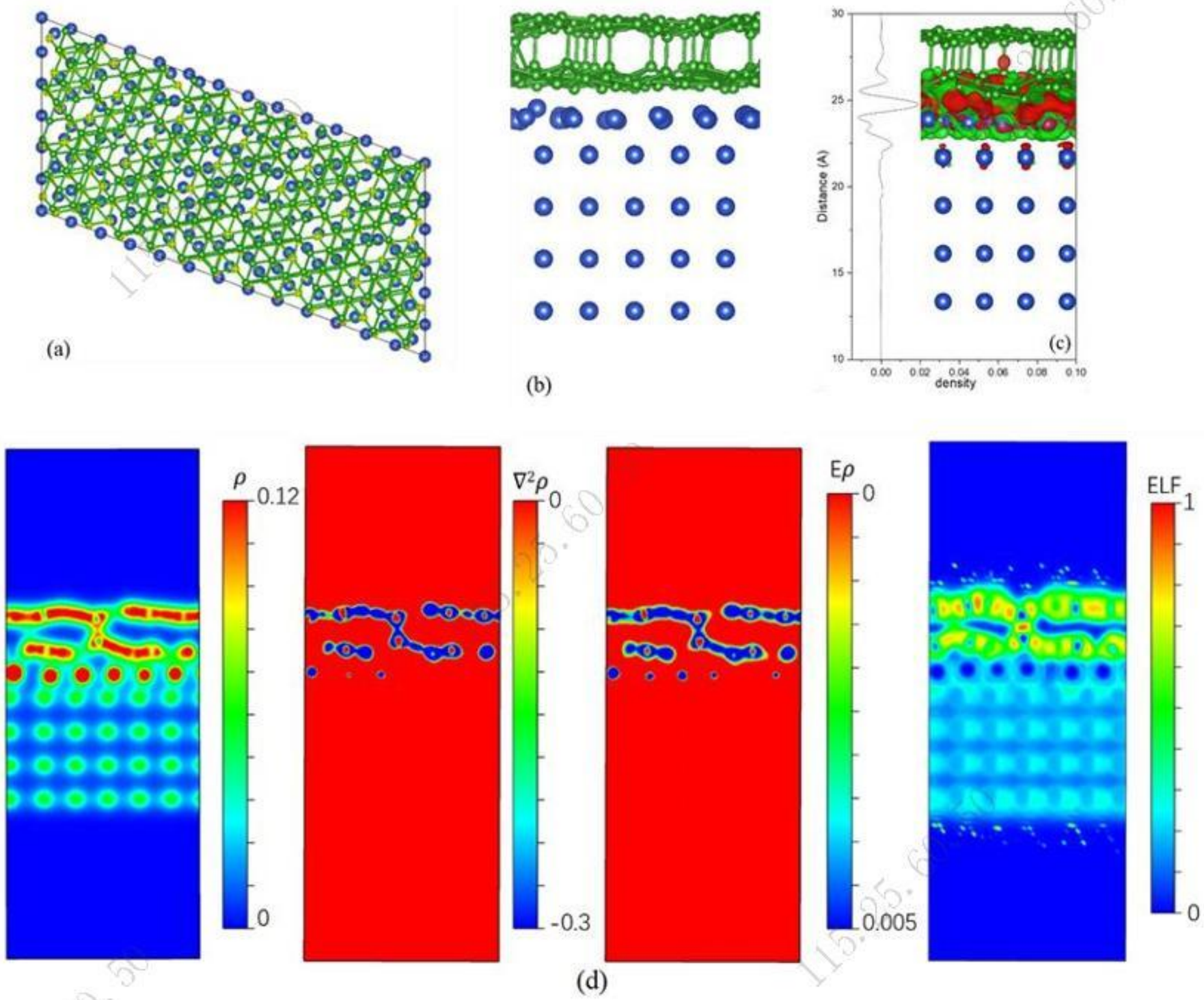


Figure 3- 1 BL β_{12}-like phase borophene on Cu (1 1 1) films, where (a) and (b) are top and side views respectively, (c) Planar-averaged electron density where red and green represent electrons and holes respectively. (d) Electron number density, Laplacian of electron number density, electron energy density and electron localization function of BL borophene.

The top and side views of BL β_{12}-like phase borophene on Cu (111) substrate are shown in Figure 3-1(a) and (b), respectively. The vacancy density of β_{12}-like phase is 11/72, coordination number are 4, 5 and 6. Cu (1 1 1) substrate and borophene are formed in the XY plane. The charge distribution at the ground state reveals strong coupling interactions between the Cu (1 1 1) substrate and the BL borophene (Figure 3-1 (c)). The electrons are mainly distributed between Cu (1 1 1) and BL borophene, and the holes are distributed in the bottom of borophene and top layer of Cu atoms. This indicates that the electrons transform from Cu to regions between Cu (1 1 1) surface and borophene layer. In this way, the sandwich structure of the hole-electron-hole promote structure stability and strong interfacial van der

Waals (vdW) interaction exists between the Cu substrate and the interfacial layer of BL borophene. Moreover, the formation of interlayer chemical bonds, facilitated by the transfer of electrons, contributes to the robust interaction observed between the top layer of borophene and the borophene of the interfacial layer. The vdW interaction is also present between the interfacial Cu atoms of a Cu film. As a result of interaction between BL borophene and Cu substrate, the Cu substrate becomes rough which is caused by the fluctuation of Cu atoms on the surface of Cu substrate. This interaction makes the Cu atoms on the surface of the Cu substrate move around.

As shown in Figure 3-1(c), the Cu atoms transfer a huge number of electrons to the borophene layer, resulting in the BL borophene. See Figure 3-1 (d), the electric density $\rho>>0$, electronic energy density E<0, and $\nabla^2\rho<0$ for the interlayer chemical bond BL borophene. This shows that it is a standard covalent bond[218, 219], as shown in Figure 3-1(d). The interlayer distance is 3.7 Å, and the interlayer bond length is 1.8 Å. The interlayer distance of the borophene grown on the Cu (1 1 1) film is less than that of BL borophene grown without a Cu substrate in a vacuum. This is due to the attraction between positive and negative charges in the Cu film and the interfacial and top layers of borophene. Figure 3-1(d) shows the electronic localization function (ELF) distributions of interlayer bonds, respectively. ELF represents the de-regionalization of the electrons[220, 221], which is the more localized the electron is, the more confined it is and as a result the electrons are less likely to escape. According to these results of ELF distribution, the Cu substrate forms a stable and strong covalent bond with BL borophene.

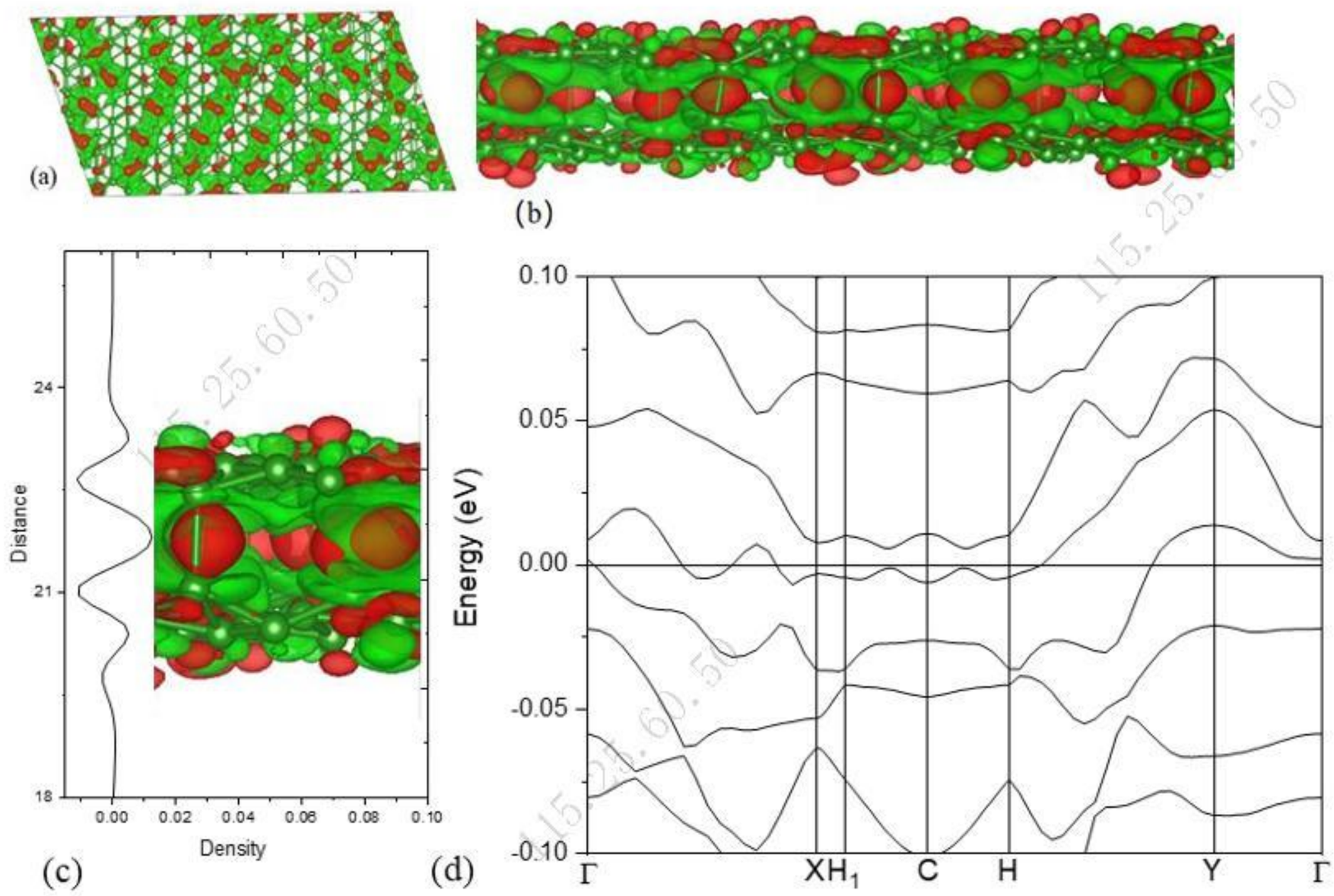


Figure 3- 2 (a) and (b) Top and side views of charge difference density of BL borophene without substrate, (c) Planar-averaged electron density, and (d) band structure of BL Borophene.

Figures 3-2(a) and (b) represent the top and side views of charge difference density of BL borophene without a substrate. In the XY plane, electrons and holes exhibit a broken line distribution pattern. Specifically, electrons are predominantly concentrated on the interlayer bonds, while holes are mainly distributed within the borophene layers. This forms a stable hole-electron-hole sandwich structure. The planar-averaged electron density of BL borophene also shows a large electron density at the interlayer bond and a large hole density at the borophene layer. The energy bands of BL borophene cross the Fermi level, which shows the metallicity of BL β_{12}-like phase borophene.

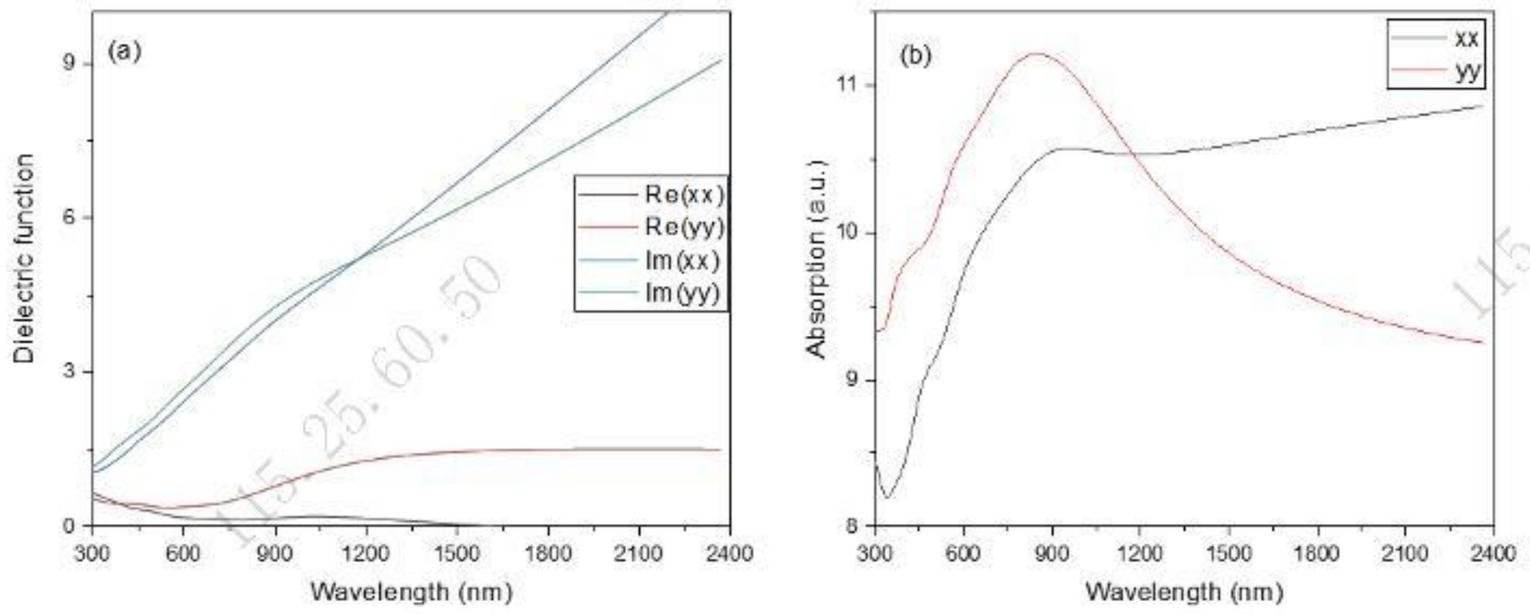


Figure 3- 3 (a) The dielectric function of BL borophene on Cu (1 1 1) film in the infrared region and visible regions. (b) Optical absorption spectra of BL borophene on Cu (1 1 1) substrate.

The optical properties of borophene and Cu are essential for researching sunlight absorption and transition into current for various applications. Sophisticated dielectric function particularly shows optical properties by Equations (3-1) and (3-2). In this work in terms of imaginary part $\varepsilon_2(\omega)$ the absorption spectra are calculated at the wavelength between 300 nm and 2400 nm as shown in Figure 3-2(a). The dielectric constant of BL borophene is valuable to explore[153], its real parts along the x and y-axis are positive in the VR which shows that there is no plasmon effect. Considering the stability and variation of electronic properties of borophene, it is valuable to interrogate the effects of borophene on optical properties, especially the optical absorbance as shown in Figure 3-2(b), which has two absorption peaks along x and y-axis. One peak is red and the other one is black. The contribution of the Y-axis peak is larger in the visible and near-infrared (NIR) regions and in the mid-infrared (MIR) region, the absorption along the X-axis is larger than that along the Y-axis, which makes β_{12}-like phase borophene a potential optical sensor. The β_{12}-like phase borophene, synthesized on the Cu (1 1 1) film exhibits remarkable optical properties, characterized by superior absorption spectrum and dielectric constant, establishing its potential as a highly efficient material for optoelectronic applications. With a unique electronic structure and strong interlayer coupling, the BL borophene demonstrates enhanced light-matter interactions, resulting exceptional light absorption across a broad range of wavelengths. The absorption spectrum of this material shows a significantly high absorption coefficient, enabling efficient harvesting of incident photons and facilitating the development

of high-performance photovoltaic devices. Additionally, the BL borophene exhibits an impressive dielectric constant, reflecting its ability to store and transport electric charges, which holds great promise for the design and fabrication of advanced electronic devices with improving capacitance and charge transport characteristics[222, 223].

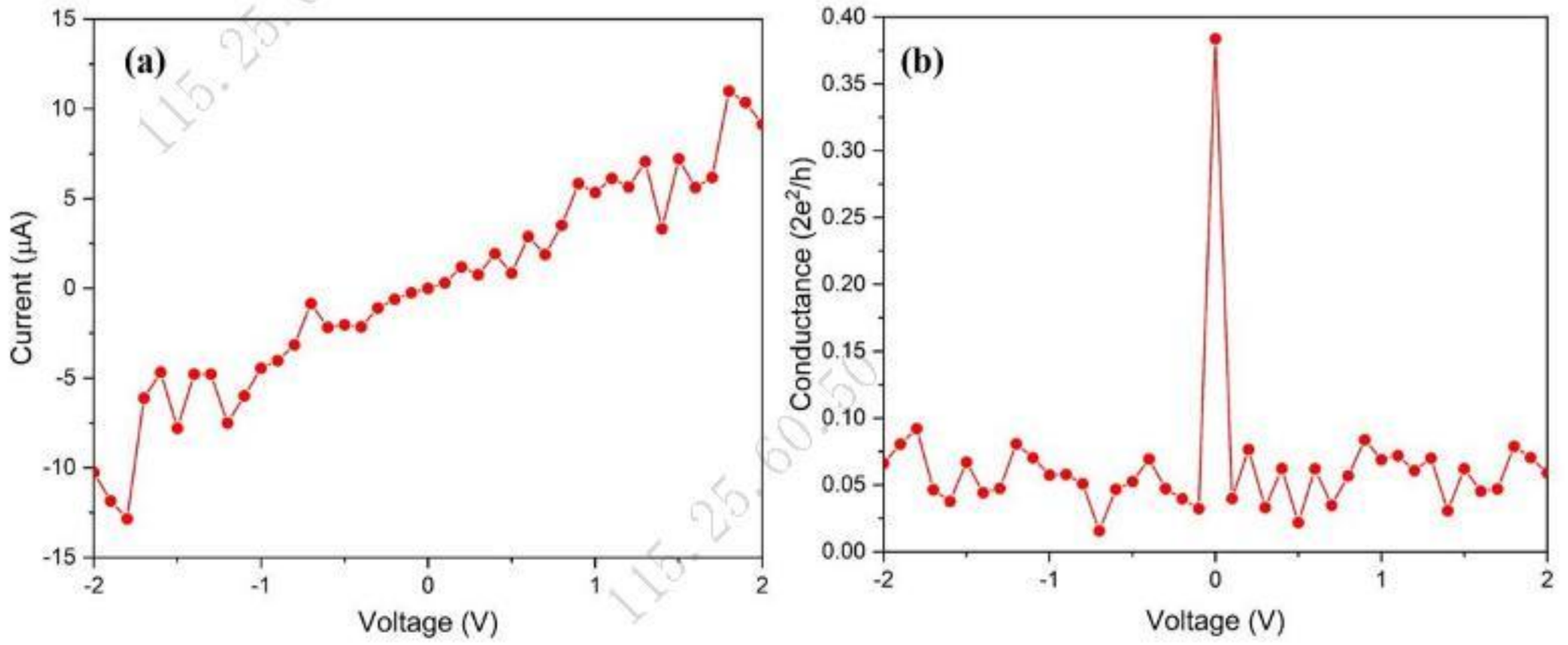


Figure 3- 4 (a) I–V curves and (b) conductance of device electrical transport. (a) I–V curve with different voltage regions, (b) Conductance with different voltage regions.

Figure 3-4(a) reveals that the characteristics of I-V curves molecular unit length. Interestingly, the distribution of I-V curves remains symmetrical at both positive and negative bias voltages. As the bias voltage increases, there is a linear growth in current across all molecules until it reaches its initial peak. Subsequently, a Negative Differential Resistance (NDR) effect becomes apparent, leading to a decrease in current. Figure 3-4(b) illustrates the voltage-dependent conductance, reflecting the electron transport capacity and aligning with the relationship $I = \sigma V$ in the DC circuit.

In the case of small molecular unit lengths ($n \leq 2$), the highest conductance is observed within the range of -0.5 to 0.5 V. Conversely, when dealing with large molecular unit lengths ($n \geq 3$), the conductance within the -0.1 to 0.1 V range is significantly lower compared to that at higher bias voltages. Furthermore, around ±0.1 V, the conductance reaches its minimum, serving as a distinct indicator of the transition in transport behavior from shorter to longer

molecular.

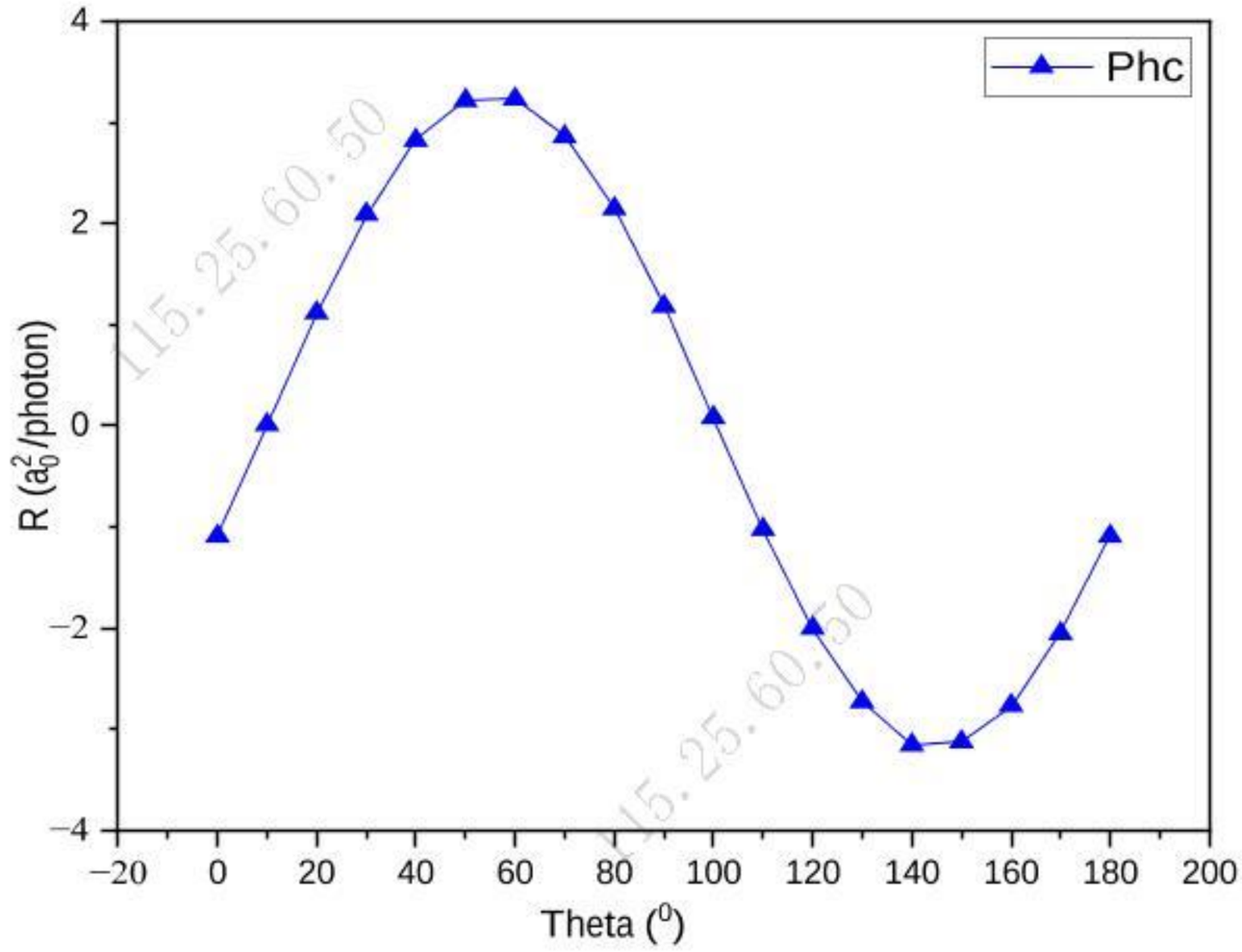


Figure 3- 5 Photocurrent at different linear polarization angles.

Figure 3-5 illustrates the dependency of photocurrent on optical absorption. In our pursuit of optimizing photocurrent, we introduce slight bias voltages of 0.1 V and 0.5 V see figure 3.4(a) and (b). The results highlight a remarkable 103-fold enhancement in photocurrent during resonant optical absorption, as visualized in Figure 3-5.

Notably, a bias voltage of 0.5 V is recommended, considering the minimal dark current and conductance observed in Figure 3-4(a) and Figure 3-4(b). This voltage configuration proves to be highly effective in maximizing the desired outcomes while minimizing unwanted electrical interference.

However, it's apparent that the increase in photocurrent intensity, while significant, remains somewhat modest in comparison to resonant optical absorption. This underscores the need for further adjustments in bias voltages to achieve more substantial gains in photocurrent. Therefore, meticulous optimization of the bias voltage settings will play a pivotal role in addressing this challenge and enhancing the overall performance of the device.

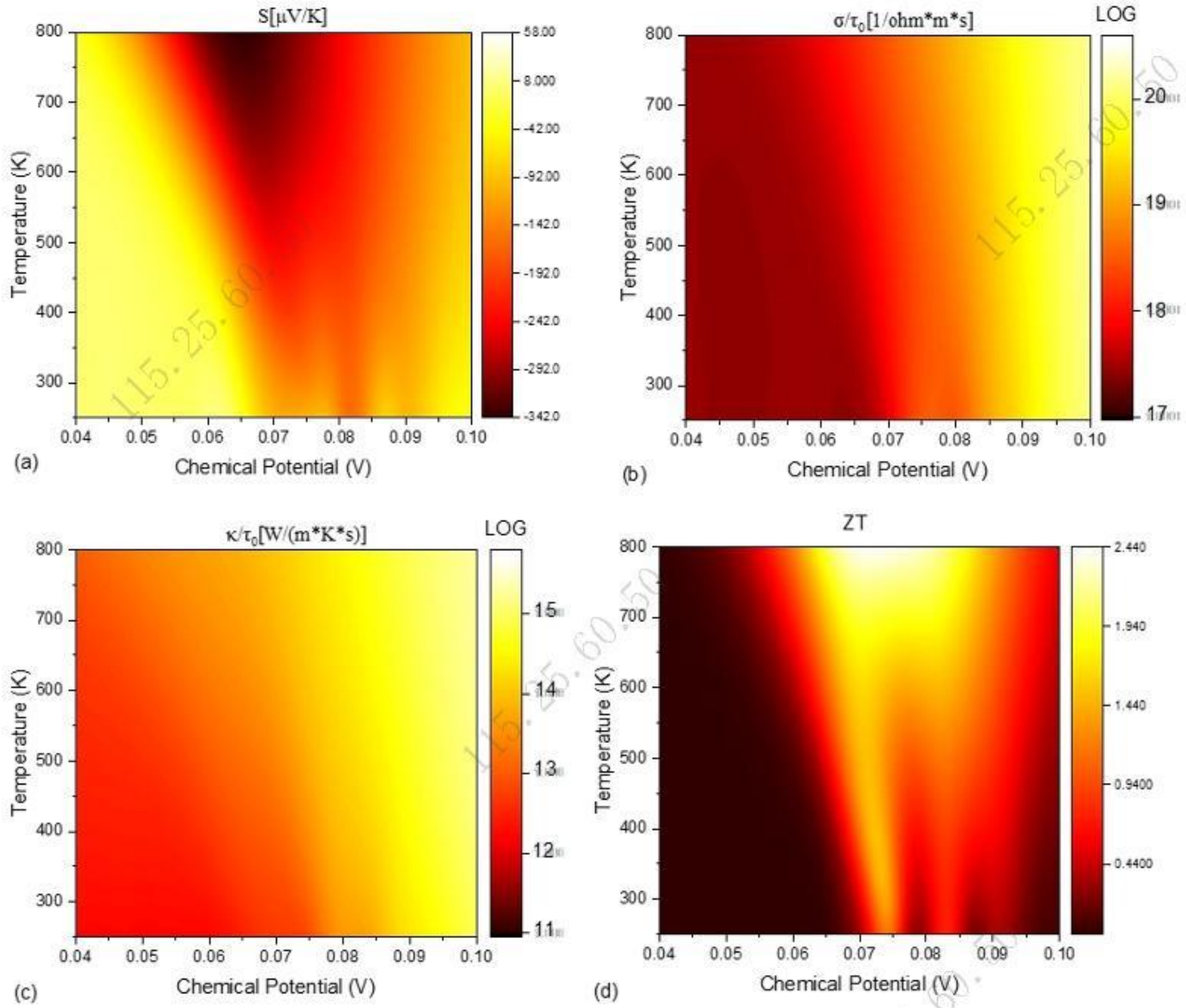


Figure 3- 6 Chemical Potential and temperature dependent (a) Seebeck Coefficient, (b) Electrical Conductivity, (c) Thermal Conductivity, (d) Thermoelectric figure of Merit.

We have studied the uses of BL borophene through thermoelectric devices and its performance can be evaluated by thermoelectric figure of merit (ZT).

$$ZT = \frac{S^2 \sigma T}{\kappa} \tag{3-8}$$

Where S is Seebeck coefficient, σ is electrical conductivity, κ is thermal conductivity and T is the temperature. Figure 3-6(a) shows the S variation with temperature (K) and chemical potential (V), from 550 to 800 K, and in 0.06 to 0.08 V, S is high and it is negative, which proves that Seebeck effect is strong in this interval, and this interval corresponds to the maximum value of the thermoelectric effect, which shows the dominant role of electrons in BL β_{12} -like phase borophene transport. Figure 3-6(b) is the σ, from which, electrical conductivity mostly depends on chemical potential because it occurs at all temperatures but not at all chemical potential. Figure 3-6(c) represents the κ of BL borophene, which increases

with the increase of chemical potential. Figure 3-6(d) shows ZT, which is high at high temperature region between 0.06 to 0.08 V. At around 800 K corresponding to 0.06 to 0.08 V it shows biggest ZT. Therefore, high σ, low κ and large S are necessary to increase the thermoelectric conversion efficiency of a thermoelectric device. The BL β_{12}-like phase borophene, synthesized on the Cu (1 1 1) film, exhibits remarkable thermoelectric properties characterized by its high thermoelectric efficiency and exceptional transport properties, including a high Seebeck coefficient, significant electrical conductivity, and low thermal conductivity, all of which contribute to its outstanding performance as a thermoelectric material. These favorable characteristics arise from the unique electronic structure and phonon dispersion of BL borophene, leading to enhanced charge carrier mobility and reduced phonon scattering, thus facilitating efficient energy conversion between heat and electricity, making it an intriguing candidate for fundamental studies and explorations in the field of thermoelectric materials[224-226].

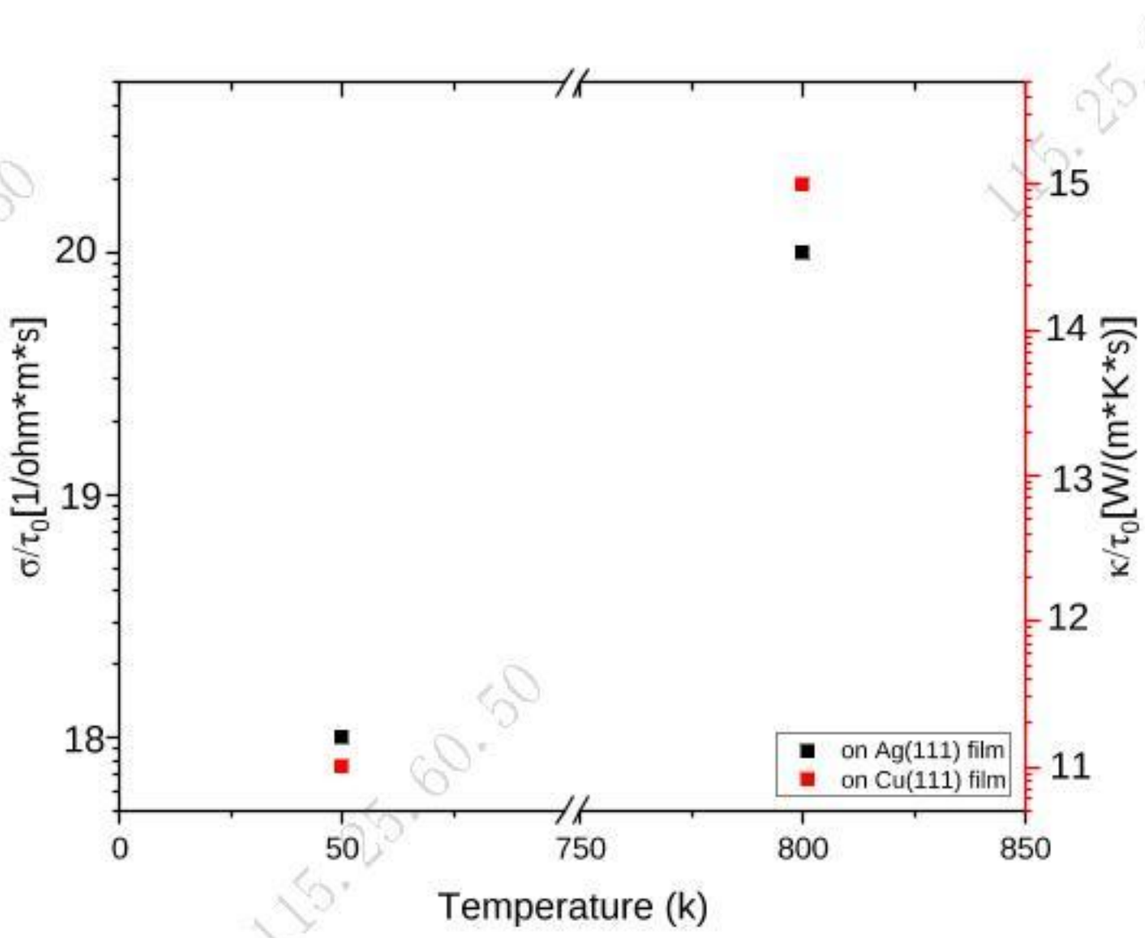


Figure 3- 7 Thermoelectric properties of two different kinds of BL borophene synthesized on Ag (1 1 1) and Cu (1 1 1) film, respectively. Data on Ag film is from Ref. [32].

Figure 3.7. Thermoelectric properties of two different kinds of BL borophene synthesized on Ag (1 1 1) and Cu (1 1 1) film, respectively. Data on Ag film is from Ref. [32].

It is intriguing to compare the thermoelectric properties of bilayer borophene synthesized on the Cu (1 1 1) film with those reported in our previous theoretical study on the Ag (1 1 1) film in Ref [32], seeing supporting information. According to formula (7), good thermoelectric materials should exhibit large S and σ and small κ. In Figure 3-7, it is found that with BL synthesized on the Ag (1 1 1) film, the κ is beneficial for ZT, but σ is relatively small in the best region of S at low temperature; while BL is synthesized on the Cu film, the σ is large, and κ is also large for the best region of S at high temperature. In addition, due to the existence of double Dirac points, there is an obvious separation of positive and negative Seebeck coefficients near a determined chemical potential of borophene synthesized on Ag substrate, and there is no Dirac point in borophene synthesis on Cu (1 1 1) substrate, so the above phenomenon does not exist. Therefore, borophene synthesized on Ag (1 1 1) substrate is suitable for low temperature regions, while borophene synthesized on Cu (1 1 1) substrate is suitable for high temperature regions. Therefore, the choice of metal films can induce variations in the structure of bilayer borophene, and consequently, their distinct geometries and electronic properties contribute to different thermoelectric properties.

3.4 Summary of the Chapter

In summary, this study has demonstrated the properties of BL borophene, synthesized on the Cu (1 1 1) film, as a promising candidate for optoelectronic devices. By analyzing its electronic, thermoelectric, and optical properties, we have gained insight into the physical characteristics of this material. The complex exciton peaks observed in both the IR and VR regions in the optical absorption spectrum highlight its unique optical properties. Moreover, the remarkable thermoelectric properties of bilayer borophene indicate its potential as an efficient energy converter. Overall, our findings highlight that BL borophene has a future in the domain of optoelectronics, with the potential to contribute to the development of high-performance devices.

4 Electronic, thermoelectric and electrical transport properties of single layer γ-graphyne

4.1 Introduction

Carbon, the most prevalent element in both the atmosphere and the Earth's crust. Over recent decades, numerous carbon allotropes showcasing unique characteristics have been unveiled. Among them, the renowned ones include the zero-dimensional (0D) fullerene[227], one-dimensional (1D) nanotube[228], and two-dimensional (2D) graphene[1]. Graphene, featuring a hexagonal lattice of carbon atoms, has garnered significant interest due to its exceptional properties such as high thermal conductivity, superior charge carrier mobility, and remarkable Young's modulus[229-231]. The remarkable properties of graphene have paved the way for its widespread application across various domains including electronics, photonics, and sensor technologies[232-234]. Graphene's properties haven't been sufficiently optimized for practical applications, leading researchers to explore alternative structures like graphyne[235-237]. Graphyne, initially introduced by Baughman et al. in 1987[238], stands as a sibling material to graphene, garnering significant attention within the realm of carbon-based materials[239-241]. It represents a novel carbon allotrope family formed by introducing acetylenic bonds between two carbon atoms in graphene's structure. The dimensionality of carbon materials is determined by their hybridization. Specifically, sp^2 hybridization results in the formation of 2D graphene, while sp and sp^3 hybridizations yield 1D and 3D carbon materials, respectively[242-244]. However, graphyne distinguishes itself as the singular 2D carbon material known to possess both sp and sp^2 hybridization[245, 246], this unique combination contributes to its versatile and flexible crystal structure. γ-Graphyne is a promising 2D material with a unique structure that includes a natural bandgap, unlike graphene. This bandgap is crucial for semiconductor applications, enabling on-off switching in devices like transistors. Additionally, γ-graphyne's structure suggests high thermoelectric efficiency[247-250] and excellent carrier mobility, which are important for energy conversion and electronic applications. These properties make γ-graphyne an attractive candidate for

advanced electronic and thermoelectric devices. This study investigates the electronic, thermoelectric, and electrical transport properties of single-layer γ-graphyne using advanced computational methods.

With the inclusion of acetylenic fragments, graphyne manifests in four distinct geometric structures alpha(α)-graphyne, beta(β)-graphyne, gamma(γ)-graphyne, and (6,6,12)-graphyne[238, 251]. Out of these configurations, only γ-graphyne and its nanostructures derived from it have been successfully synthesized in experiments using a cross-coupling reaction involving hex ethynylbenzene[252, 253]. These innovative carbon allotropes not only boast unique mechanical and chemical attributes akin to graphene[254-257] but also showcase a diverse array of remarkable electronic and thermal transport capabilities, positioning them as formidable contenders to graphene[258, 259]. The γ-graphyne exhibits characteristics of a narrow gap semiconductor, comparable to bulk silicon[260-262] in terms of electronic conductivity, with an intrinsic carrier mobility reaching up to 10^5 cm / Vm^{-1} at room temperature[263]. Moreover, due to the presence of acetylenic linkages and nano-sized pores, the lattice thermal conductivity of γ-graphyne is notably lower than that of graphene. Theoretical predictions suggest that the conductance of γ-graphyne is approximately 10–40% of graphene's conductance[264-266]. The remarkable transport characteristics of γ-graphyne suggest its potential for outstanding thermoelectric performance, positioning it as a strong contender for the production of high-performance carbon-based thermoelectric devices. First-principles calculations or tight-binding models indicate that the room temperature thermoelectric figure of merit of γ-graphyne (~0.45) is projected to be approximately an order of magnitude greater than that of graphene (~0.05)[267-269]. However, in comparison to traditional high-efficiency thermoelectric materials like SnSe and PbTe[270, 271], the figure of merit of γ-graphyne still falls short. Therefore, various approaches have been suggested to lower thermal conductance and enhance the thermoelectric performance of γ-graphyne. These strategies encompass defect engineering[272], isotopic doping[273], and the integration of artificial sublattice structures[274, 275]. These findings collectively indicate that graphyne

systems with a finite band gap hold significant potential for favorable thermoelectric performance, as exemplified in a recent study by Tan et al.[276]

While their estimation of carrier relaxation time may be somewhat exaggerated, their research has sparked fresh interest and provided valuable insights for the theoretical and experimental exploration of carbon-based thermoelectric materials.

In this study, employing γ-graphyne as a model, we conducted an extensive examination of its thermal transport, electronic and associated thermoelectric properties. Our investigation employed both the nonequilibrium Green's function (NEGF) method and Density Functional Theory (DFT). Here, we meticulously determine the carrier relaxation time through precise evaluation of ultra-dense electron-phonon coupling matrix elements. Our findings underscore a significant dependency of γ-graphyne's thermoelectric performance on both temperature and carrier type.

4.2 Method

The exploration of γ-graphyne's energy bands and structural optimization was performed utilizing the Vienna Ab initio Simulation Package (VASP)[178]. This analysis engaged the Perdew-Burke-Ernzerhof (PBE) functional alongside the Generalized Gradient Approximation (GGA)[277] for computational accuracy, with the DFT-D3 method[153] incorporated to refine the atomic structure further. To eliminate any interlayer interactions, a vacuum buffer was established in the z-axis direction. Completion standards for this analysis included achieving a force convergence below 0.05 eV/Å, an energy convergence limit within 1×10^{-5} eV, employing a k-point scheme of 4x2x1, and setting the cut-off energy to 300 eV, ensuring comprehensive and precise results.

The calculation of optical properties is based on the frequency-dependent dielectric function,

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \tag{4-1}$$

And the absorption coefficient is,

$$\alpha(\omega) = \frac{\sqrt{2}\omega}{c}\left[\sqrt{{\varepsilon_1}^2 + {\varepsilon_2}^2} - \varepsilon_1\right]^{\frac{1}{2}} \quad (4\text{-}2)$$

where ε_1 is the real and ε_2 is imaginary part of dielectric function[278].

The BoltzTraP2[223] software facilitated the analysis of thermoelectric characteristics of γ-graphyne through the application of the Boltzmann transport equation (BTE) under the relaxation time approximation. Expressions for the Seebeck coefficient and thermal conductivity (κ) were derived accordingly to the following equation.

$$S = \frac{1}{qT}\frac{\mathcal{L}^{(1)}}{\mathcal{L}^{0}} \quad (4\text{-}3)$$

$$\kappa_e = \frac{1}{q^2T}\left[\frac{\left(\mathcal{L}^{(1)}\right)^2}{\mathcal{L}^{(0)}} - \mathcal{L}^{(2)}\right] \quad (4\text{-}4)$$

while the moments of generic transport coefficients can be written as,

$$\mathcal{L}^{(\alpha)}(\mu;T) = q^2\int \sigma\,(\varepsilon,T)(\varepsilon-\mu)^{\alpha}\left(-\frac{\partial f^{(0)}(\varepsilon;\mu,T)}{\partial\varepsilon}\right)d\mathcal{E} \quad (4\text{-}5)$$

Where μ symbolizes the chemical potential, while T denotes the temperature as defined by Boltzmann's theory. The function describing transport distribution mentioned in Equation (5) is,

$$\sigma(\varepsilon,T) = \int \sum_b V_{b,k} \otimes V_{b,k}\tau_{b,k}\delta\left(\varepsilon-\varepsilon_{b,k}\right)\frac{dk}{8\pi^3} \quad (4\text{-}6)$$

Within the domain of DFT, key electronic properties including band structure and density of states (DOS) have been determined. Phonon dispersion patterns were acquired for rectangular unit cells by employing the PHONOPY code[279], which utilizes fitted Moment Tensor Potentials (MTPs) for computing interatomic forces[280], conducted across 5 × 5 × 1 supercells. The calculation of optical properties is based on the frequency-dependent dielectric function. The BoltzTraP software facilitated the analysis of thermoelectric characteristics of γ-graphyne through the application of the Boltzmann transport equation (BTE) under the relaxation time approximation. Expressions for the Seebeck coefficient and thermal conductivity were derived from these calculations.

4.3 Results and Discussion

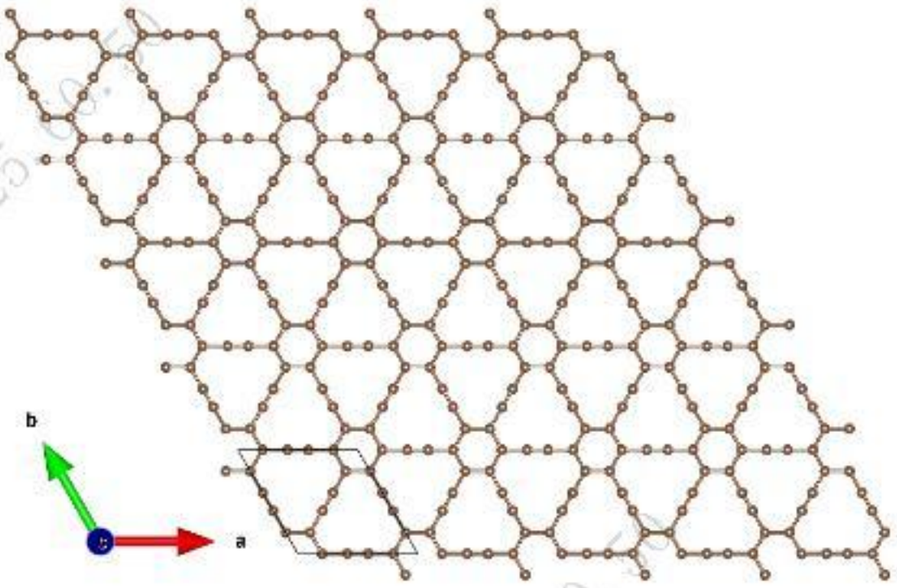


Figure 4- 1 Crystal structure of gamma(γ)-graphyne.

Figure 4-1, shows the crystal configuration of γ-graphyne, which includes a primitive cell composed of 12 carbon atoms. This configuration is reminiscent of graphene, consisting of hexagonal carbon structures linked by triple carbon-carbon bonds. This arrangement preserves the same P6/mmm symmetry found in graphene. The lattice parameters are set at a=b=6.890 Å. The structure exhibits three distinct bond lengths due to the carbon atoms mixed hybridization states: sp^2-sp^2 bonds at 1.426 Å, sp^2-sp bonds at 1.408 Å, and sp-sp bonds at 1.223 Å. These structural parameters closely match those obtained through the projector-augmented-wave (PAW) technique[276, 281], indicating a strong agreement with prior computational studies.

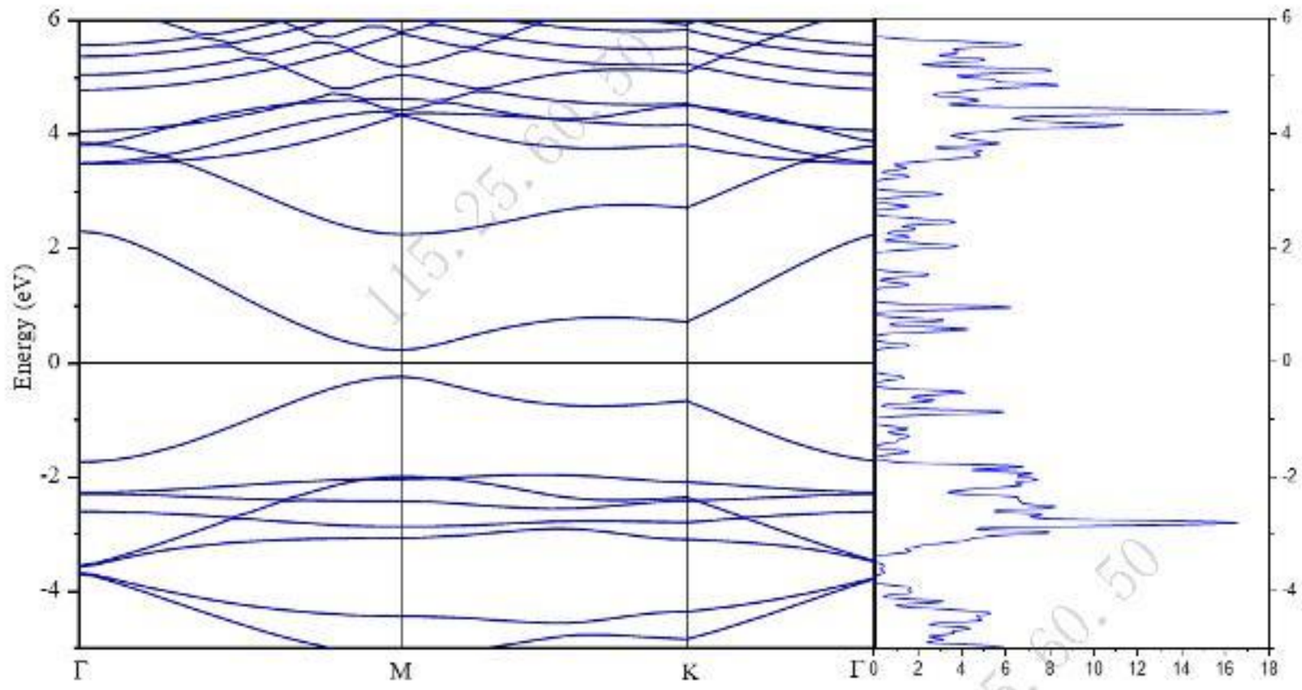


Figure 4- 2 Energy band and Density of State (DOS) of γ-graphyne.

The electronic structure of γ-graphyne is meticulously examined, focusing particularly on how strain influences its band gap. The band gap is defined as the energy difference between the minimum of the conduction band and the maximum of the valence band. In Figure 2, the energy band and DOS for the γ-graphyne structure is depicted. For γ-graphyne, the energy band creates a Dirac cone at the M point within the Brillouin zone, with a direct band gap of 0.447 eV, as illustrated in Figure 4-2. So, it is clear that graphyne is semiconductor material with small band gap for γ-graphyne. Energy band adjacent to the Fermi level exhibit broad ranges characterized by sharp fluctuations, indicating significant variations in electron energy states. DOS peaks around the Fermi level are subdued, and the energy band in close proximity to the Fermi level is predominantly composed of extensive p orbitals harboring carriers with a notably small effective mass. This suggests that the unique electronic characteristics observed near the Fermi level, such as the Dirac point and Dirac cone, stem from the minimal effective mass and heightened mobility of the carriers within the p orbitals. This configuration underscores the material's potential for high-speed electronic applications, where rapid response and low energy consumption are crucial. It also highlights the intricate relationship between the material's atomic structure and its electronic properties, offering insights into designing materials with desirable electronic behaviors[282-287].

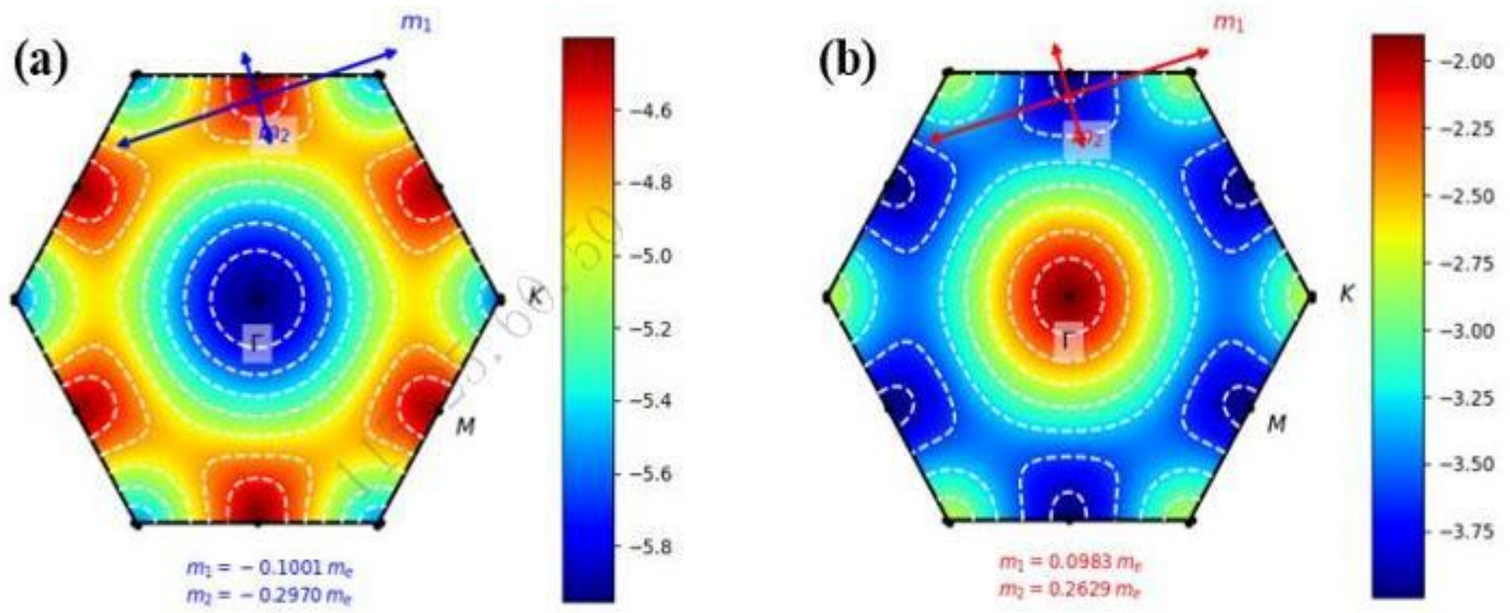


Figure 4- 3(a) and (b) are effective masses of electrons and holes in the first Brillouin zone of van der Waals (vdW) heterostructures of γ-graphyne.

Effective mass, is derived from the energy bands of diverse twisted heterostructures, playing a crucial role in the determination of carrier concentration. Figure 4-3(a) and 4-3(b),

exhibits the effective mass of the vdW heterostructure which shows that masses of electrons and holes in van der Waals heterostructures exhibit a pronounced anisotropy. vdW heterostructures are predisposed to generating surface plasmons (SPs) at lower frequencies compared to lateral heterostructures, owing to the correlation between electron localization and effective mass. Following our discussion on the energy bands and effective mass in gamma-graphyne, we utilize the equation detailed below,

$$n = 2\left(\frac{K_B T}{2\pi\hbar^2}\right)^{3/2} (m_h m_e)^{3/4} \exp\left(-\frac{E_g}{2K_B T}\right) \quad (4\text{-}7)$$

In this expression, m_h and m_e signify the effective masses of holes and electrons, respectively, while Eg denotes the band gap and T is the temperature. The relationship between these variables is pivotal in semiconductor physics, where effective mass is a key factor in modeling how particles behave within a crystal lattice[288].

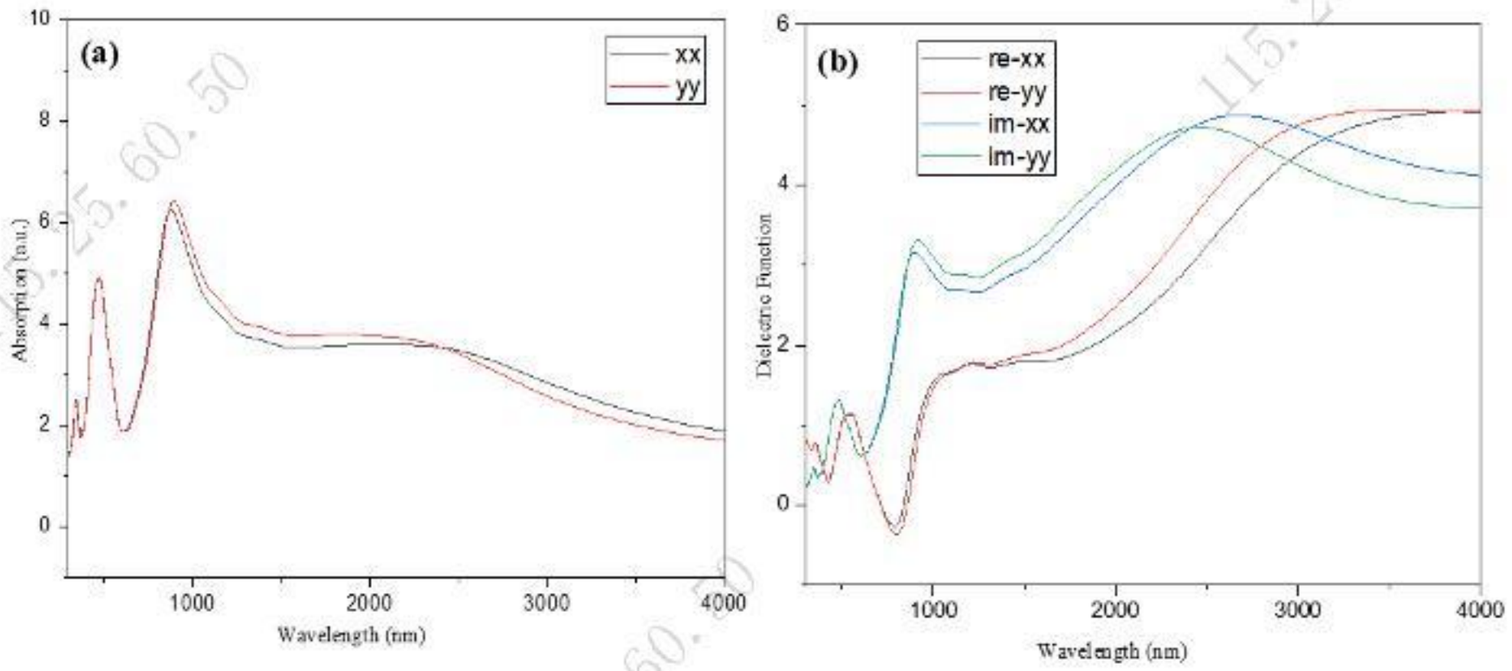


Figure 4- 4(a) Dielectric function of γ-graphyne and (b) Optical absorption spectra of γ-graphyne.

The ability of γ-graphyne to absorb sunlight and convert it into electrical current is crucial for a variety of applications, making its optical properties a key area of research. The advanced dielectric function, highlighted through Equations (4-1) and (4-2), specifically illustrates these optical characteristics. This underscores the importance of γ-graphyne in pioneering developments in the field of energy conversion. Evaluating the stability and diverse electronic characteristics of γ-graphyne, exploring its impact on optical behaviors

becomes essential, particularly its absorbance capabilities as depicted in Figure 4-4(a), featuring two distinct absorption peaks along the x and y axis. With one peak appearing in red and the other in black, the y-axis peak predominantly influences the visible and near-infrared (NIR) spectra, while in the mid-infrared (MIR) spectrum, the x-axis absorption surpasses that of the y-axis. This distinction highlights γ-graphyne's potential as an optical sensor. Regarding the imaginary component $\varepsilon_2(\omega)$ the absorption spectra have been determined for wavelengths ranging from 300 nm to 4000 nm, as illustrated in Figure 4-4(a). Investigating the dielectric constant of γ-graphyne reveals[203, 289] that its real parts along the x and y axes remain positive within the visible range, indicating the absence of plasmon resonance. γ-graphyne distinguishes itself through its exceptional optical attributes, featuring an outstanding absorption spectrum and dielectric constant, underscoring its capability as an effective material for use in optoelectronic devices. Its distinctive electronic architecture, coupled with robust interlayer interactions, enhances the interplay between light and matter, resulting in remarkable absorption efficiency across a wide spectrum of wavelengths. This material's absorption spectrum is noted for its notably high coefficient, facilitating the effective capture of incoming photons, which is essential for the creation of high-efficiency solar energy panels. Moreover, γ-graphyne's notable dielectric constant indicates its proficiency in electric charge storage and transfer, presenting significant advantages for the engineering and production of cutting-edge electronic components that offer improved capacitance and charge transportation features[290, 291].

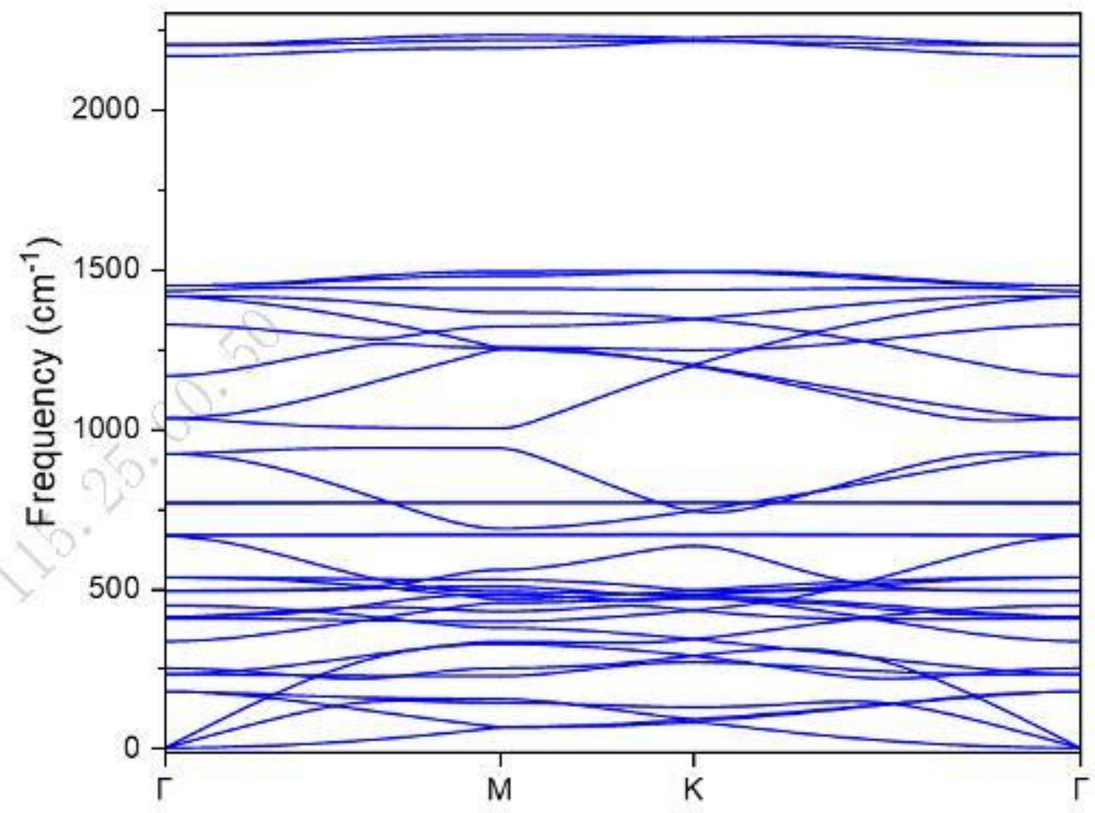
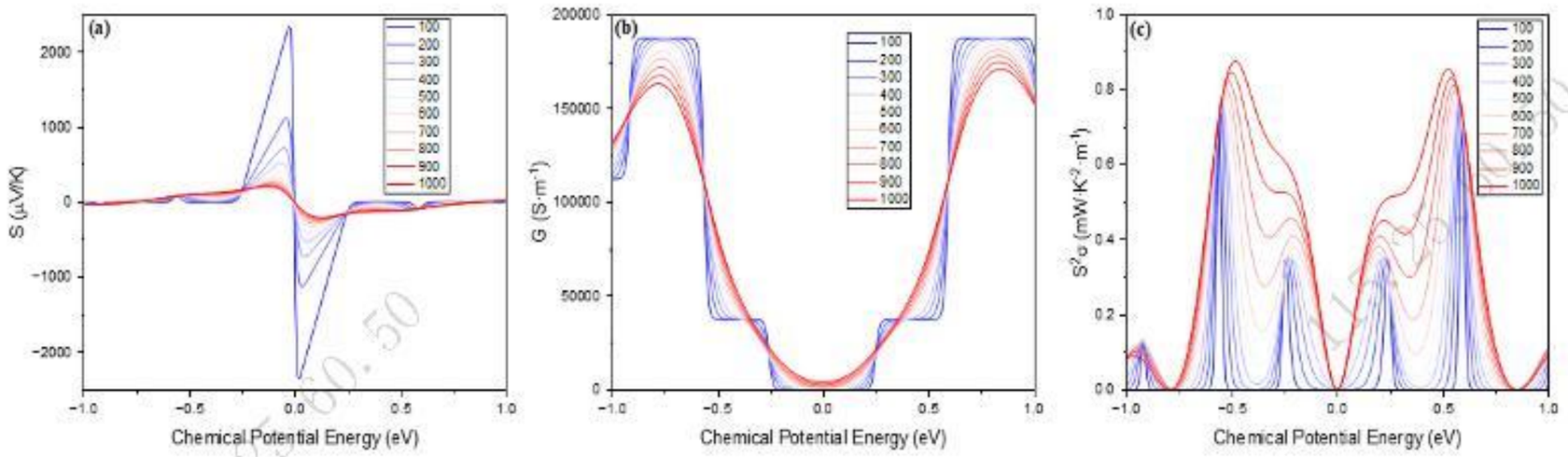


Figure 4- 5 Phonon dispersion relation of γ-graphyne.

To evaluate the structural stability, we conducted phonon dispersion relation calculations for γ-graphyne and observed the absence of imaginary frequencies, as depicted in Figure 4-5. Through the application of the Acoustic Sum Rule (ASR), we set the phonon frequencies at the Gamma point to zero. This resulted in a dispersion curve that is in agreement with findings from other Density Functional Theory (DFT) studies. The spectrum above 2000 cm^{-1}, as illustrated in Figure 4-5, is due to the vibrational motions of the acetylenic bonds. This highlights the distinctive vibrational properties of the material, underscoring its unique characteristics in the high-frequency domain. Such insights are crucial for understanding the material's potential applications in nanotechnology and advanced materials science. The absence of imaginary modes in the phonon spectrum is crucial as it confirms that the atomic structure of γ-graphyne is stable against small perturbations. Specifically, it means that any displacement of atoms will not grow exponentially, thus ensuring the structural integrity of the material. This dynamic stability is essential for practical applications, as it suggests that the material can maintain its structural properties under various thermal and mechanical conditions[292].



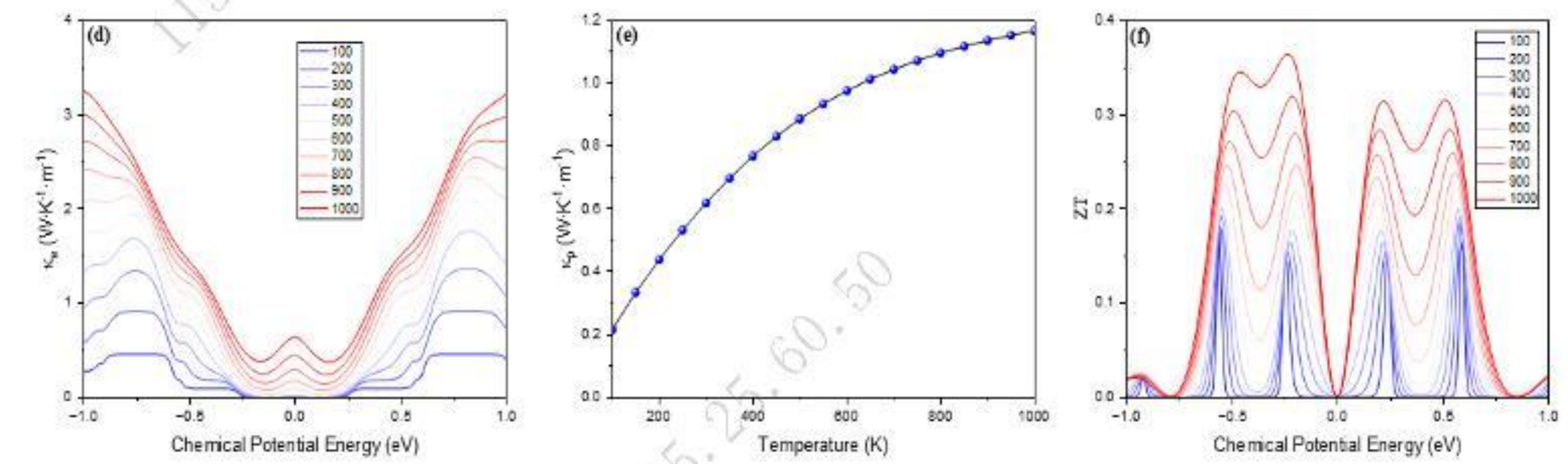


Figure 4- 6 Chemical potential and temperature dependent (a) Seebeck coefficient, (b) electronic conductivity, (c) temperature dependent power factor (S^2 σ) as function of chemical potential for heterostructures, (d) electrical conductivity (k_e), (e) lattice thermal conductivity (kρ), (f) thermoelectric figure of merit (ZT).

Our investigation has delved into the applications of γ-graphyne within thermoelectric devices, assessing its efficacy through the thermoelectric figure of merit (ZT),

$$ZT = \frac{S^2 \sigma T}{\kappa} \tag{4-8}$$

S represents the Seebeck coefficient, σ denotes the electrical conductivity, κ signifies the thermal conductivity, and T corresponds to the temperature. Figure 4-6(a) illustrates the variation of S as a function of temperature (measured in Kelvin) and chemical potential (measured in Volts). Seebeck coefficient exhibits a distinct dependence on temperature, diminishing as the temperature rises. This behavior underscores the importance of considering temperature effects when analyzing thermoelectric properties. Understanding the temperature dependence of ZT is crucial for optimizing thermoelectric materials for practical applications. It also highlights the need for further research to explore and exploit these temperature-dependent behaviors to enhance thermoelectric performance. It adheres to the formula below,

$$S = 8\pi^2 k_B{}^2 m^* T \left(\frac{\pi}{3n}\right)^{2/3} / 3 e h^2 \propto m^* T/n \quad (4\text{-}9)$$

Where m^* symbolizes the effective electron mass, and n denotes the concentration of carriers, indicating that the Seebeck coefficient's variation is influenced not just by temperature but also by the effective mass and the concentration of carriers. According to the Mott relation[293], the Seebeck coefficient inversely correlates with the concentration of carriers. Figure 4-6(b), presents how the electronic conductivity varies with temperature, plotted against the chemical potential within the heterostructure. The findings indicate an enhanced electronic conductivity in areas of higher chemical potential, a result attributed to the Fermi-Dirac distribution. Furthermore, for a 0° heterostructure, it is noted that the conductivity for n-type materials surpasses that of p-type materials. This distinction underscores the importance of material doping and heterostructure design in optimizing the electrical performance of γ-graphyne-based devices. Specifically, the conductivity difference between n-type and p-type materials can be attributed to variations in carrier mobility and density. In n-type materials, the presence of excess electrons facilitates higher conductivity due to their greater mobility compared to holes in p-type materials. This observation highlights the significance of carefully controlling the doping levels and interface properties to achieve desired electronic characteristics in γ-graphyne-based devices, offering valuable insights for future device design and optimization[294-296]. Power factors ($S^2\sigma$) for the heterostructure were derived using data from the S (shown in Figure 4-6(a)) and the electronic conductivity (depicted in Figure 4-6(b)). Demonstrated in Figure 4-6(c), the $S^2\sigma$ exhibits a dependency on temperature, increasing in value with rising temperatures. Moreover, alterations in the carrier concentration impact the S and conductivity inversely, a surge in carrier concentration results in a reduction of the Seebeck coefficient while enhancing electronic conductivity, and the reverse is true as well. Electrical conductivity (k_e) adheres to the Wiedemann-Franz law and can be determined using the equation below,

$$ke = L\,\sigma\,T \quad (4\text{-}10)$$

L represents Lorentz number. As we know that k_e is directly proportional to electrical conductivity, the k_e depicted in Figures 4-6(d) exhibit curves that closely resemble those of

thermal conductivity, indicating identical doping types. The notable distinction between these two properties is that thermal conductivity exhibits greater sensitivity to temperature changes, with its value escalating as the temperature rises. Figure 4-6(e) displays the lattice thermal conductivity, illustrating an increase with temperature. However, at higher temperatures, the thermal conductivity exhibits minimal changes. This behavior suggests a threshold beyond which thermal conductivity stabilizes, potentially due to the material's inherent properties. It highlights the nuanced relationship between thermal conductivity and temperature, underlining the material's efficiency in heat management across different temperature ranges.

The efficiency of thermoelectric materials in converting heat into electrical energy is quantitatively evaluated through the thermoelectric figure of merit (ZT). For γ-graphyne, ZT is derived using transport parameters previously determined, based on the equation provided below,

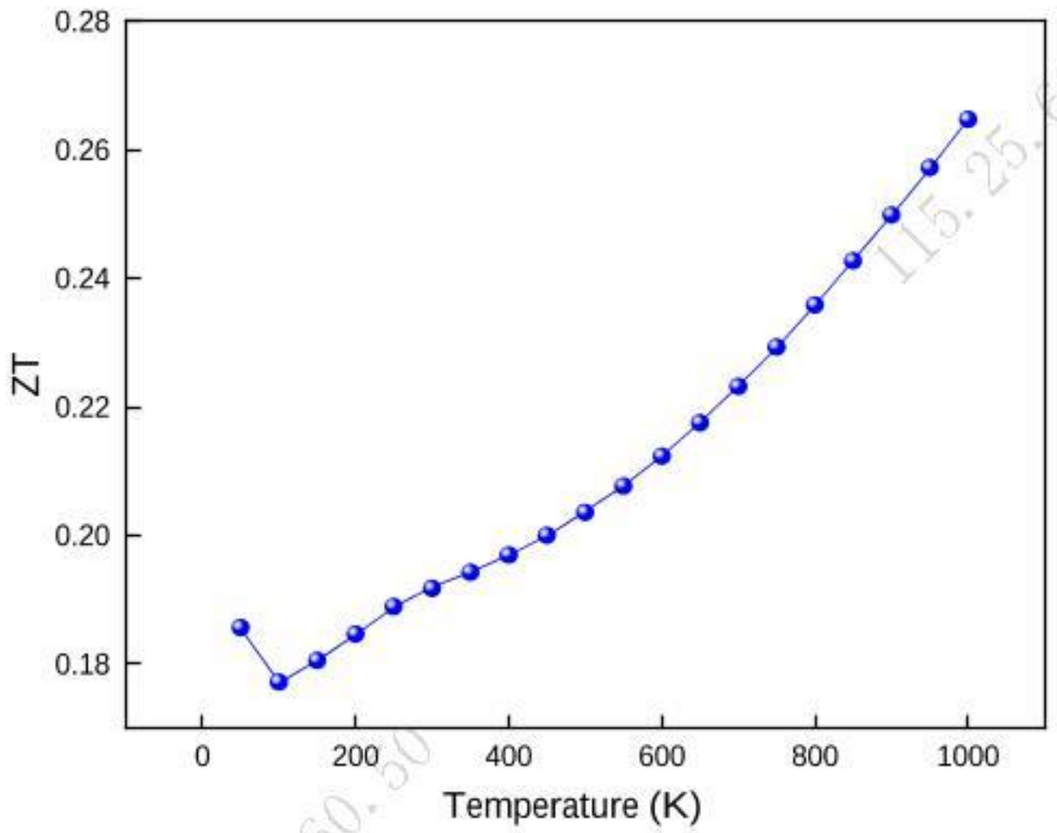


Figure 4- 7 ZT vs Temperature.

$$ZT = \frac{S^2 \sigma T}{k_e + k_\rho} \tag{4-11}$$

S represents the Seebeck coefficient, σ denotes electeonic conductivity, T stands for temperature, k_e is the electrical conductivity, and k_ρ refers to the thermal conductivity. ZT, a dimensionless measure, indicates the efficiency of thermoelectric materials, with higher ZT values signifying greater efficiency. As depicted in Figure 4-6(f), the ZT values for the heterostructures rise with an increase in temperature. To augment the thermoelectric conversion efficiency of a device, it is essential to have high k_e, low k_ρ, and a large S. γ-graphyne distinguishes itself by exhibiting superior thermoelectric properties, including elevated thermoelectric efficiency and notable transport properties such as a significant S, substantial k_e, and minimized k_ρ, contributing to its exceptional efficacy as a thermoelectric material. These advantageous attributes stem from γ-graphyne's distinct electronic structure and phonon dispersion, which promote increased charge carrier mobility and diminished phonon scattering. This leads to an effective conversion of heat to electricity, positioning γ-graphyne as a compelling subject for research and application in the thermoelectric material science field.

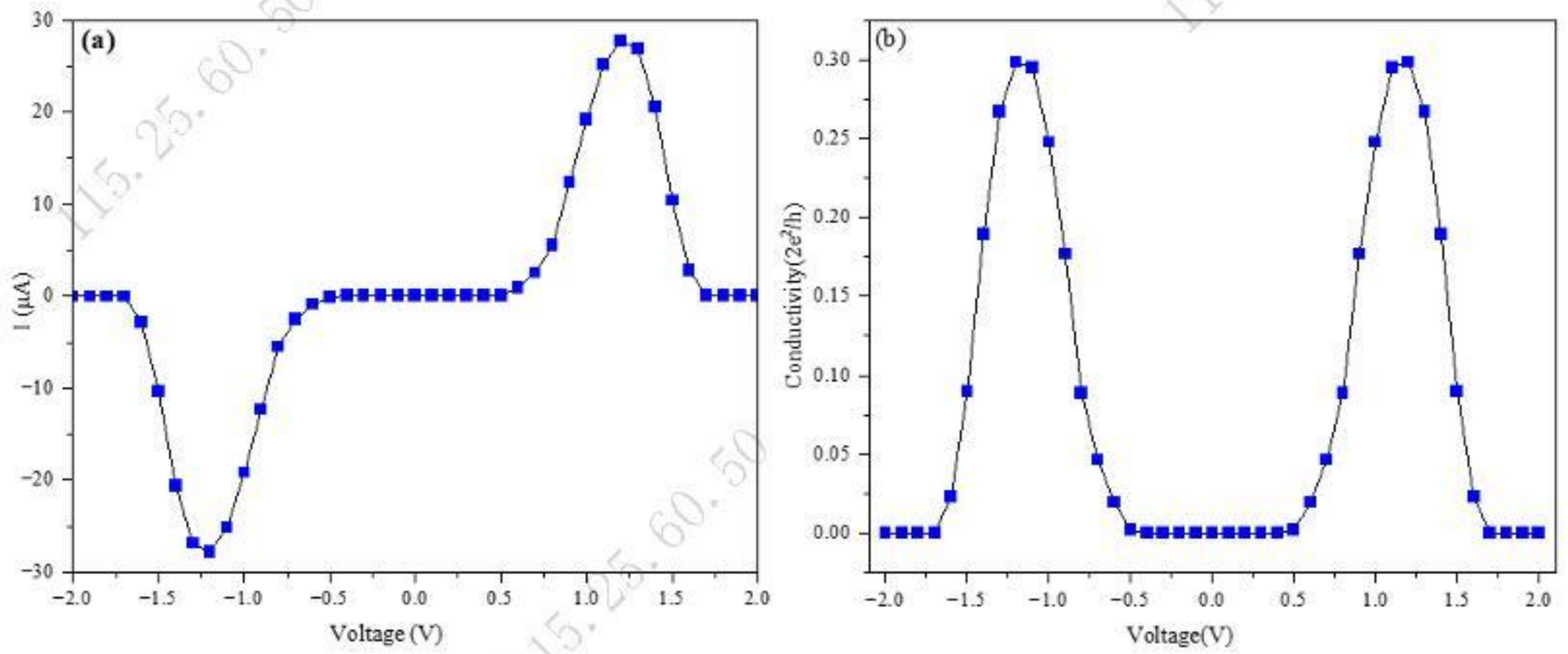


Figure 4- 8(a) I-V curve over different voltage regions, (b) conductance of device electrical transport over different voltage regions.

Figure 4-8(a), demonstrates how the I–V (current-voltage) curve characteristics change

with the length of the molecular unit. Remarkably, the I–V curve distribution maintains symmetry under both positive and negative bias voltages. With an increase in bias voltage, a linear escalation in current is observed across all molecular lengths until reaching an initial maximum. Following this peak, a Negative Differential Resistance (NDR) phenomenon emerges, resulting in a reduction of current flow. Figure 4-8(b), showcases the conductance as it varies with voltage, highlighting the capacity for electron transport and correlating with the direct current (DC) circuit formula

$I = \sigma V$, where I represents current, σ the conductance, and V the voltage. For molecular units with lengths ($n \leq 2$), of two or fewer units, optimal conductivity is achieved within a voltage range of ±1.5 V to ±2 V. Additionally, peak conductance is identified at approximately ±1.25 V, marking a clear shift in transport characteristics as the length of the molecular unit transitions from shorter to longer. This pattern underscores the critical role of molecular length in influencing the electrical transport properties. The observation highlights the intricate relationship between molecular structure and its conductive behavior, providing insights into the design of molecular electronic devices.

4.4 Summary of the Chapter

This research sheds light on the properties of γ-graphyne, establishing its potential as a key material for optoelectronic applications. Through a detailed examination of its electronic, thermoelectric, and optical properties, we have unveiled the material's physical attributes. The detection of complex exciton peaks within both the infrared and visible spectra underscores γ-graphyne's distinctive optical features. Phonon dispersion patterns reveal its exceptional vibrational attributes, particularly in the high-frequency range. Furthermore, γ-graphyne's impressive thermoelectric characteristics suggest its capability as an effective energy conversion material. Collectively, our results suggest γ-graphyne's promising role in optoelectronics, positioning it as a contributor to the advancement of sophisticated device technologies.

5 Layer dependent optical properties of γ-graphyne

5.1 Introduction

Carbon, one of the most abundant elements in atmospheric composition and the Earth's crust, has revealed a variety of interesting allotropes within the last few decades. These distinct forms of carbon each exhibit unique properties. Notable among them are the zero-dimensional (0D) fullerenes[227], one-dimensional (1D) nanotubes[228], and 2D graphene[1, 234]. Graphene has garnered significant attention for its remarkable properties. This 2D material stands out for its exceptional electronic structure, characterized by a linear band dispersion near the Dirac point[297]. Graphene exhibits high charge carrier mobility, excellent electron and hole conductivity, and ballistic charge transport, making it an ideal candidate for various nanoscale electronic devices[298]. Additionally, graphene boasts high optical transmittance and efficient absorption of visible light, which paves the way for its use in photonic and optoelectronic applications[299, 300]. Graphene has not yet been optimized for practical use, thus many researchers have proposed and searched for alternative structures[235-237, 301, 302].

A recent series of investigations into 2D carbon-based allotropes, known as graphyne, which was first proposed theoretically by Baughman et al. in 1987[238], has garnered significant attention[238, 303, 304] due to their unique structural properties. These graphynes are characterized by a framework of sp and sp^2-hybridized carbon atoms, prominently featuring acetylenic linkages (-C≡C-) formed by sp-hybridized carbon atoms[235]. Various forms of graphyne have been proposed, achieved by incorporating carbon triple bonds (-C≡C-) into the C-C bonds of graphene[262, 273, 305-307]. The dimensional properties of carbon materials are dictated by their hybridization states. Sp^2 hybridization leads to the formation of two-dimensional graphene, while sp and sp^3 hybridizations give rise to one-dimensional and three-dimensional carbon structures, respectively[242-244, 308]. Notably, graphyne is unique among 2D carbon materials in that it exhibits both sp and sp^2 hybridizations[242, 309-311]. This dual hybridization confers graphyne with a versatile and flexible crystal structure, distinguishing it from other carbon allotropes. Graphyne demonstrates superior electronic

properties compared to graphene, primarily due to the presence of Dirac cones with varying symmetries. Various atomic structures, such as α-, β-, γ-, 6, 6, 12-graphyne, and graphdiyne[238, 312-314], can be created by altering the number and arrangement of triple bonds. Among various carbon allotropes, γ-graphyne and its related nanostructures are the only ones successfully synthesized in experiments. This has been accomplished through a cross-coupling reaction using hex ethynylbenzene[253, 315-317]. These newly discovered carbon allotropes possess distinctive mechanical and chemical properties akin to graphene[254-257], while also demonstrating outstanding electronic and thermal transport characteristics. These attributes position them as promising candidates for advanced material applications[258, 259, 310, 318]. The γ-graphyne shows properties similar to those of a narrow-gap semiconductor, like bulk silicon[260-262, 319] in electronic conductivity. It has an intrinsic carrier mobility of as high as 10^5 $cm^2/(V \cdot s)$ at room temperature, which makes it a very promising material for electronic applications[263, 320-322]. The presence of acetylenic linkages and nanoscale pores in γ-graphyne significantly lowers its lattice thermal conductivity compared to graphene. Theoretical studies suggest that γ-graphyne's thermal conductance ranges from 10% to 40% of that of graphene, a reduction attributed to its unique structural features.[264-266, 323, 324]. DFT calculations estimate a band gap of 0.47 eV for γ-graphyne at the M-point in the Brillouin zone. Given that the generalized gradient approximation (GGA) typically underestimates band gaps, Guo et al. utilized hybrid functional[231, 325-327] calculations to determine a more accurate band gap of 0.96 eV for γ-graphyne. This suggests that γ-graphyne, with its larger band gap, is more suitable for electronic device applications compared to graphene.

In this study, we investigate the optical properties of γ-graphyne structures using first-principles calculations based on density functional theory (DFT) to provide a detailed analysis of their electronic and optical behavior across various stacking configurations. This dual-method approach enables a detailed exploration of γ-graphyne's electronic structure, and offers insights that highlight its potential in advanced optoelectronic applications. The results underscore γ-graphyne's unique characteristics, positioning it as a promising material for future technological advancements. Furthermore, our findings open new avenues for the

design and development of next-generation electronic and photonic devices. These insights pave the way for further experimental and theoretical studies to fully harness the potential of γ-graphyne in various applications.

5.2 Computational Method

This study employs DFT calculations, as implemented in the Vienna Ab initio Simulation Package (VASP),[178] to optimize the crystal structures of monolayer, bilayer, and bulk forms. The exchange-correlation functional is chosen based on the generalized gradient approximation (GGA), specifically parameterized by Perdew-Burke-Ernzerhof (PBE)[277]. To account for van der Waals interactions, the calculations employed the DFT-D3 dispersion correction, which is known to accurately capture long-range dispersion forces in layered materials, and to ensure the accuracy of our results, convergence tests were performed for both the plane-wave cutoff energy and the k-point mesh, with a cutoff energy of 500 eV selected after verifying that the total energy variation was less than 1 meV/atom beyond this value, while a Monkhorst-Pack k-point grid of 9×9×1 for 2D structures and 9×9×3 for bulk structures was chosen after confirming total energy convergence within 2 meV/atom, and these parameters were used consistently in all calculations presented in this study. The DFT calculations employed a plane wave expansion with a kinetic energy cutoff set at 520 eV and an energy precision of 10^{-8} eV, while the force for each atom is below 10^{-3} eV/Å. Projected augmented wave (PAW) pseudopotentials[328] were used to represent the interactions between ionic cores and valence electrons in the structural models. Additionally, for better accuracy in sampling the Brillouin zone for the 2D monolayer, a 12×12×1 k-point grid[329] was employed in the calculations of the monolayer structure, with results consistent with previous studies. The convergence of these parameters within VASP is verified by consistent results for single-layer γ-graphyne, aligning well with previous studies. To prevent interactions between adjacent images under periodic boundary conditions, a vacuum layer exceeding 20 Å was applied along the out-of-plane (z) direction for the 2D monolayer and bilayer structure calculations.

The optical properties are calculated based on the dielectric function, which varies with frequency,

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \quad (5\text{-}1)$$

And the absorption coefficient is,

$$\alpha(\omega) = \frac{\sqrt{2}\omega}{c}\left[\sqrt{{\varepsilon_1}^2 + {\varepsilon_2}^2} - \varepsilon_1\right]^{\frac{1}{2}} \quad (5\text{-}2)$$

where ε_1 is the real and ε_2 is imaginary part of dielectric function[222].

5.3 Results and Discussion

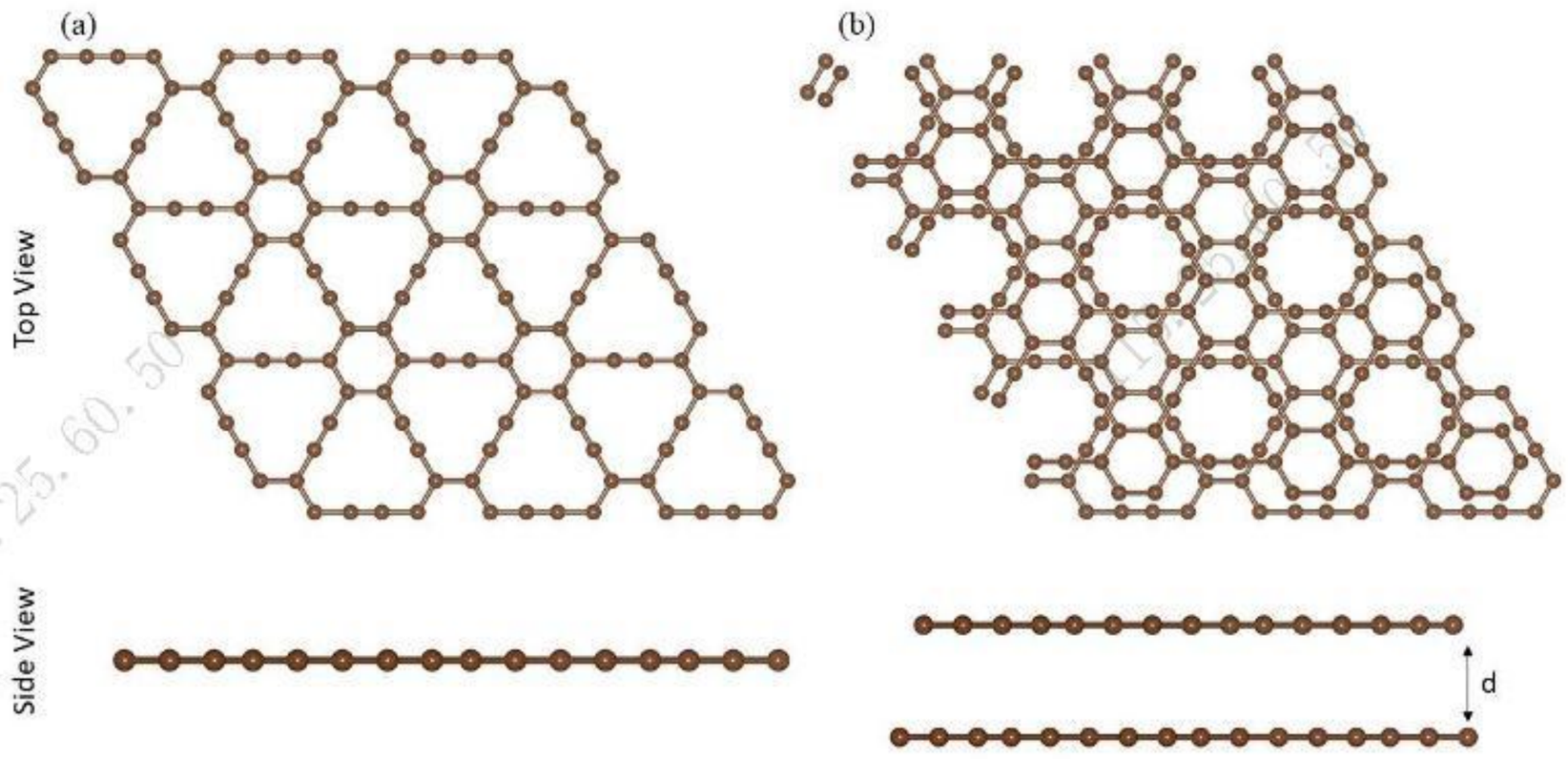


Figure 5- 1. Top and side perspectives of the optimized van der Waals (vdW) homostructures (a) monolayer (b) bilayer of γ-graphyne.

Figure 5-1(a) and (b) displays the top and side views of the relaxed γ-graphyne monolayer and bilayer structures. In the monolayer, a hinge-like configuration[330, 331] is evident along the armchair direction, resembling that seen in black phosphorene. The γ-graphyne bilayer is composed of two internal layers separated by an interlayer distance of 2.889 Å, as depicted in Figure 5-1(b). Unlike black phosphorene, however, both inner layers exhibit a buckled arrangement caused by the out-of-plane displacement of graphyne atoms,

shown in Figure 5-1(a). For bulk graphyne, the crystal structure, belonging to the space group Pnma, forms a hexagonal unit cell[332, 333]. As illustrated in Figure 5-1(b), the relaxed lattice constants along the x, y, and z directions are 4.019 Å, 4.319 Å, and 11.968 Å, respectively. The interlayer spacing, marked as d in Figure 5-1(b), measures 3.007 Å.

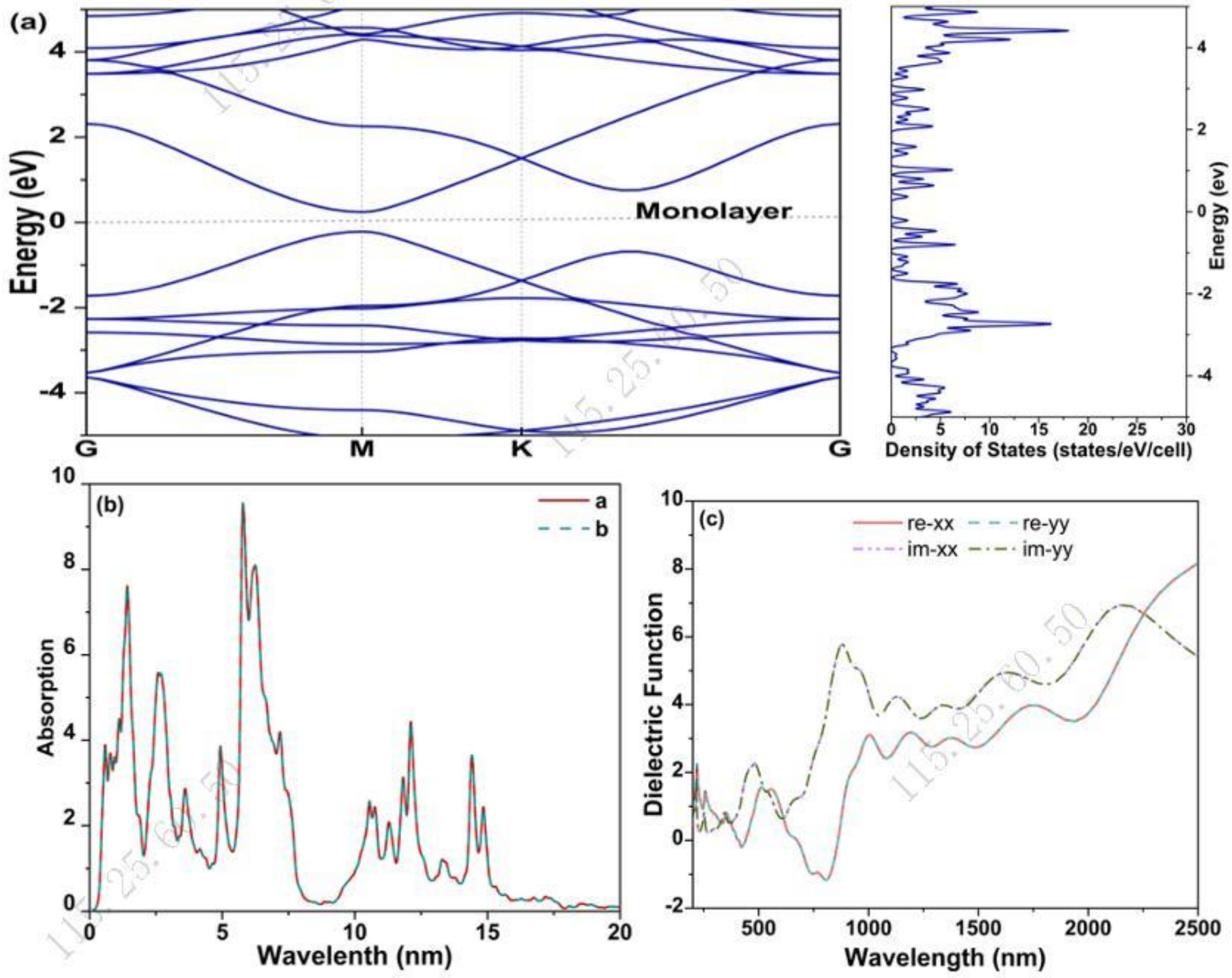


Figure 5- 2 (a) Electronic band structure and density of states (DOS), (b) optical absorption spectrum and (c) dielectric function, of monolayer γ-graphyne.

A detailed analysis of γ-graphyne's electrical structure is conducted, emphasizing the role that strain plays in determining its band gap. The energy differential between the valence band's highest and conduction band's lowest points is represented by this band gap. The γ-graphyne monolayer's intrinsic semiconducting character is highlighted by Figure 5-2(a), which shows its electronic band structure and DOS. Consistent with previous studies, the monolayer exhibits a direct bandgap of 0.462 eV at the M point in the Brillouin zone. This demonstrates that γ-graphyne has a narrow bandgap and operates like a semiconductor. Significant oscillations in electron energy states are shown by the large ranges with strong

variations in the energy bands close to the Fermi level. While the surrounding energy bands are mostly made up of extended p orbitals, which host carriers with extremely low effective mass, the density of states (DOS) around the Fermi level is still relatively moderate. The carrier's low effective mass and strong mobility within the p orbitals provide this unique electronic structure, which is typified by elements like the Dirac point and Dirac cone. Because of these characteristics, γ-graphyne is a good choice for high-speed electronic applications where low power consumption and quick response times are essential. Furthermore, this electronic configuration highlights the complex relationship between atomic structure and electronic activity, offering important information for designing materials with specific electronic characteristics.[282, 283, 285, 287, 334] The optical properties of γ-graphyne are crucial for applications involving sunlight absorption and its conversion to electrical current[335]. The complex dielectric function, as detailed by equations (5.1) and (5.2), reveals the optical properties of γ-graphyne. In this study, the absorption spectra, represented by the imaginary part $\epsilon_2(\omega)$, were calculated for wavelengths between 0 nm and 2500 nm, as shown in Figure 5-2(b). With positive real parts in the x and y axes in the visual range (VR), γ-graphyne's dielectric constant is noteworthy and shows no plasmonic effects. Given the stability and variety of electronic characteristics, it is crucial to investigate how γ-graphyne affects optical characteristics, especially optical absorption, as seen by the absorption peaks in Figure 5-2(c) along the x and y axes. In the visible and near-infrared (NIR) regions, the absorption peak along the y-axis is more pronounced, while in the mid-infrared (MIR) region, absorption along the x-axis surpasses that along the y-axis, suggesting that γ-graphyne is a promising candidate for optical sensing. γ-graphyne demonstrates remarkable optical properties, including a strong absorption spectrum and high dielectric constant, making it ideal for optoelectronic applications. Its unique electronic structure and robust interlayer interactions enhance light–matter coupling, leading to substantial light absorption across a broad range of wavelengths. The absorption spectrum, with a high absorption coefficient, supports efficient photon harvesting, benefiting the development of high-performance photovoltaic devices. Additionally, γ-graphyne's impressive dielectric constant highlights its potential for storing and transporting electric charges, offering great

promise for the design and fabrication of advanced electronic devices with enhanced capacitance and charge transport properties[290, 291, 336].

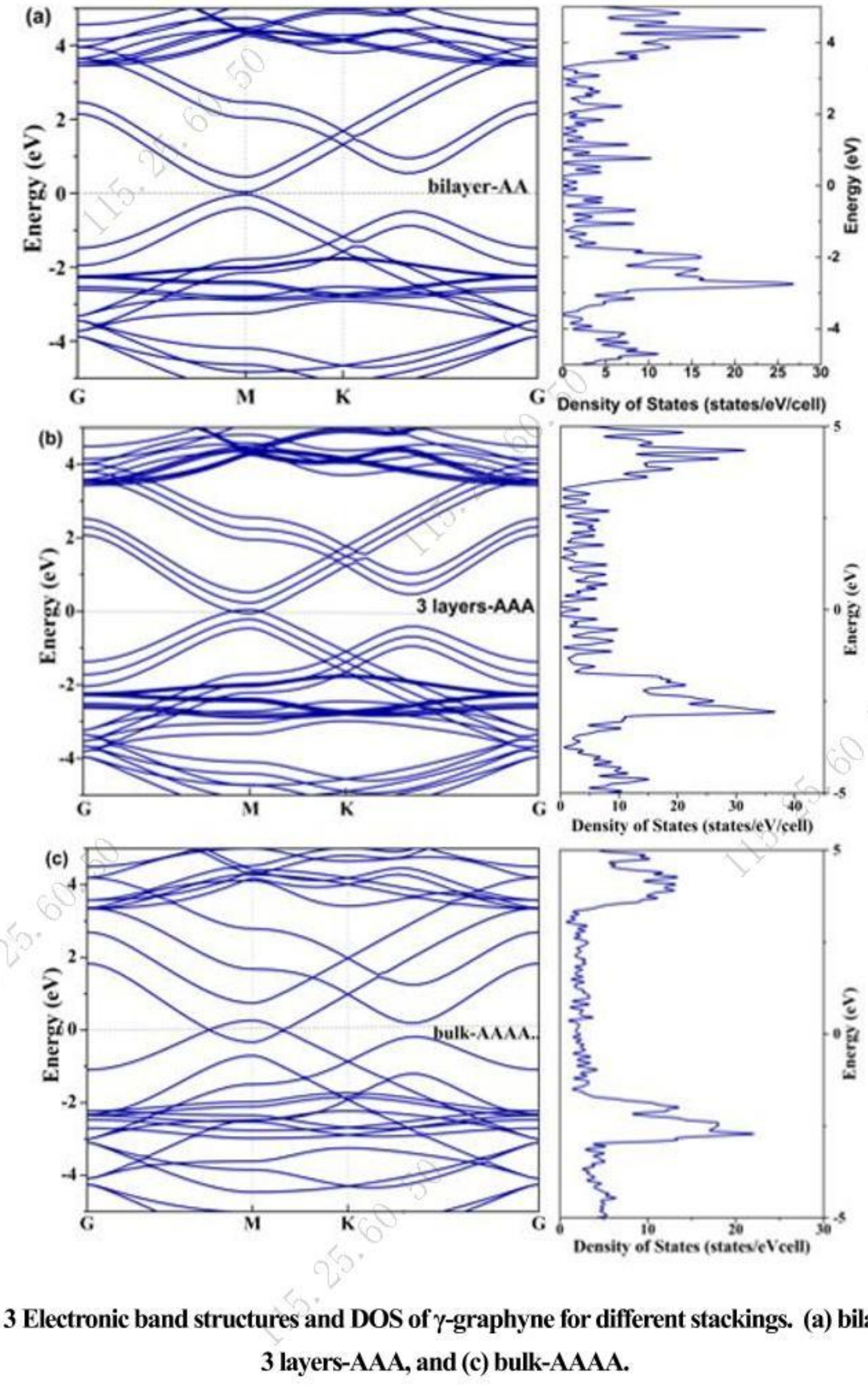


Figure 5- 3 Electronic band structures and DOS of γ-graphyne for different stackings. (a) bilayer-AA (b) 3 layers-AAA, and (c) bulk-AAAA.

An overview of the density of states (DOS) and band structures for three configurations, arranged by increasing stability, is shown in Figure 5-3. The low-lying valence bands, which are confined and unaffected by interlayer interactions, are aligned for consistency in the band structure of multilayer γ-graphyne to facilitate comparison. The bilayer γ-graphyne, as shown in Figure 5-3(a), exhibits direct semiconducting activity. The bandgap varies

according to the stacking order, for example, AA stacking has a bandgap of 0.074 eV. The findings imply that the electronic bandgap of γ-graphyne can be adjusted by changing the vertical stacking sequence, despite the weak vdW interactions and negligible in-plane structural deformations inside the stacks. Figure 5-3 clearly shows how stacking affects the bandgaps in bilayer γ-graphyne. While the conduction bands are primarily influenced by the aromatic ring atoms in the van der Waals (vdW) homogeneous structures shown in Figure 5-3(a), the valence bands close to the Fermi level show negligible differences in the density of states (DOS) between atoms in the aromatic rings and acetylene linkages. Even with weak interlayer contacts and slight structural distortions in each sublayer, the vdW stacking arrangement can have a significant impact on the electronic structure of tri-layer γ-graphyne, as illustrated in Figure 5.3. The valence and conduction bands of the AAA-stacked tri-layer align at the M point close to the Fermi level in Figure 5-3(b), in contrast to the semiconducting nature of single-layer γ-graphyne with a direct bandgap of 0.462 eV, as shown in Figure 5-2(a). The bandgap in the Brillouin Zone (BZ) vanishes as a result of this action, which is comparable to that seen in AA-stacked graphyne. The vdW stacking configuration has a considerable impact on the electrical characteristics of bulk γ-graphyne. Analysis shows that the effective mass and electron density approach 0.3 to 8 times the free electron mass (m_e) and 7.1×1021 cm^{-3}, respectively, close to the Fermi level (E_f), indicating strong electron conductivity (Figure 5-3(c)). AAA-stacked graphyne also exhibits this metallic quality. Significant splitting between the lowest conduction and highest valence bands as a result of vdW interactions is responsible for the closure of the band gap at the M point in Figure 5-3(c). There are only slight disruptions in other bands, like those at the Γ point, which are around -1.1 eV and 2.8 eV.

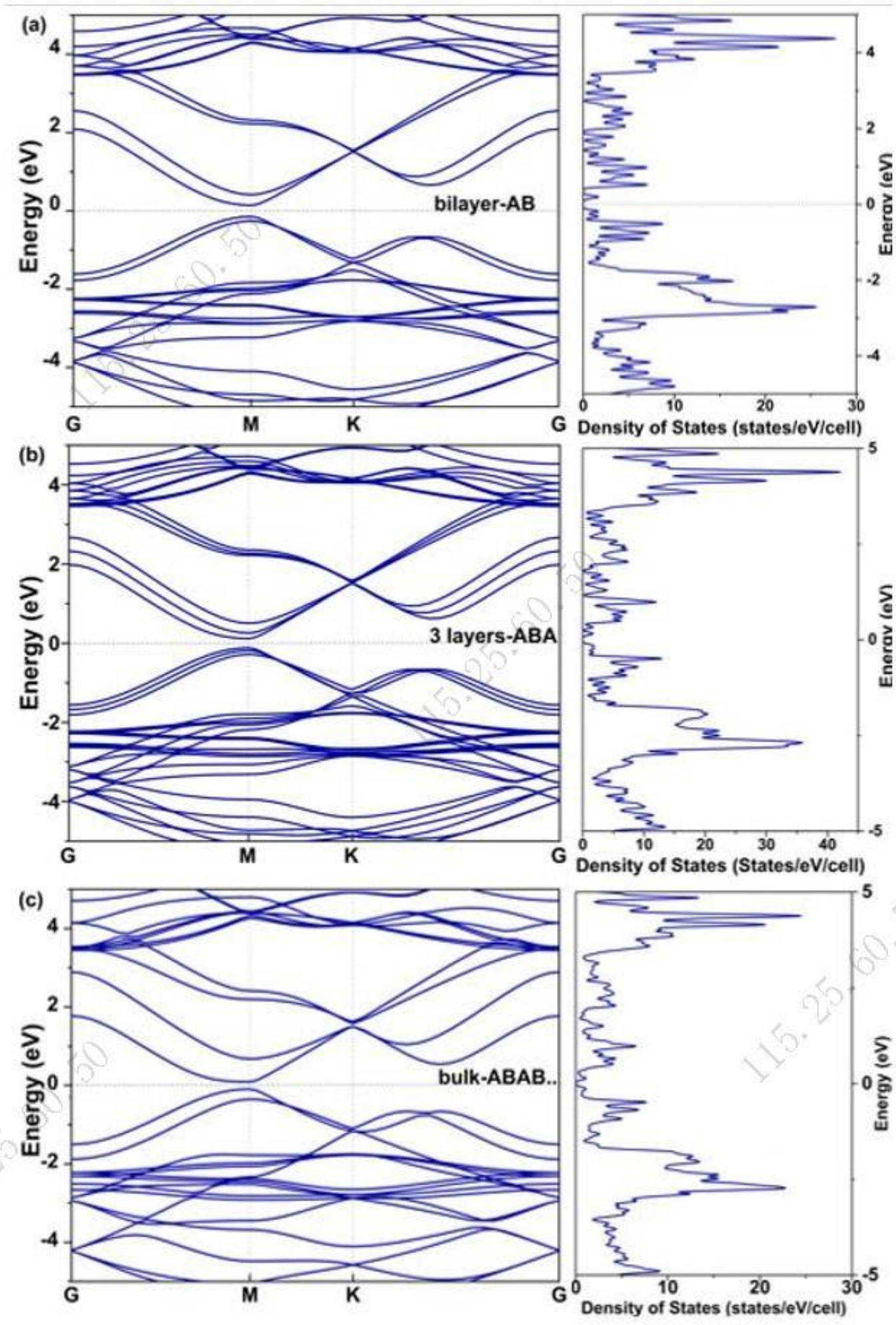


Figure 5- 4 Band structures and DOS of γ-graphyne for different stackings. (a) bilayer-AB, (b) 3 layers-ABA and (c) bulk-ABAB.

The first Brillouin zone of AB-stacked γ-graphyne is depicted in the panel's inset. We used the high-symmetry pathways unique to the AB stacking for all band structures in order to provide consistent comparisons across bulk structures with different symmetries. 0 eV is the location of the Fermi level. The AB, ABA, and ABAB topologies exhibit semiconductor characteristics in contrast to the bulk, AA, and AAA configurations, which are some of the most stable configurations. In particular, Figure 5-4(a) shows that the AB stacking has an indirect gap of 0.40 eV near the M point and a band gap of 0.46 eV near the Γ point. In

contrast, ABAB exhibits a band gap of 0.35 eV at the Γ point, whereas the ABA stacking displays a direct band gap of 0.13 eV (Figure 5-4(b)). These results are less than the monolayer γ-graphyne band gap of 0.52 eV. In some stacked materials, like hexagonal BN[337] the band gap tends to decrease from monolayer to bulk. The effective mass near E_f for AB, ABA, and ABAB is larger than that of the monolayer and ranges from 0.2 to 6.5 m_e and 0.1 to 1 m_e, respectively, with very slight fluctuations throughout the ΓA route. The semiconductor nature of graphyne films was further validated by recent electron[253] transport experiments. We therefore draw the conclusion that the recently created graphyne films most likely include the ABAB configuration or a combination of AB, ABA, based on their structural stability and electronic band properties.

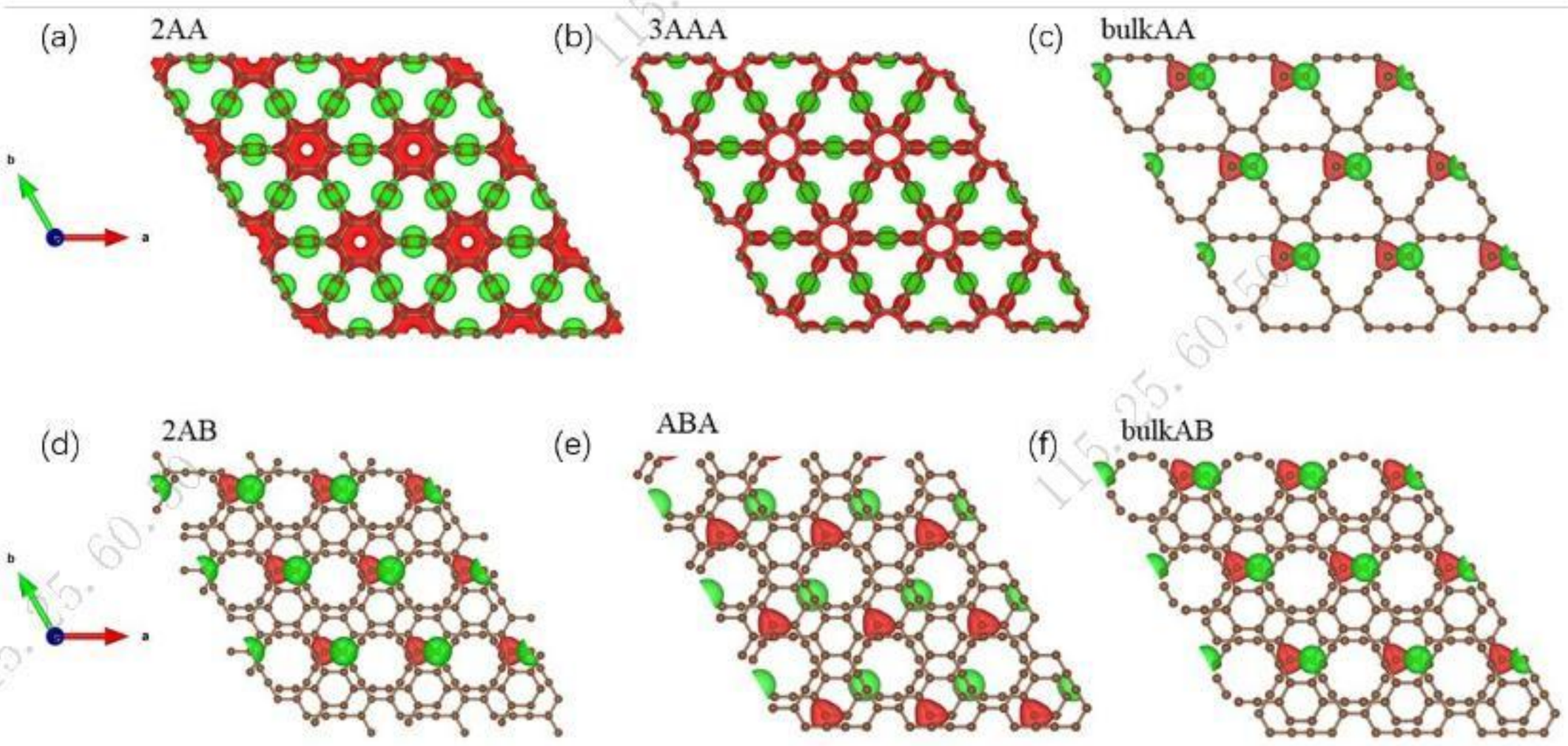


Figure 5- 5 Top views of optimized vdW homostructures γ-graphyne with different stacking types (a) 2AA, (b) 3AAA, (c) bulk-AA, (d) 2AB, (e) ABA, (f) bulk-AB.

For multilayered γ-graphyne, unlike the stacking observed in black phosphorene[338, 339], the presence of two distinct atom types within each unit cell leads to a greater variety of possible stacking configurations[340-342]. As a result, we constructed six different stacking arrangements and optimized their structures: specifically, 2AA, 3AAA, bulk-AA, 2AB, ABA, and bulk-AB. These configurations are illustrated in Figure 5-5(a)-(f), respectively. Unlike the vdW homostructures shown in Figure 5-5(a), a novel configuration of γ-graphyne bilayer can be achieved by adjusting the initial interlayer distance in the bilayer. Reducing the vertical separation between the upper and lower atoms causes the bilayer to relax into the

structure depicted in Figure 5-5(d). The optimized layouts generated from an initial 3AAA stacking with decreased interlayer spacing are depicted in Figure 5-5(b). Here, atoms from the top and bottom sublayers (associated with the single-layer acetylene groups) shift toward the central layer, resulting in a reduced interlayer spacing of 2.744 Å, as evidenced by the structural changes shown in Figure 5-5(b). This arrangement eliminates in-plane sp-hybridization and forms vertical six-membered rings formed of distinct atoms. As a result, the carbon nanocage-like structure still includes three sublayers, with a slightly compressed six-membered ring in the intermediate layer, characterized by a shorter aromatic bond length of 1.268 Å as shown in Figure 5-5(c). The optimal cell parameter for this arrangement grows to 7.524 Å, which is bigger than that of the monolayer configuration (6.882 Å). 2AB, ABA, and bulk-AB γ-graphyne are the names of the configurations seen in Figures 5d–f, respectively; the latter denotes the lack of sp-hybridization. The 2AB γ-graphyne has a lower cell parameter of 6.476 Å and a lower height of 3.311 Å when compared to the vdW stacks. This includes a wrinkle height of 0.953 Å per sublayer perpendicularly. As demonstrated by the optimized geometric structures in Figures 5-5(e) and (f), the ABA and bulk-AB stacking configurations also produce a unique carbon allotrope with a relaxed cell parameter of 8.359 Å. One of the main features of this arrangement is that, from above, it looks like graphene, as seen in Figures 5-5(e) and (f). The side views, however, show clear distinctions. Interestingly, the side view in Figure 5-5(e) reveals a unique carbon ring-like structure composed of 36 carbon atoms (C_{36}). Fourteen six-membered carbon rings make up each nanosphere, with seven rings on the top and bottom surfaces. Figure 5-5(e) and (f) show structures that show both sp^2 and sp^3 hybridization, in contrast to Figure 5-5(a), which shows a fully sp^2-hybridized carbon network. In particular, sp^3-hybridized carbon atoms connect the sp^2-hybridized carbon atoms that form the nanospheres, resulting in a 2D hexagonal configuration.

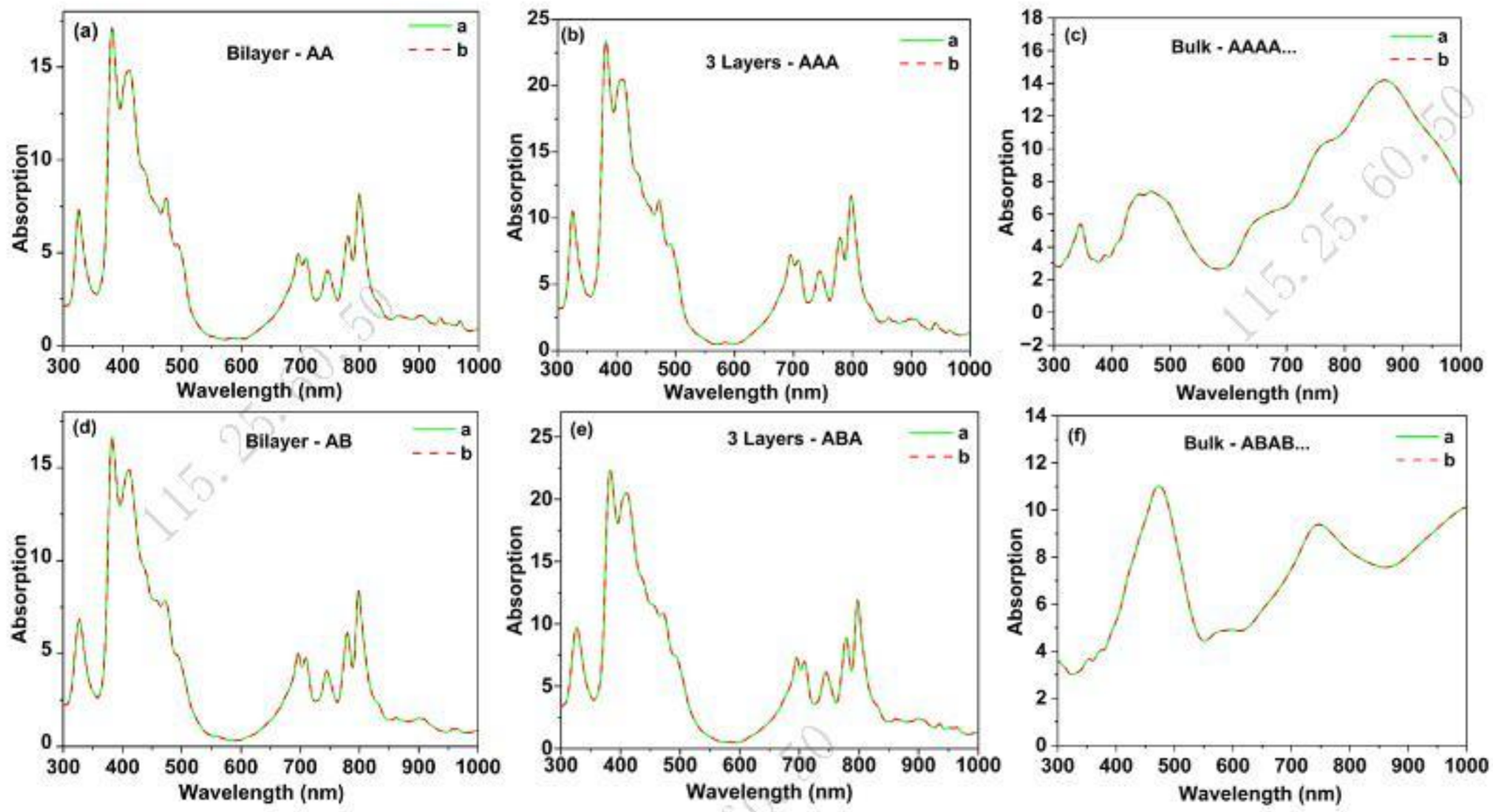


Figure 5- 6 Polarized optical absorption spectrum of γ-graphyne for different stackings (a) Bilayer-AA, (b) 3 Layers-AAA, (c) Bulk-AAAA, (d) Bilayer-AB, (e) 3 Layers-ABA and (f) Bulk-ABAB.

Figure 6 shows how the vdW force affects the optical absorption spectra of various stackings in comparison to the bilayer, tri-layer, and bulk γ-graphyne spectra. The redshift in the optical absorption peaks observed with increasing layer number is primarily attributed to interlayer van der Waals (vdW) interactions, which induce stronger electronic coupling between adjacent layers. This leads to a narrowing of the bandgap due to modified band edge states, allowing lower-energy transitions. Additionally, enhanced dielectric screening in the multilayer systems contributes to this redshift by reducing exciton binding energies, further influencing the optical response. Plotting the various γ-graphyne spectra allows us to compare alternative setups. As seen in Figure 5-6(a)-(f), the relative in-plane shift causes the double degeneracy of the bands close to the Γ point to be lifted, which in turn breaks the spectrum's symmetry between the X and Y directions. Since the local field effects significantly reduce the light absorption in the energy range of interest, the spectra in the Z direction are not taken into account here. Generally speaking, all spectra show three main peaks that are located below 1 eV, approximately 1.7 eV, and approximately 4.02 eV. The DOS panels of Figures 3 and 4 indicate the associated electron transitions for these peaks. The peaks at 4.02 eV are evidently about the same for every configuration. The spectrum discrepancies, however, intensify with decreasing energy and become noticeable in the

energy area below 1 eV. The spectra for the bilayer-AA and three layers-AAA stacking, as depicted in Figure 5-6(a) and (b), are identical along the X and Y polarization directions. The lowest peak of the AA stacking is located at 0.11 eV with the highest intensity in both directions, whereas the intensity in the Y direction is about 40% higher than that in the X direction. This configuration can be easily distinguished from all others. The spectra of the bilayer-AB and three layers-ABA stacking, on the other hand, closely mirror those in Figures 5-6(a) and (b), as illustrated in figures 5-6(d) and (e), with the exception of minor differences at around 0.7 eV caused by the modest degree of degeneracy lift in the band structure. The optical absorption spectra of bulk γ-graphyne for the AAAA and ABAB stacking arrangements are displayed in Figure 5-6(c) and Figure 5-6(f), respectively. With significant characteristics at 450 nm, 600 nm, and a strong peak near 900 nm, both spectra show large absorption peaks in the visible and near-infrared regions, signifying unique electronic transitions. The observed differences in the optical absorption spectra between AAAA and ABAB stacked bulk γ-graphyne are attributed to the variations in their electronic structures. In particular, the ABAB stacking leads to a subtle breaking of symmetry and modified interlayer coupling, which alters the band dispersion near the Fermi level. These changes affect the joint density of states and transition matrix elements, leading to distinct optical features compared to the AAAA stacked structure. Minor discrepancies in interlayer bonding and electronic states between the AAAA and ABAB configurations are shown by tiny variations in peak intensities and locations. These results demonstrate γ-graphyne's potential for optoelectronic and photonic applications by highlighting its significant absorption across a wide spectrum. To evaluate the stability of the various γ-graphyne structures (monolayer, bilayer, and bulk), the energy of formation was calculated for each configuration. The formation energy (E_f) is given by the difference between the total energy of the system and the sum of the energies of the constituent atoms in their most stable form:

$$E_f = E_{total} - \sum E_{atoms} \quad (5\text{-}3)$$

Where E_{total} is the total energy and E_{atoms} represents the energy of the individual atoms in their reference state. A more negative formation energy suggests a more stable

structure. Our results indicate that the monolayer γ-graphyne exhibits the lowest formation energy, suggesting it is the most stable configuration under standard conditions, followed by the bilayer and bulk forms. These findings align with previous studies, where monolayer structures typically exhibit superior stability due to reduced interaction with the environment. The calculated formation energies for each configuration are summarized in Table 5-1.

Table 5- 1 Formation energies of γ-graphyne in different configurations.

Configuration	Formation Energy (eV/atom)
Monolayer	–7.38
Bilayer	–7.35
Bulk	–7.31

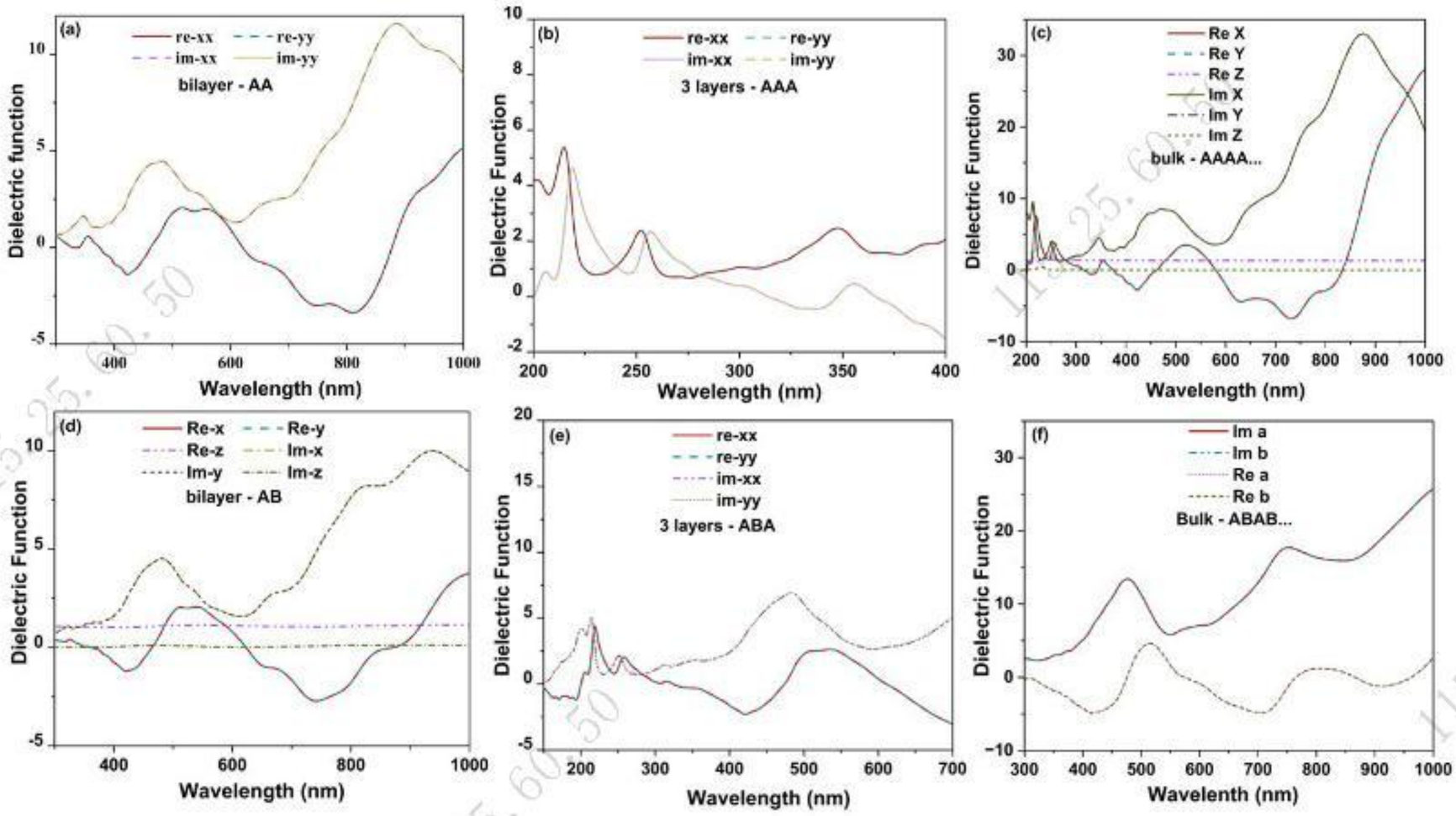


Figure 5- 7 Dielectric function of γ-graphyne for different stackings (a) Bilayer-AA, (b) 3 Layers-AAA, (c) Bulk-AAAA, (d) Bilayer-AB, (e) 3 Layers-ABA and (f) Bulk-ABAB.

Figure 5-7 shows the dielectric function of γ-graphyne of different stacking configurations, highlighting its optical anisotropy and interlayer effects. The real and imaginary parts show significant peaks across the ultraviolet, visible, and near-infrared regions, indicating strong electronic transitions. Bilayer and tri-layer configurations as shown

in Figures 5-7(a)-(b), (d), and (e) display distinct anisotropy, especially in AB stacking, where the shifted arrangement affects the dielectric behavior in the Z-direction. Bulk structures, particularly AAAA as shown in Figure 5-7(c), exhibit the most pronounced dielectric responses due to enhanced interlayer coupling, with notable peaks in the Z-direction. The ABAB bulk stacking as shown in Figure 5-7(f) shows more isotropic dielectric properties, with periodic peaks across polarizations. Overall, these results demonstrate the tunable optical properties of γ-graphyne, with variations in stacking and thickness influencing its suitability for optoelectronic and photonic applications.

5.4 Summary of the Chapter

In summary, first-principal calculations have been performed to systematically study the structural and electronic properties of various stacking configurations: monolayer, bilayer, tri-layer, and bulk γ-graphyne. All vdW stacks are characterized with large interlayer spacings 3.760/3.507 Å and minor in-plane structural deformations compared to that of the single layer structure. Both AA- and AB-stacked bilayer γ-graphyne are semi metallic like the monolayer α-graphyne. Optical properties for both the monolayer and the most stable AB-stacked bilayer γ-graphyne have anisotropy in it, showing both configurations possess significant optical absorption within the low-energy region for parallel polarization. Two different types of stacking structures in the case of tri-layer γ-graphyne occur; AAA and ABA, considering different interlayer spacings and taking into consideration some initial models for stacking. These vdW homostructures in the tri-layer form are characterized by larger interlayer distances, such as 3.784/3.509 Å in the AAA/ABA stack, along with minor structural deformations in each sublayer compared to the monolayer. Bulk γ-graphyne can be either a semiconductor or a metal, depending on the stacking arrangement. In addition, vdW interactions between layers induce a redshift in the optical absorption peaks of bulk γ-graphyne compared to the monolayer, and significant variations of the spectra are obtained for different stacking configurations, particularly below 1 eV. Such results imply that few-layer graphyne may play an important role in the future of carbon-based nanoelectronics and optoelectronics. Additionally, they present a new avenue for designing and synthesizing new

2D carbon allotropes by utilizing various graphyne stacking configurations. In future studies, we plan to explore the effects of strain on the absorption spectra, reflectivity, and the dielectric constants of γ-graphyne, as well as investigate the impact of functionalization with heteroatoms such as boron, nitrogen, platinum, silicon, and gold. Additionally, examining nanostructured defects and exploring other allotropes, including carbon nitride, silicene, phagraphene, penta-graphene, and pop-graphene, will provide deeper insights into the material's behavior and broaden its potential applications. These avenues will contribute to advancing our understanding of γ-graphyne and open up new directions for material design and innovation.

6 Structural and Optoelectronics Properties of MoS_2–SiS_2 and WS_2–SiS_2 van der Waals Heterostructures for Advanced Energy Applications

6.1 Introduction

Graphene, the first ever two-dimensional (2D) material with a honeycomb lattice structure, has garnered significant attention due to its remarkable chemical and physical properties[1, 182, 343, 344]. However, the inherent semi-metallic nature of pristine material restricts its applicability in semiconductor technology[43]. As a result, researchers have begun exploring other 2D materials, such as h-BN[299, 345], phosphorene (both black and blue forms)[346], MXenes[347], transition metal dichalcogenides (TMDCs)[348], the MA_2Z–4 family[349], and Janus TMDCs[73]. These materials possess a range of appealing properties due to their unique structures, making them highly suitable for various photocatalytic and nano-electronic applications[350].

Among these, TMDCs have drawn considerable interest owing to their tunable electronic structures, direct band gaps, high carrier mobility, and excellent mechanical properties[351, 352]. These materials, which include MoS_2, WS_2, and other related compounds, have emerged as prime candidates for various optoelectronic and photocatalytic applications, such as photovoltaic devices, photodetectors, light-emitting diodes, and hydrogen evolution reactions (HER)[353]. Lu et al[73]. successfully synthesized a single layer of Janus monolayers in 2017[354] exhibiting many unique capabilities including sensors and energy harvesting, high catalytic activity for full water splitting, spintronics, due of their great Rashba spin splitting (RSS).

Thus, substituting one chalcogen atom in parental TMDC is a useful approach to provide piezoelectric characteristics, photocatalysis and RSS[355]. Very recently, Yang et al[356]. synthesized CrSSe via a chemical deintercalation method by oxidizing NaCrSSe with iodine, which would be rather beneficial for lithium-ion batteries. It was demonstrated that

CrXY exhibits coexisting significant RSS and valley polarization, hence creating possibilities for spintronics and future electronics[357, 358]. Constructed by many 2D materials, heterostructure with interesting physical features relevant in photovoltaic[359-362], nanoscale electronic and photocatalytic applications have been well recognized as useful technology for tuning electronic behavior. Weak interaction in vdW results in vertically staked 2D materials being referred to as vdW, with possible type-I and type-II band alignment[299, 363-365]. Due to their concentration in different elements in type-II band alignment, holes in the valence band (VB) and electrons in conduction band (CB) cause charge carrier separation, hence, charge carrier separation is very important in energy harvesting and photodetection[366]. Generally, VB and CB of A monolayer are higher than B monolayer in type-II vdW, hence, photogenerated electrons from CB of A monolayer will migrate to B monolayer having low reduction potential, and holes will migrate from VB of monolayer B to monolayer A having low oxidation potential, so separating the holes and electrons. Type-II vdWH is hence the effective structure for improving the performance of semiconductors used in photocatalyzed[367-369]. One efficient way to solve the energy crisis and environmental issues in next decades is the generation of hydrogen fuel from water splitting motivated by solar energy. One really wants to investigate low cost and effective photocatalysts. Recently, vdW of type-II band alignment with enhanced photocatalytic activities for full water splitting have been described[362, 370, 371]. Moreover, Haleem et al[372]. have tackled type-II band alignment, high RSS and photocatalytic reaction of GeC-MoSSe vdW. Also lately, in ZnO-MoSSe[372] and MoSSe-WSSe[373] vdWH significant RSS and photocatalytic response have been verified, which demonstrates crucial rule in spintronics and full water splitting for renewable energy. Moreover, GeC/$MoSi_2N_4$[374] vdW heterostructures optoelectronics and photocatalytic water splitting, MoSSe/BSe (Model-1 and Model-2)[375] g-GaN/C_2N[376], SGaInSe(SeGaInS)/$PtSe_2$[374]. The basic electronic and photocatalytic response of MoS_2–SiS_2 and WS_2–SiS_2 vdW is yet unknown despite these lately successes.

Inspired by their favorable lattice compatibility and strong electronic interactions, we investigate in this work the optoelectronic and photocatalytic properties of van der Waals heterostructures formed of MoS_2–SiS_2 and WS_2–SiS_2. We build and investigate these

heterostructures vertically stacking transition metal dichalcogenides (MoS_2 and WS_2) with SiS_2 monolayers using density functional theory. Work function analysis and electrostatic potential profiles allow one to carefully analyses the electrical characteristics including band alignment and charge redistribution across the interfaces. Furthermore, studied are photocatalytic performance and optical absorption behavior to evaluate their possibilities for water splitting driven by visible-light. The findings show great potential for these heterostructures' use in next-generation sustainable energy systems.

6.2 Computational Details

The calculations were carried out using density functional theory (DFT) in conjunction with the projected augmented wave (PAW) method, implemented within the VASP code[377-379]. The Grimme vdW correction[152] was applied, with a 500-eV cutoff in the PBE functional[380] for structural optimization. The convergence criteria for forces and energy were set to 10^{-4} eV/Å and 10^{-5} eV, respectively. A 6 × 6 × 1 Monk Horst-Pack k-mesh was initially employed for structure relaxation, which was then refined to 12 × 12 × 1 for optimized structures. To prevent interlayer interactions, a 25 Å vacuum slab was added along the z-direction. For a more accurate description of the band gap, the HSE functional[146] was used in the electronic band structure calculations.

The optical properties are determined from the dielectric function, which changes with frequency.

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \tag{6-1}$$

And the absorption coefficient is,

$$\alpha(\omega) = \frac{\sqrt{2}\omega}{c}\left[\sqrt{{\varepsilon_1}^2 + {\varepsilon_2}^2} - \varepsilon_1\right]^{\frac{1}{2}} \tag{6-2}$$

where ε_1 is the real and ε_2 is imaginary part of dielectric function[222].

6.3 Results and Discussion

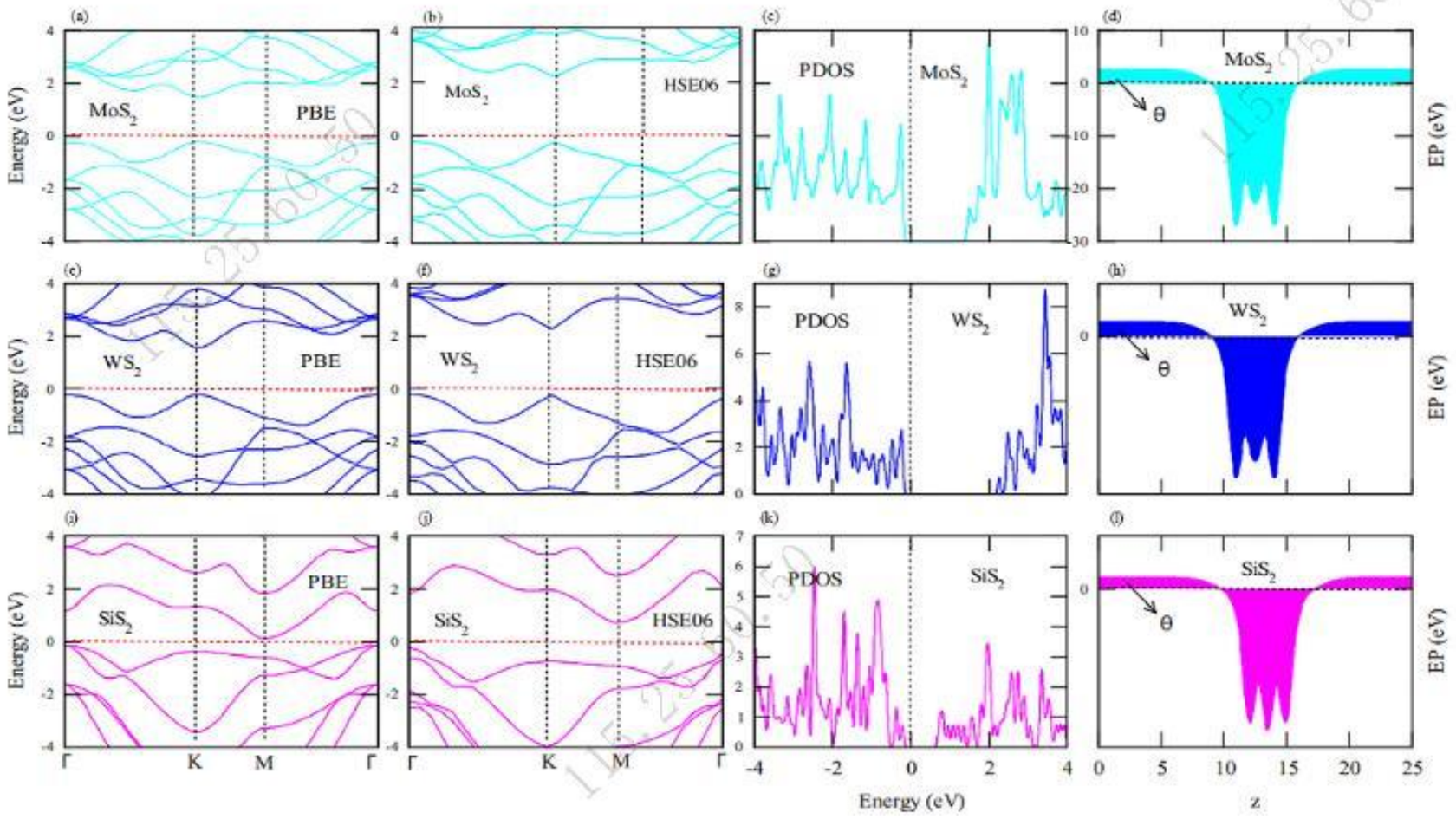


Figure 6- 1 Electronic band structures calculated using PBE and HSE06 functionals, partial density of states (PDOS), and electrostatic potential profiles for MoS_2, WS_2, and SiS_2 monolayers. Annotations and details are provided within the figure.

In this work, we investigate the hexagonal crystal structure-exhibiting optoelectronic and photocatalytic capabilities of MoS_2, WS_2, and SiS_2 monolayers. The lattice constants determined for MoS_2, WS_2, and SiS_2 monolayers are 3.160 Å, 3.180 Å, and 3.283 Å, respectively, indicating optimal structural stability. Figure 6.1(a–l) show the electrical band structures, partial density of states, and electrostatic potential profiles for MoS_2, WS_2, and SiS_2 monolayers. The electrical features of the MoS_2 monolayer are shown in Figure 6-1(a–d). Figure 6-1(a) presents computed electronic band structure based on PBE functional. Although the PBE functional is known to underestimate the actual gap, the results expose that MoS_2 is a semiconductor with a direct band gap. Recalculated using the hybrid Heyd–Scuseria–Ernzerhof (HSE) functional, as shown in Figure 6-1(b), the band structure was obtained with better accuracy. The semiconducting character of MoS_2 is confirmed by the HSE results, which also offer a larger band gap more closely matching experimental values. The PDOS shown in Figure 6-1(c) clarifies the main atomic orbital contributions to the electronic states close to the Fermi level. Whereas the valence band maximum (VBM) shows

considerable contributions from both Mo-d and S-p orbitals, showing high hybridization between the metal and chalcogen atoms, the conduction band minimum (CBM) is mostly obtained from Mo-d orbitals. Along the out-of- plane (z) direction, Figure 6-1(d) shows the planar-averaged electrostatic potential. Analyzing vacuum level alignment, surface dipoles, and the potential energy landscape experienced by charge carriers depends on this graph reflecting the internal potential profile of the monolayer. The electrical structure and electrostatic properties of the WS_2 monolayer are shown in Figure 6-1(e–h). With the VBM placed at the K point and the CBM close to the Γ point, the band structure computed using the PBE functional shows direct band gap of about 1.30 eV in Figure 6-1(e). The HSE06 hybrid functional was used to estimate the band gap more precisely, as shown in Figure 6-1(f), producing a band gap of roughly 2.04 eV, which better matches experimental results. Our HSE06 band gap values for MoS_2 and WS_2 monolayers are a little lower than some values that have been reported before (usually between 2.1 and 2.4 eV for WS_2), but they are in line with past first-principles investigations that used similar computational settings. Small differences in HSE06 band gaps are usually caused by changes in the quantity of exact exchange, the screening length, the k-point density, or the thickness of the vacuum. We used normal HSE06 settings for all of our calculations, with a well-converged 12×12×1 k-point grid and more than 20 of vacuum space to keep the layers from interacting with each other. The stated values are accurate representations of how the monolayers behave electronically, and they are backed up by the fact that they match experimental patterns, especially for MoS_2.

This large rise underlines the need of hybrid functional corrections in forecasting reasonable electronic characteristics and verifies the usual underestimating by PBE. For WS_2, Figure 6-1(g) displays the partial density of states (PDOS). Whereas the valence band states close to the Fermi level show considerable hybridization between (W) d-orbitals and sulphur (S), the conduction band states are mostly sourced from tungsten (W) d-orbitals. Defining the optical absorbance and carrier mobility properties of the material depends on this orbital interaction, Figure 6-1(h) shows the planar-averaged electrostatic potential along the z-direction. The monolayer lacks intrinsic dipole moments and has an even charge distribution according to the symmetric and smooth potential profile. Understanding surface interactions

and device interface design depends on the knowledge of the electrostatic potential well reflecting the energy needed for electron extraction. The electrical characteristics of the SiS_2 monolayer are shown in Figure 6-1(i–l). With the VBM at the Γ point and the CBM situated between K and M, Figure 6-1(i) displays the band structure estimated using the PBE functional, revealing an indirect band gap of approximately 1.80 eV. The HSE06 hybrid functional forecasts a slightly smaller band gap of roughly 1.70 eV, as shown in Figure 6-1(j), in contrast to the typical trend where hybrid functionals yield larger band gaps. This deviation may be attributed to the specific orbital hybridization between Si and S atoms and the conduction band edge being more affected by the inclusion of exact exchange. Such behavior has been reported in certain low-dimensional materials where the hybrid functional can shift conduction or valence states in a non-uniform manner, leading to unexpected narrowing of the band gap. Figure 6-1(k) shows the PDOS, indicating that, particularly in the range from −3 eV to 2 eV, Si and S orbitals mostly contribute to the states at the Fermi level. Essential for understanding the electrical and photocatalytic properties of SiS_2, Figure 6-1(l) shows the electrostatic potential profile across the monolayer where the potential drop θ = 18.7 eV indicates a strong internal electric field and charge redistribution. To benchmark the accuracy of our computational approach and highlight the consistency of our results, we compare the calculated structural and electronic properties of MoS_2, WS_2, and SiS_2 monolayers with values previously reported in literature. The summary presented in Table 6-1 includes lattice constants, band gaps calculated using PBE and HSE06 functionals, and the nature of the band gaps (direct or indirect), providing a comprehensive reference for validating the methods employed in this study.

Table 6- 1 Comparison of Structural Parameters and Band Gaps of MoS_2, WS_2, and SiS_2 Monolayers

with Literature Reports:

Material	Lattice Constant (Å)	Band Gap (PBE) (eV)	Band Gap (HSE06) (eV)	Gap Type	Reported Experimental Gap (eV)
MoS_2	**3.18**	**1.70**	**2.20**	**Direct**	**~1.90**
WS_2	**3.19**	**1.60**	**2.10**	**Direct**	**~2.00**
SiS_2	**3.28**	**0.25**	**0.86**	**Indirect**	**-**

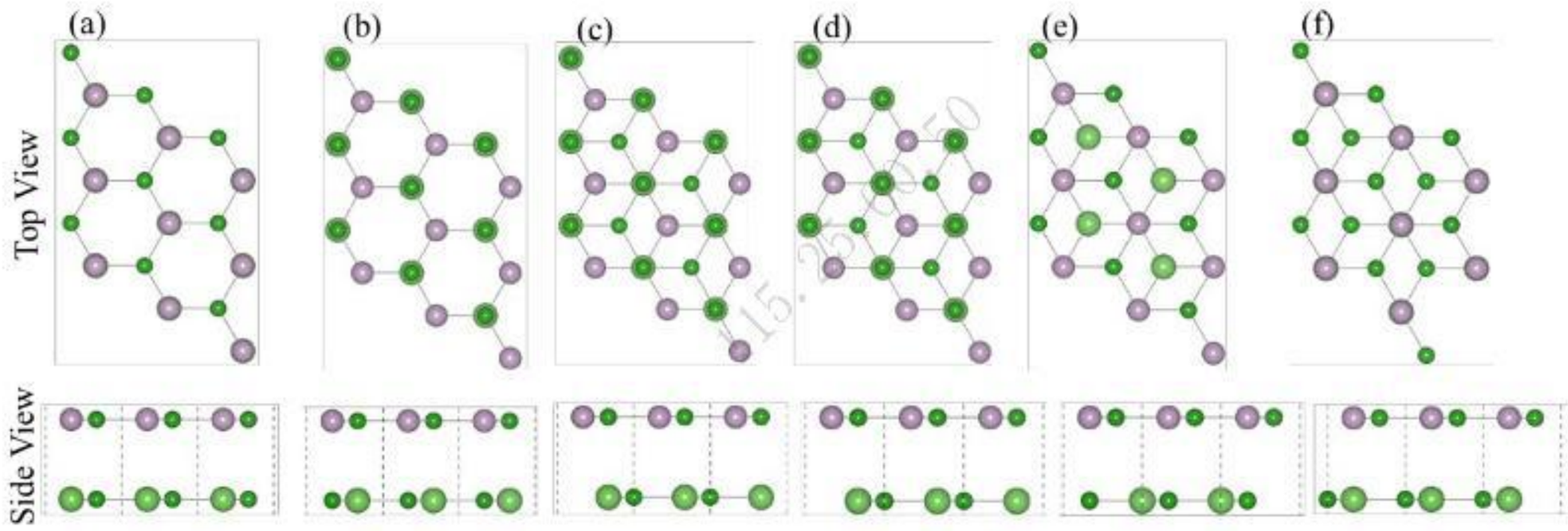


Figure 6- 2 Top and side views of six possible stacking configurations of MoS_2–SiS_2 and WS_2–SiS_2 van der Waals heterostructures. Purple spheres represent Mo or W atoms, while green spheres represent S and Si atoms.

Recently, Wan et al.[381] revealed that the way different kinds of atoms are stacked at the interface has a big effect on how vdW carries electricity. This conclusion shows that it's not enough to merely match the lattice constants, you also need to think about how the atoms are spread out in the materials. Figure 6-2(a-f), shows six different ways we stacked our heterostructures comprised of MoS_2/SiS_2 and WS_2/SiS_2. Each configuration represents a distinct stacking arrangement based on the vertical alignment of atoms—for example, configuration (a) has Mo (or W) atoms aligned above Si atoms, while (b) aligns S atoms over S atoms; other configurations vary by lateral shifts or inversion of atomic sublayers. This picture shows the atomic layers in the wrong order from what Zhang et al.[382] indicated they should be. We found the binding energy as shown in Table 6-3 for each stacking configuration to see how stable these heterojunctions are from a thermodynamic point of view. Usually, a greater negative binding energy means that the arrangement is more stable. One fundamental approach to tell how stable a system is to look at its binding energy. Many

studies have shown that binding energy and structural stability are linked[383]. For example, structures with higher negative binding energies are less likely to change shape or break apart into different phases. The negative binding energy reveals that constructing MoS_2/SiS_2 and WS_2/SiS_2 van der Waals heterostructures gives off heat and is energetically beneficial. The layers are more stable and have stronger van der Waals interactions when the negative binding energy is bigger and the interlayer gap is less (see Table 6-3). Our study shows that the most stable configuration is when an atom in the top monolayer is directly above an atom in the bottom monolayer. The stacking pattern shown in Figure 6-2(b) makes it easier for atomic orbitals to overlap. This makes interactions between layers stronger and the structure of the heterostructure more stable. These results indicate how essential the stacking order is for modifying the electrical and optoelectronic properties of MoS_2/SiS_2 and WS_2/SiS_2 vdW heterostructures. Table 6-2 presents the calculated lattice constants, electronic band gaps (from PBE and HSE06 functionals), vacuum level difference (ΔV), and work function (Φ) for these heterostructures.

Table 6- 2 Calculated electronic properties of the MoS_2–SiS_2 and WS_2–SiS_2 heterostructures, including PBE and HSE06 band gaps, potential difference ΔV (in eV), and work function Φ (in eV).

Parameters a (Å)	MoS_2/SiS_2	WS_2/SiS_2
E_9 (PBE) (eV)	0.92	1.67
E_9 (HSE06) (eV)	0.81	1.51
V	3.45	3.91
Φ	5.35	5.71

To evaluate the thermodynamic stability of the MoS_2/SiS_2 and WS_2/SiS_2 heterostructures, the binding energy ($E_{Binding}$) was calculated using the formula:

$$(E_{Binding}) = H_{eterostructures} - (E_{monolayer\,1} + E_{monolayer\,2}) \quad (6\text{-}3)$$

Where, $H_{eterostructures}$ is the total energy of relaxed heterostructures, $E_{monolayer\,1}$ and $E_{monolayer\,2}$ are the total energies of the isolated MoS_2 (or WS_2) and SiS_2 monolayers, respectively,

determined within the identical unit cell parameters. A lower binding energy means that the layers interact more strongly with each other and the structure is more stable. Table 6-3 shows a summary of the results for several stacking arrangements.

Table 6- 3 Calculated binding energies of MoS_2/SiS_2 and WS_2/SiS_2 vdW Heterostructures for different possible stacking.

Stacking	E_bA	E_bB	E_bC	E_bB	E_bE	E_bF
MoS_2–SiS_2	-0.959	-0.920	-0.991	-0.901	-0.922	-0.971
WS_2–SiS_2	-0.881	-0.819	-0.972	-0.891	-0.839	-0.903

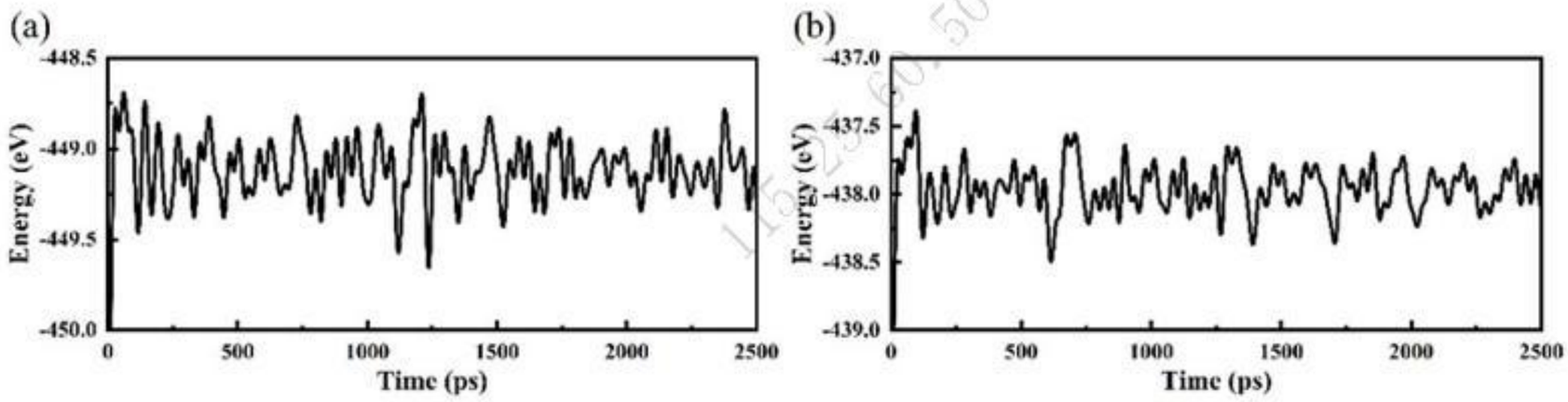


Figure 6- 3 Thermal stability analysis of (a) MoS_2/SiS_2 and (b) WS_2/SiS_2 vdW heterostructures, respectively.

Thermal stability under ambient circumstances is crucial to guarantee the practical relevance of the intended heterostructures. We investigated the thermal stability of the MoS_2–SiS_2 and WS_2–SiS_2 van der Waals heterostructures by ab initio molecular dynamics (AIMD) simulations[384, 385]. No bond breakage or structural distortion was seen as illustrated in Figure 6-3 following exposing the systems to a simulation run of **5 picoseconds** with a time step of **1 femtosecond**, at 300 K. The designs kept their symmetry all around, and their free energy variances were few. These results verify that room temperature thermally stable vdW for both MoS_2/SiS_2 and WS_2/SiS_2. For future nanoelectronics and optoelectronic device integration, these heterostructures so show interesting prospects. Bader charge analysis was done on both heterostructures to learn more about the electrical interactions at the interface. The results show that the SiS_2 layer gives the MoS_2 layer around 0.18 |e| of charge and the WS_2 layer about 0.22 |e| of charge in the two systems. This transfer of charge between layers shows that an internal electric field has formed. This makes it easier for charges to separate and supports

the type-II band alignment that was seen. These results add to the evidence that MoS_2/SiS_2 and WS_2/SiS_2 heterostructures could be useful for photocatalytic water-splitting.

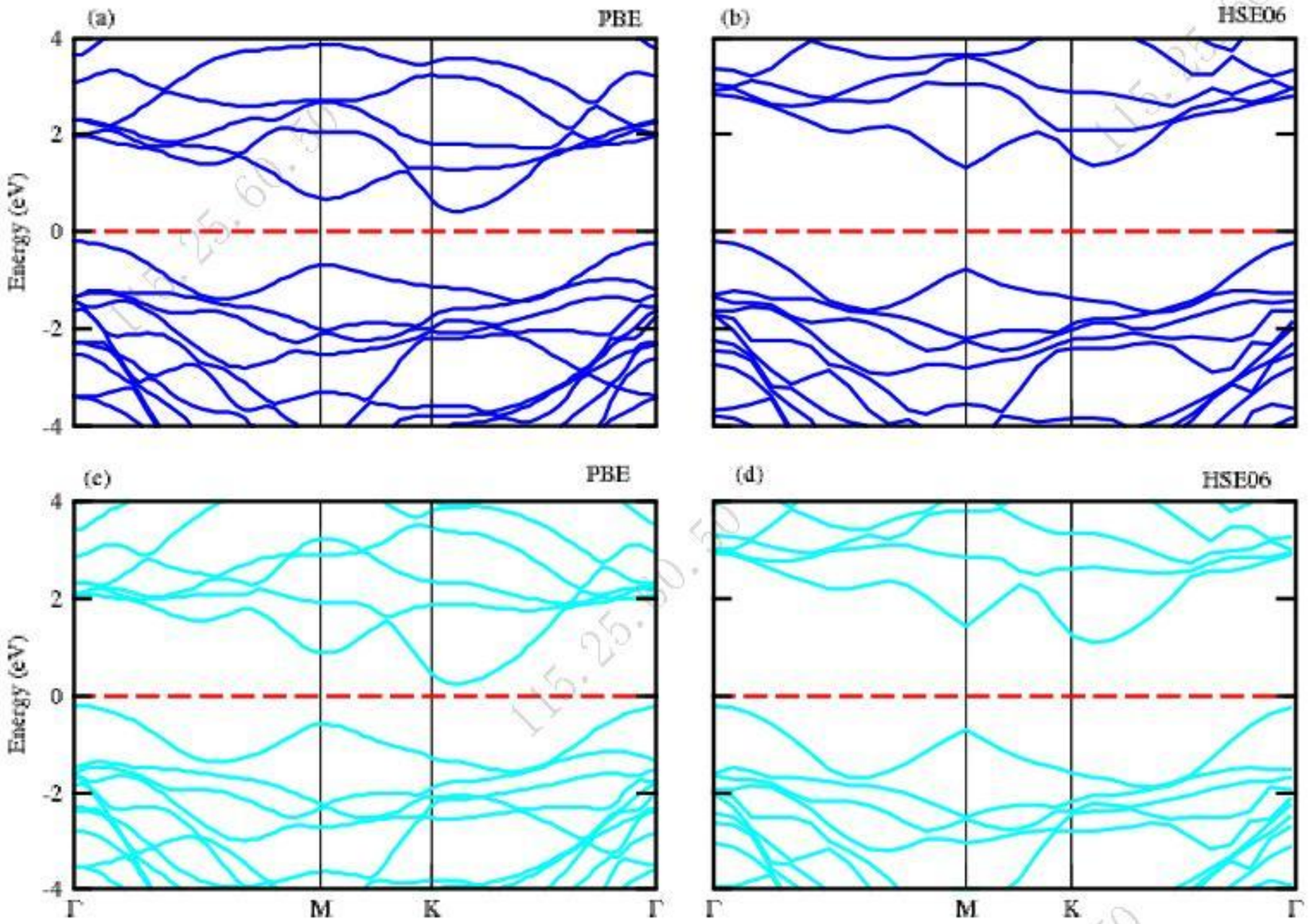


Figure 6- 4 Electronic band structures of van der Waals heterostructures: (a, b) MoS_2–SiS_2 and (c, d) WS_2–SiS_2, calculated using PBE and HSE06 functionals as indicated within the figure.

Band structures computed utilizing both semi-local and hybrid functionals helped one to obtain understanding of the electrical behavior of the suggested heterostructures. Computed using PBE and HSE06 functionals, the electronic band structures of the MoS_2/SiS_2 vdW heterostructures are illustrated in Figure 6-4(a–d). At the Γ-point, both heterostructures show a direct bandgap despite the indirect bandgap character of one of the individual monolayers, this is advantageous for effective radiative recombination of charge carriers. Their possible usage in optoelectronic devices is improved by this direct character. Particularly, Figure 6-4(a, c) shows the PBE-calculated band structures, Figure 6-4(b, d) shows the more exact HSE06 results. The change in bandgap values shows even more the tunability and band alignment control possible by vdW heterostructure development. We figured out how well the MoS_2/SiS_2 and WS_2/SiS_2 heterostructures fit together by looking at the lattice misfit between the monolayers. The best in-plane lattice constants were about 3.18 Å for MoS_2, 3.15 Å for WS_2, and 3.24 Å for SiS_2. The lattice mismatch for MoS_2/SiS_2 is

roughly 1.86%, and for WS_2/SiS_2 it is 2.79%. This is significantly below the 5% threshold that most people agree on for van der Waals heterostructures. These low mismatch values make sure that the heterostructures can be modelled without putting a lot of stress on them, which keeps the structure stable and the interlayer interaction realistic.

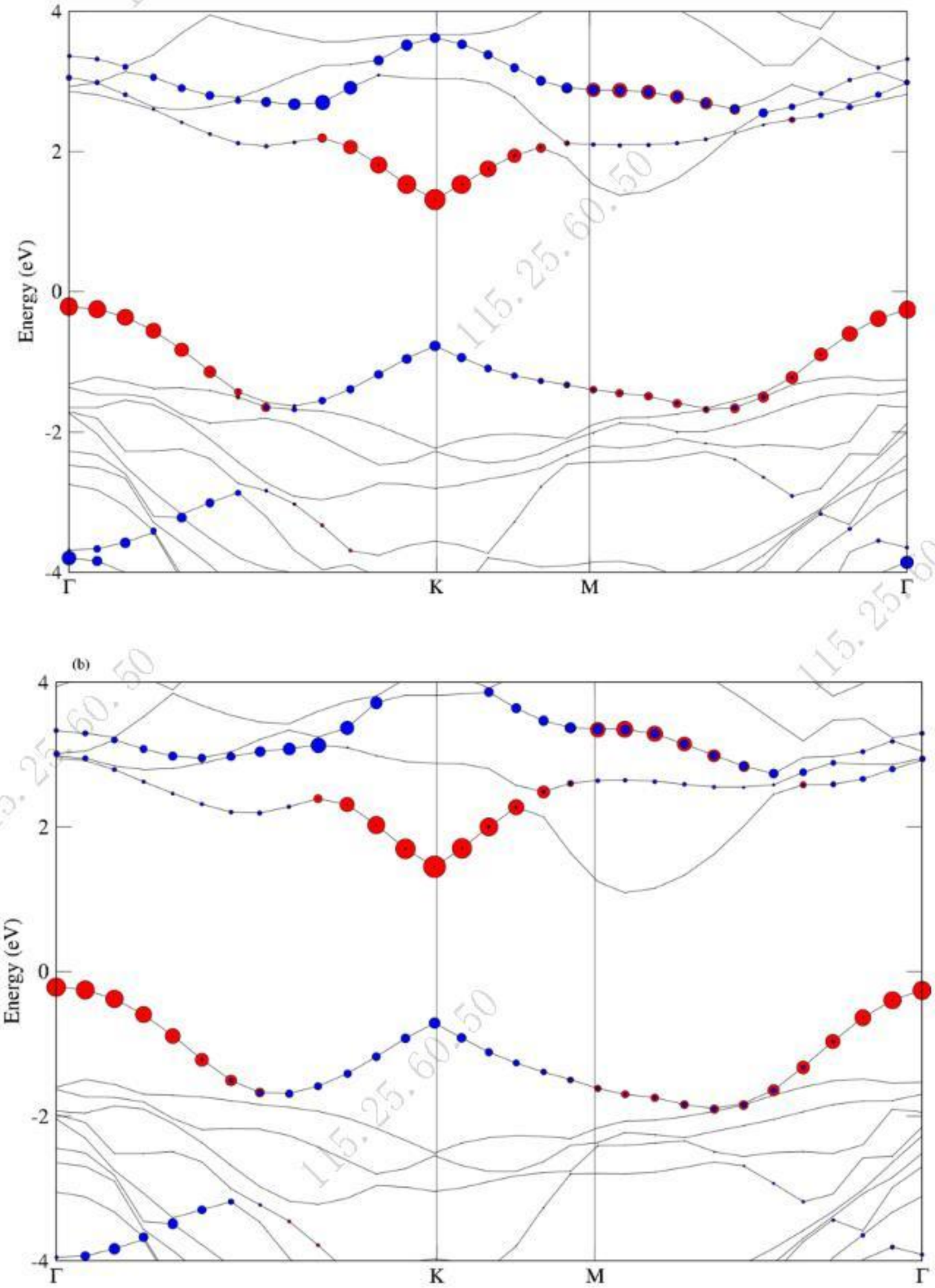


Figure 6- 5 Weighted band structures of MoS_2/SiS_2 and WS_2/SiS_2 van der Waals heterostructures, with the red dotted lines representing the Mo (W) dz^2 states and blue indicating the Si pz states.

The computed weighted band structures displayed in Figure 6-5(a, b) clearly reveal the thorough representation of type-II band alignment for the MoS_2/SiS_2 vdW heterostructures. These findings validate the existence of type-II band alignment in both heterostructures whereby CBM originates from separate constituent monolayers and VBM. Effective separation of photogenerated charge carriers is made possible by this spatial separation at the interface[386] producing an internal electric field. In particular, the associated holes travel towards the valence band of MoS_2 whereas the photogenerated electrons migrate from the MoS_2 or WS_2 layer to the conduction band of the SiS_2 layer. Under this arrangement, the layer causing the VBM acts as the hole donor while the one causing the CBM operates as the electron acceptor. The possible use of MoS_2–SiS_2 heterostructures in solar energy conversion and light-harvesting technologies is supported by this charge separation mechanism.

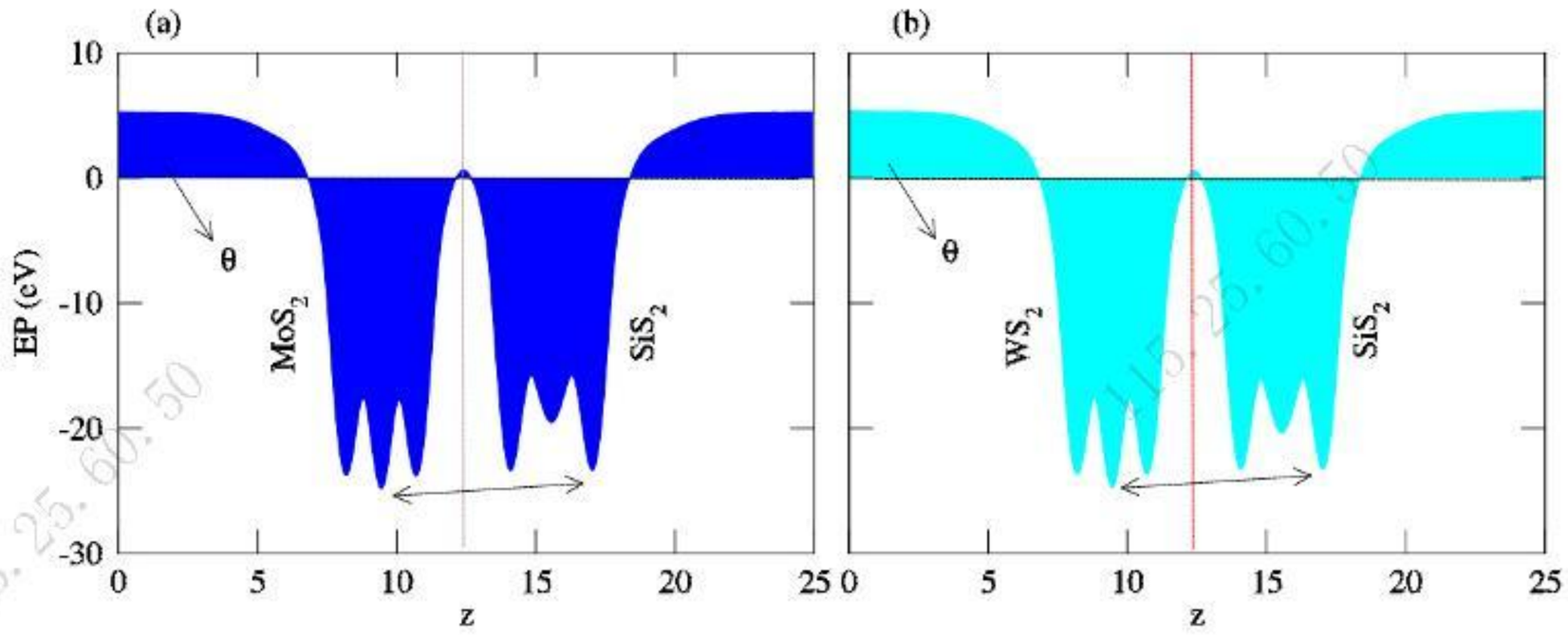


Figure 6- 6 Electrostatic potential profiles of (a) MoS_2/SiS_2 and (b) WS_2/SiS_2 vdW heterostructures, with the work function values also presented.

Weak vdW interactions between MoS_2 (or WS_2) and SiS_2 layers are verified by the slow change in electrostatic potential from one layer to the other. In figure 6-6 We assessed the electrostatic potential profiles along the out-of- plane direction in order to investigate charge redistribution more closely. Indicating a directional charge transfer from SiS_2 to the TMD layers, the MoS_2 and WS_2 layers in both heterostructures show deeper potential wells than the corresponding SiS_2 layers. An inherent electric field within the heterostructure comes from this interlayer charge transfer, which helps to spatial separate photogenerated electron–hole pairs a necessary ability to improve photocatalytic efficiency. Furthermore, the possible

drop (ΔV) at the interface suggests that the excitonic characteristics of the heterostructures vary from those of the single monolayers, hence enhancing charge carrier dynamics in the system. These results confirm the possibilities of MoS_2/SiS_2 and WS_2/SiS_2 heterostructures in optoelectronic and energy-harvesting use. Under light irradiation, the built-in electric field aids efficient carrier separation even more.

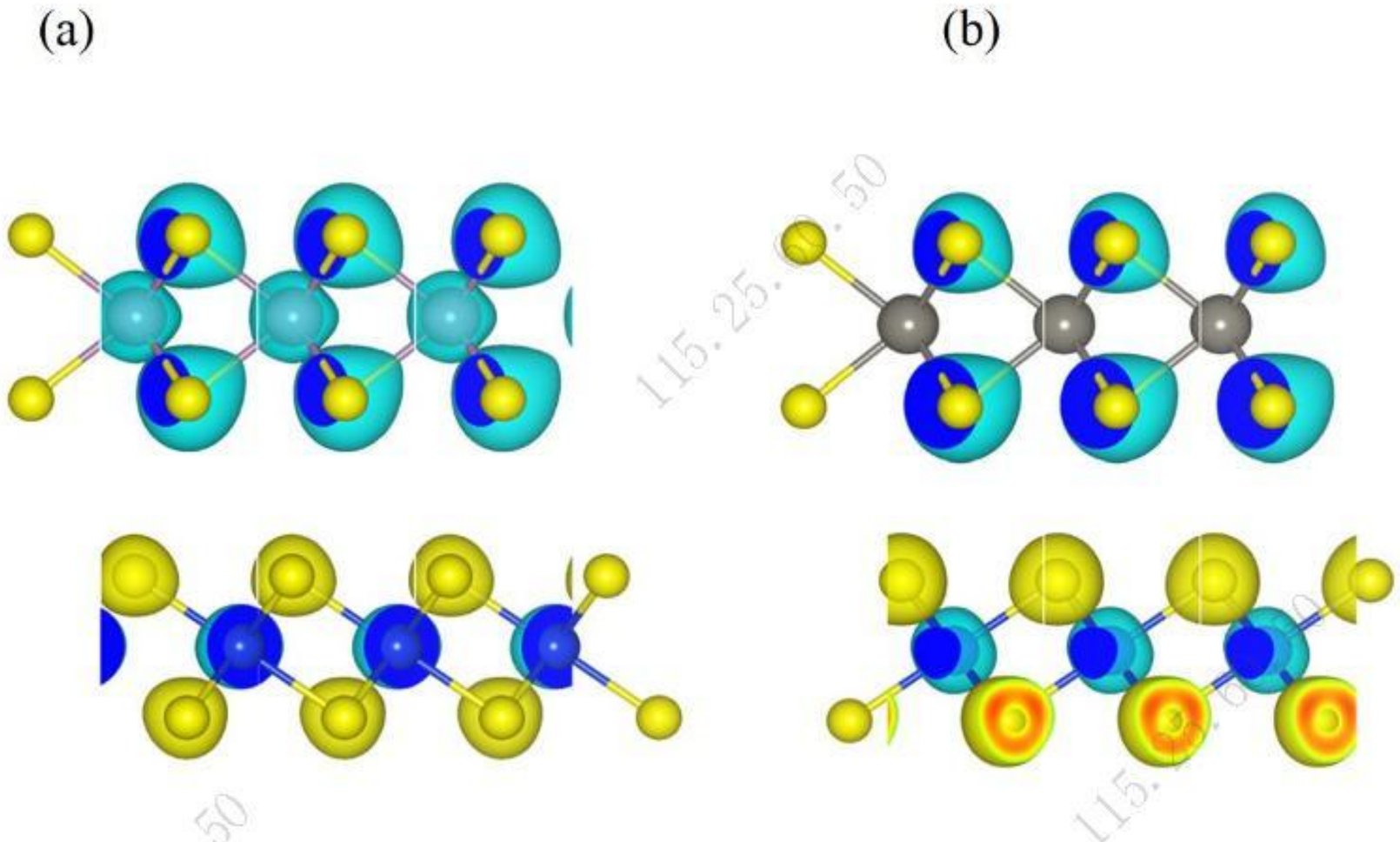


Figure 6- 7 Charge density difference for (a) MoS_2/SiS_2 and (b) WS_2/SiS_2 vdW heterostructures.

The charge redistribution noted at the interface in Figure 6-7 verifies that the SiS_2 layer provides electrons to the MoS_2 and WS_2 layers in developing heterostructures. This produces n-type doping in the TMD layers and p-type doping in the SiS_2 layer, therefore generating internal electric dipole increasing charge separation across the interface. Using Bader charge analysis, one found a minor electron transport from SiS_2 to MoS_2 and WS_2 that quite well matched the spatial pattern displayed in the graphs. The band alignment and carrier dynamics are much influenced by this charge mobility, hence increasing the potential of these heterostructures in optoelectronic and photocatalytic uses. Although the cyan zones surrounding Si atoms suggest areas of electron loss, the yellow lobes around the Mo and W atoms in Figure 6-7 hint to areas of great electron accumulation. This asymmetric distribution helps the built-in electric field to assist effective separation of photogenerated charge carriers by visually checking the direction and degree of interfacial charge transfer.

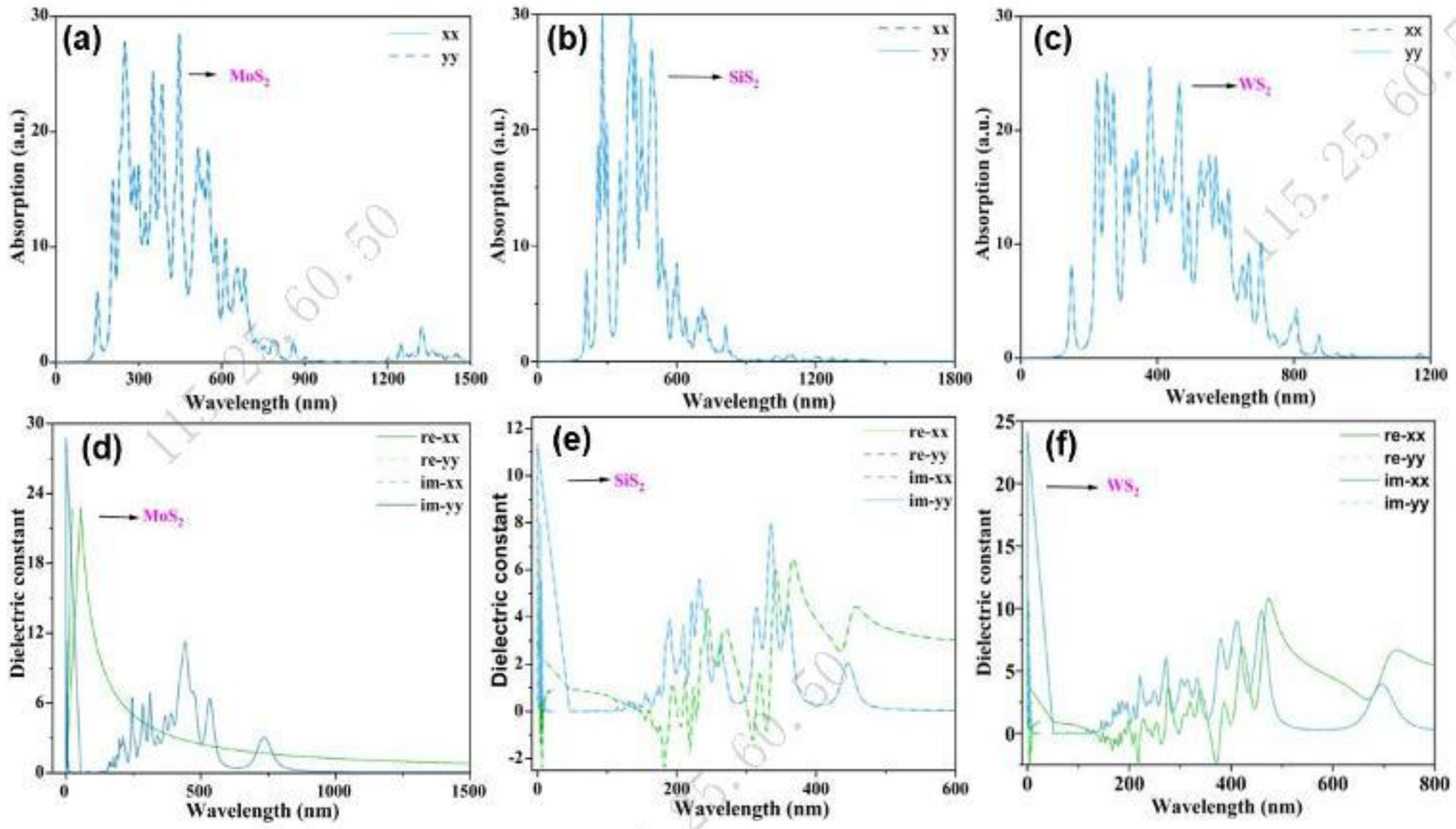


Figure 6- 8 Optical properties of monolayers WS_2, SiS_2, and MoS_2. Panels (a–c) illustrate the optical absorption spectra of every monolayer in both visible and UV bands, Panels (d–f) contain the dielectric function components, real and imaginary parts that define the polarization response of the materials to approaching light.

The possible photonic and energy uses of 2D materials are strongly influenced by their optical response. Figure 8 shows the optical characteristics of MoS_2, SiS_2, and WS_2 monolayers by means of their dielectric function components 6-8(d–f) and absorption spectra 6-8(a–c). SiS_2 shows clear absorption mostly in the UV area, implying low visible light activity, MoS_2 and WS_2 show significant absorption peaks in the visible band (400–700 nm), thereby suggesting their possible use in solar energy collecting in panels 6-8(a–c). In-plane isotropic optical response is reflected in similar absorption behavior of the xx and yy-polarisation orientations. The dielectric function has both real and imaginary components found in panels 6-8(d–f). While coinciding with interbond electronic transitions, MoS_2 and WS_2 show significant values of the imaginary component in the visible spectrum, SiS_2 has wider and more strong dielectric responses in the UV area. Actually, the real components are pointing out important polarizing effects at shorter wavelengths. Although SiS_2 would be more suited for UV-sensitive applications, our results usually show the fit of MoS_2 and WS_2 for visible-light-driven optoelectronic devices. The unique optical signatures reveal the

several light-matter interaction properties of every monolayer, therefore providing adaptability in the construction of wavelength-specific optoelectronic devices.

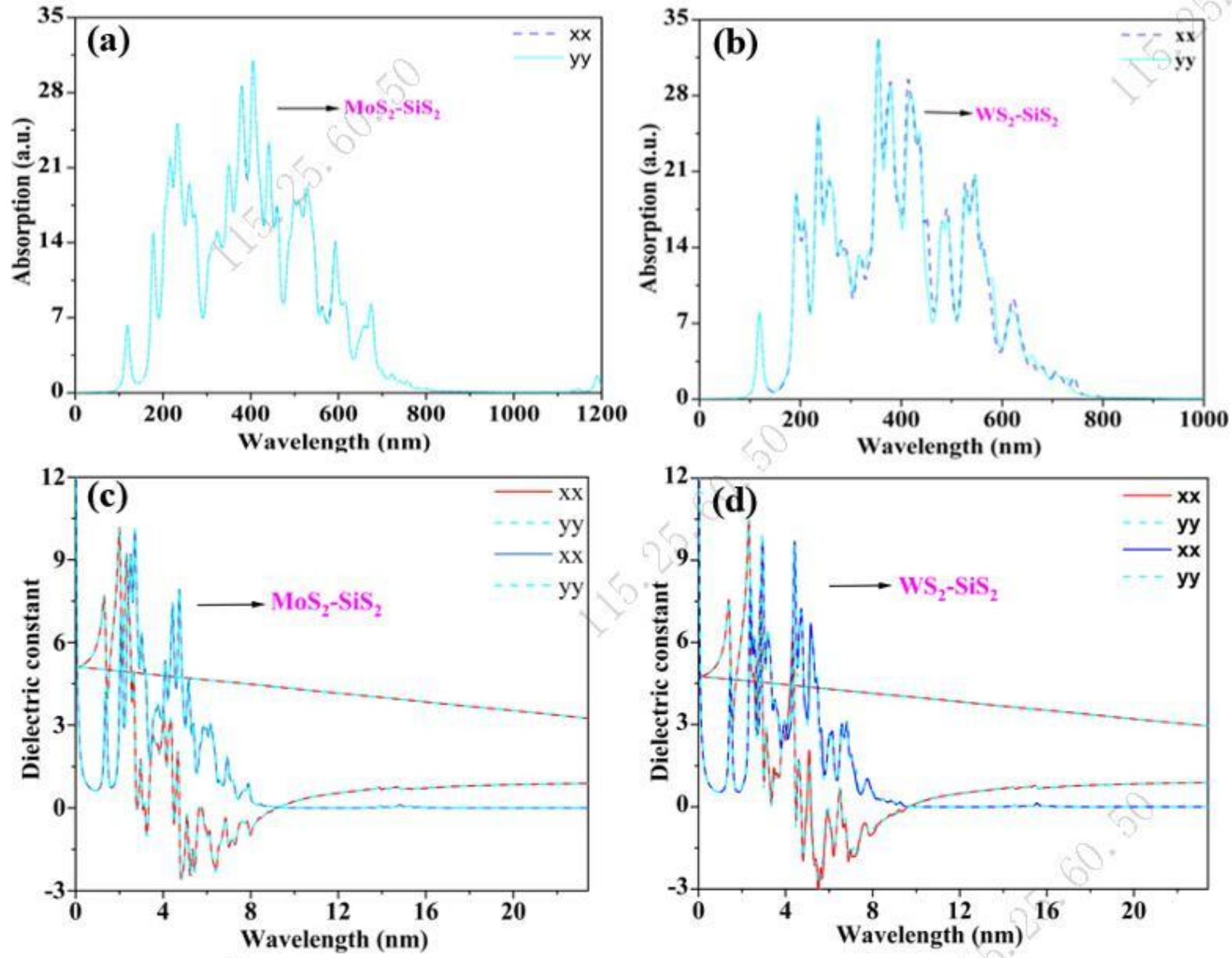


Figure 6- 9 Optical properties of MoS_2/SiS_2 and WS_2/SiS_2 based on vdW heterostructures. Reflecting significant absorption in the visible light spectrum, (a, b) absorption spectra of MoS_2/SiS_2 and WS_2/SiS_2 respectively highlight their frequency-dependent polarization response and interbond transitions for the same heterostructures, (c, d) dielectric function components, real and imaginary parts.

Using MoS_2/SiS_2 heterostructures and WS_2/SiS_2, the light interaction behavior spanning many wavelengths has been investigated. Particularly below 700 nm, the absorption spectra in Figures 6-9(a) and 6-9(b) remarkably demonstrate UV to visible range absorption for both MoS_2/SiS_2 heterostructures. Their amazing light-harvesting characteristics suggest their suit for photodetectors and solar energy conversion systems. Reflecting significant interbond electronic transitions, Figures 6-9(c) and 6-9(d) exhibit several peaks in the real and imaginary sections at lower wavelengths in the dielectric function graphs. The expected component is energy dissipation by absorption; the actual amount reveals material electric energy storage. Especially after 6 nm, both components stabilize gradually with increasing wavelength, implying little optical activity in the near-infrared range. These results support

the rather sensitive to incident light attractive candidates for utilization in optoelectronic and photocatalytic systems. Their shifting optical characteristics thus offer enormous opportunities for integration into next photonic and energy systems.

6.4 Summary of the Chapter

We used first-principles simulations to look into the structural, electrical, and optical aspects of MoS_2/SiS_2 and WS_2/SiS_2 van der Waals heterostructures. The monolayers have the right band alignments and optical responses. MoS_2 and WS_2 absorb a lot of light in the visible range, whereas SiS_2 absorbs a lot of light in the UV range. Ab initio molecular dynamics simulations show that the heterostructures stay stable at room temperature. Binding energy study shows that certain stacking patterns create combinations that are energetically favorable. Even though the interfacial electric field (ΔV) is minimal, these heterostructures could be useful in the future for nanoscale optoelectronic and solar devices. Our results add to the increasing body of research on 2D heterostructure materials for systems that use energy.

7 Conclusions and Future Prospects

7.1 Conclusion

This thesis discussed the applications of optical, electrical, photoelectric, and thermoelectric properties of 2D materials in electronic, optoelectronic and energy conversion systems, which is essential in the development of technologies. A careful study of several 2D materials, such as bilayer 12-like phase borophene, single-layer $\boldsymbol{\beta}_{12}$-like graphyne, and van der Waals heterostructures, has brought major discoveries that demonstrate how these materials can be utilized in next-generation devices. Optical and thermoelectric properties of bilayer $\boldsymbol{\beta}_{12}$-like phase borophene deposited on Cu (111) films showed excellent results and indicated that it can be used in energy-efficient uses. The study of γ-graphyne, likewise, showed that each layer has its own layer-specific optical characteristics, providing new opportunities in optimizing material properties to a certain optical and electronic task. Moreover, the study of MoS_2/SiS_2 and WS_2/SiS_2 van der Waals heterostructures demonstrated their potential structural and optoelectronic characteristics, which have the potential to be applied to new energy features, specially creating high-performance devices. Taken together, these results underline the multifunctionality of 2D materials and their high potential in the technological progress in the future. The information obtained in this study creates new opportunities in the material development and optimization of materials which are not only efficient in terms of their performance but also sustainable in their energy usage. The tendency of the future work will most probably be directed at the incorporation of these materials in real equipment, its scaling, stability, and compatibility with the existing technologies.

7.2 Innovation

This study has presented a number of unconventional contributions to the 2D materials research especially in optical, electrical, photoelectric, and thermoelectric characteristics. An important one is the enhanced synthesis of bilayer $\boldsymbol{\beta}$12-like phase borophene on Cu (111)

films, which has a high optical and thermoelectric quality, and opens up new opportunities in the creation of thermoelectric devices that use less energy. Also, the finding of layer-dependent optical characteristics in γ-graphyne also offers new opportunities on how the properties of materials can be accommodated to particular optical and electronic purposes and opens up the possibilities in photodetectors and optoelectronics. Moreover, the studies of MoS2/SiS2 and WS2/SiS2 van der Waals heterostructures have shown that the combination of complementary materials can improve the performance in other advanced energy applications, including high-efficiency solar cells and energy storage systems. All these innovations will help to the creation of high-performance, scalable 2D materials with optimized properties, next-generation optoelectronic and thermoelectric technologies.

7.3 Future Prospects

In the future, the study of 2D materials offers important prospects of expansion of the research in both basic science and in practical works. The future research agenda is to advance and enhance the synthesis methods further to enhance the scalability and stability of 2D materials especially on industrial applications. Also, further investigations of the atomic-level interactions, and defect engineering of these materials will be essential in order to optimize the performance of these materials in practical devices. Material integration and interface engineering Future studies will be motivated by the possible energy uses of van der Waals heterostructures, including high-efficiency thermoelectric devices and solar cells of the next generation. In addition, the long-term stability and environmental effects of these materials need to be known to make sure that these materials are used sustainably in the energy and electronic industries. With increasing demand of high-performance and energy efficient technologies, the 2D materials will increasingly be significant in determining the future of electronics, energy harvesting and optoelectronics.

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作者简历及在学研究成果

一、主要教育经历/工作经历（从大学起，到博士入学止）

起止年月	学习或工作单位	备注
2011 年 9 月至 2013 年 6 月	University of Malakand, Pakistan.	B.Sc.
2016 年 9 月至 2018 年 6 月	University of Malakand, Pakistan.	M.Sc.
2019 年 9 月至 2022 年 6 月	RIPHAH International University, Islamabad Pakistan	M. Phil
2022 年 9 月至今	在北京科技大学物理学专业攻读博士学位	Ph. D

二、在学期间所获的科研奖励

（1）两次荣获**北京科技大学优秀国际学生奖学金**

（2）荣获**北京科技大学提前毕业荣誉证书（Early Graduation Award）**

（3）**北京科技大学国际研究生十佳学术科研奖提名**（Top 10 International Research Academic Awards Nomination）

三、在学期间发表的论文

[1] Ali, S., Rahman, A. U., & Sun, M. (2024). Superior optical and thermoelectric properties of bilayer β 12-like phase borophene synthesized on Cu (111) film. Materials Advances, 5(7), 3029-3036.

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四、经历

Lecturer in Physics, Higher Education Department, Pakistan (2018–2022)

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